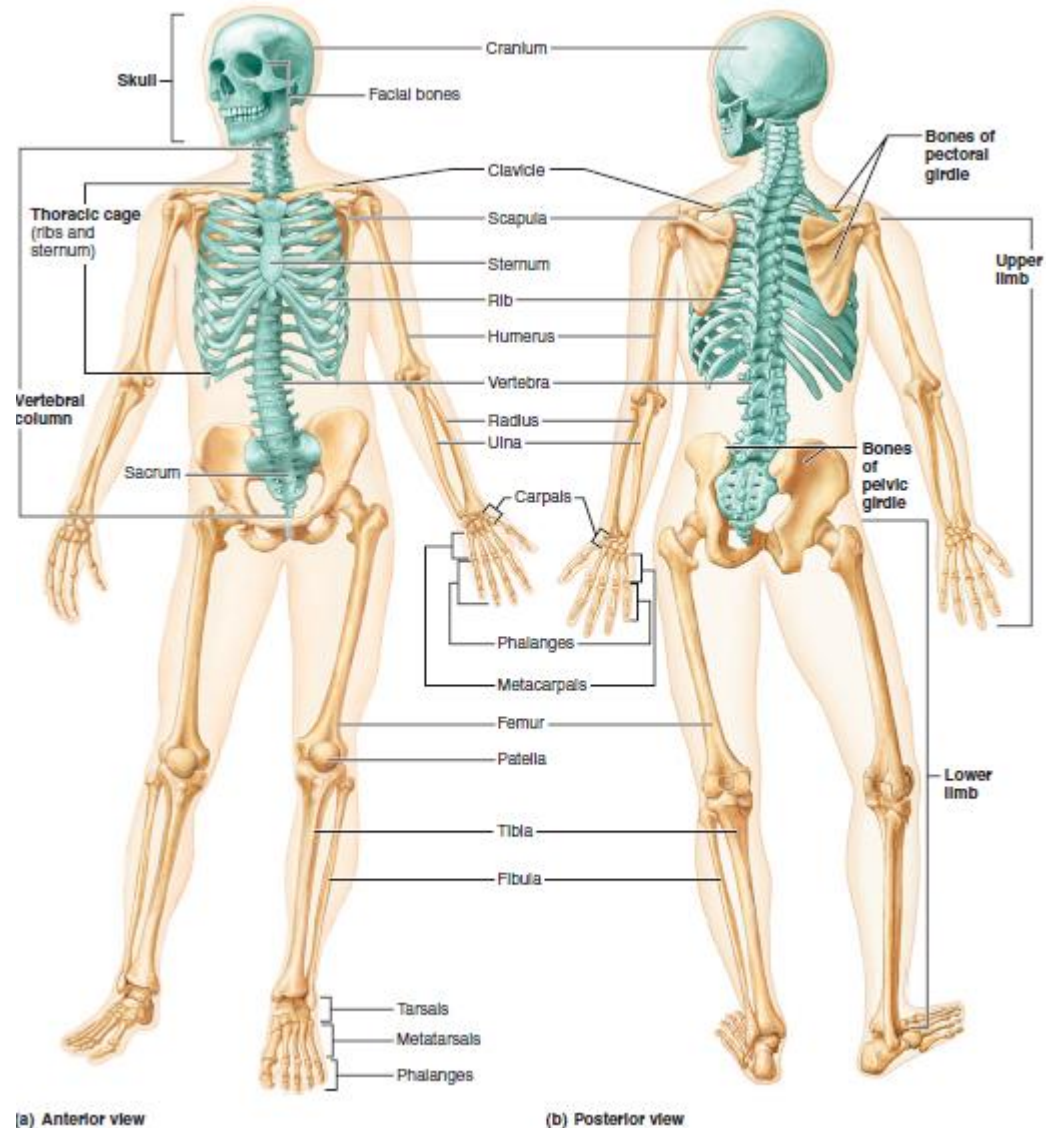


• **Anatomy of upper limb:**

- ✓ Introduction to the upper limb
- ✓ Bones of the upper limb
- ✓ Superficial structures of upper limb
- ✓ Axioappendicular Muscles
- ✓ Axilla
- ✓ Brachial plexus
- ✓ Arm
- ✓ Cubital fossa
- ✓ Forearm
- ✓ Hand
- ✓ Surface anatomy the upper limb
- ✓ Joints of the upper limb

Musculoskeletal and Integumentary system module!



Skeletal muscle

Temporalis

Masseter

Sternocleidomastoid

Pectoralis major

Biceps brachii

Abdominal muscles

Sartorius

Quadriceps femoris

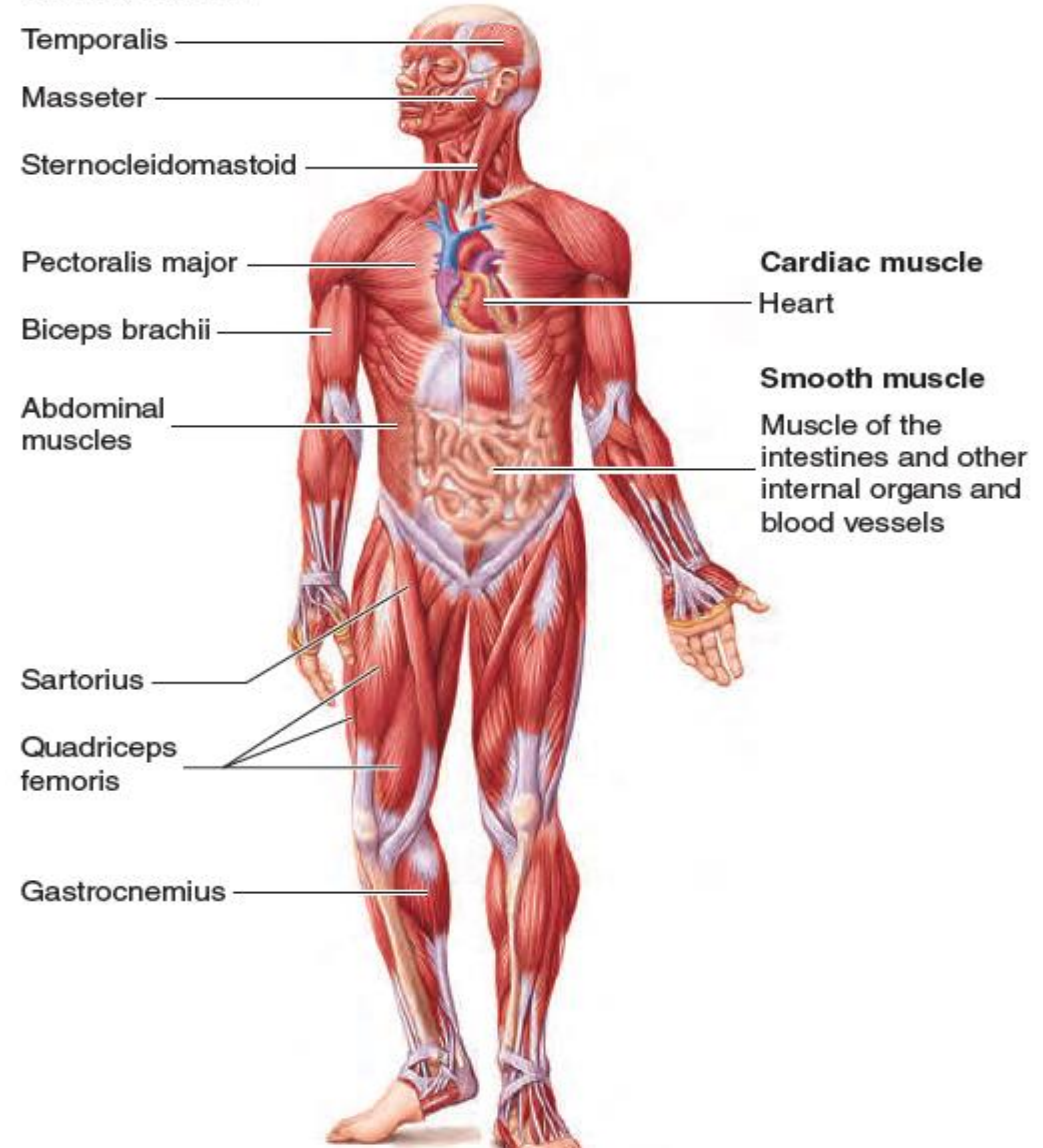
Gastrocnemius

Cardiac muscle

Heart

Smooth muscle

Muscle of the intestines and other internal organs and blood vessels



Superficial structures of upper limb

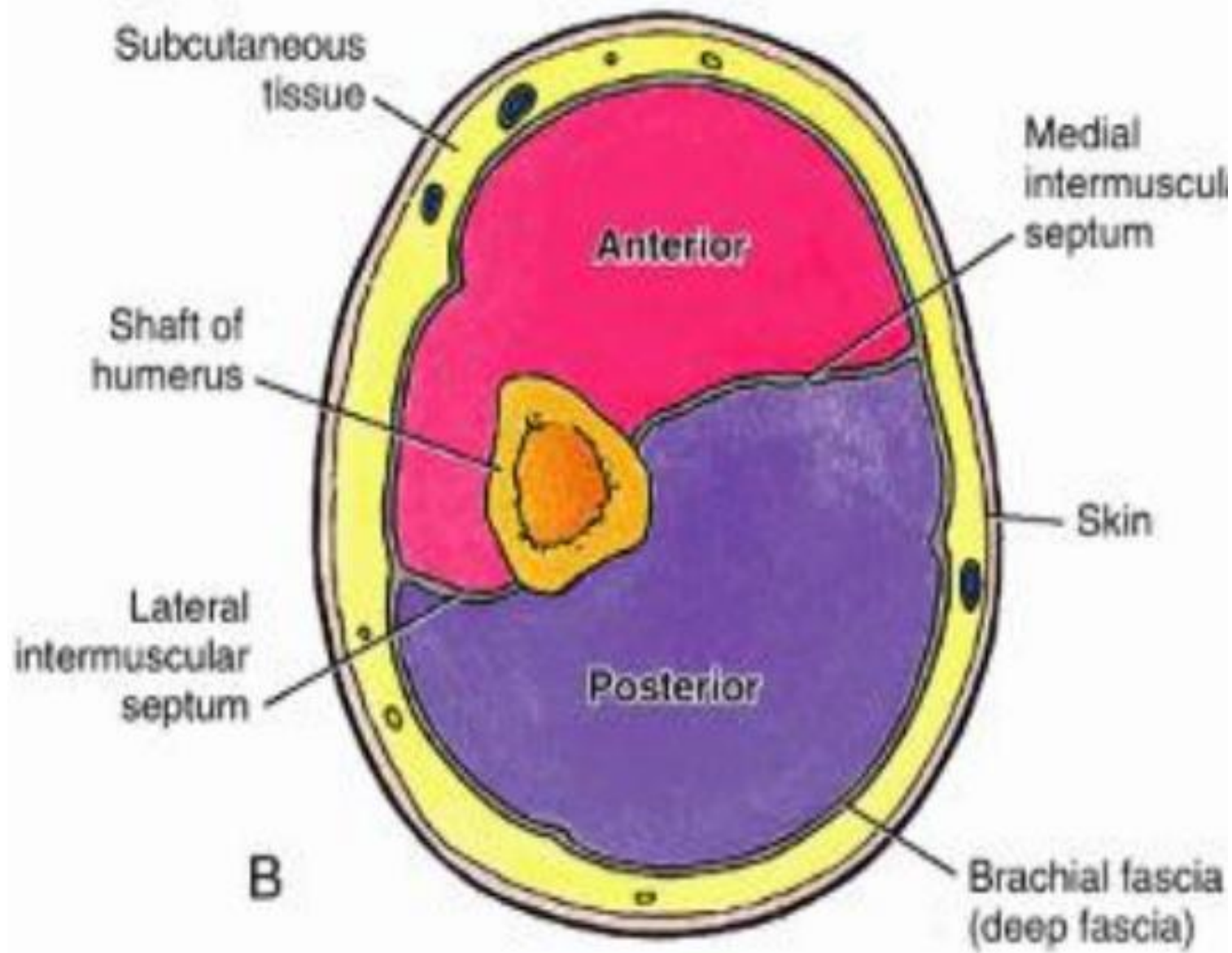
❑ **Superficial Structures of Upper Limb includes:**

- ✓ Fascia
- ✓ Cutaneous nerves
- ✓ Superficial veins
- ✓ Lymphatic drainage

Fascia of the Upper Limb

- Fascia is a tissue below the dermis layer of skin.
- The superficial fascia is a layer of connective tissue that underlies the skin and envelops the entire body, including the upper limb.
- **The fascia has two layers**
 - **Subcutaneous tissue** (superficial fascia) containing fat.
 - **It is the site used for subcutaneous injections**
 - **Deep fascia** compartmentalizing and investing the muscles.
- Deep fascia covers structures like muscle & tendon, if these are absent the deep fascia is usually attached to bone.

Fascia of the Upper Limb,,,



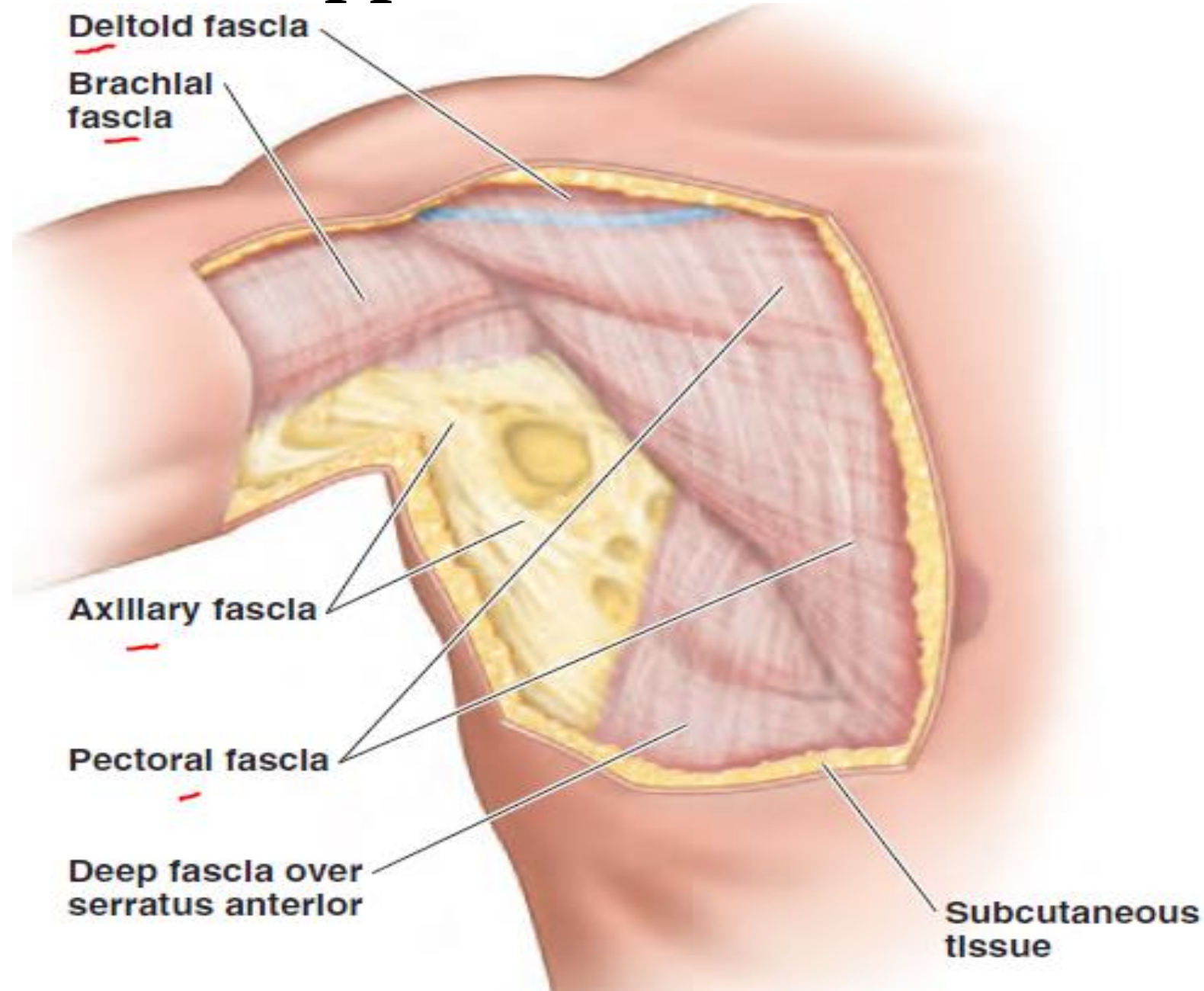
Fascia of the Upper Limb,,,

- Note! Dividing the fascia into two terms, superficial and deep is more traditional and not practicable at this time.
- **Instead**, it is better to group fasciae according to embryological origins and modes of development but this system is impractical in terms of communication in research, education, and clinical practice so it is **important a functional classification system that divides fascia** into four categories; **linking, fascicular, compression, and separating fascia.**

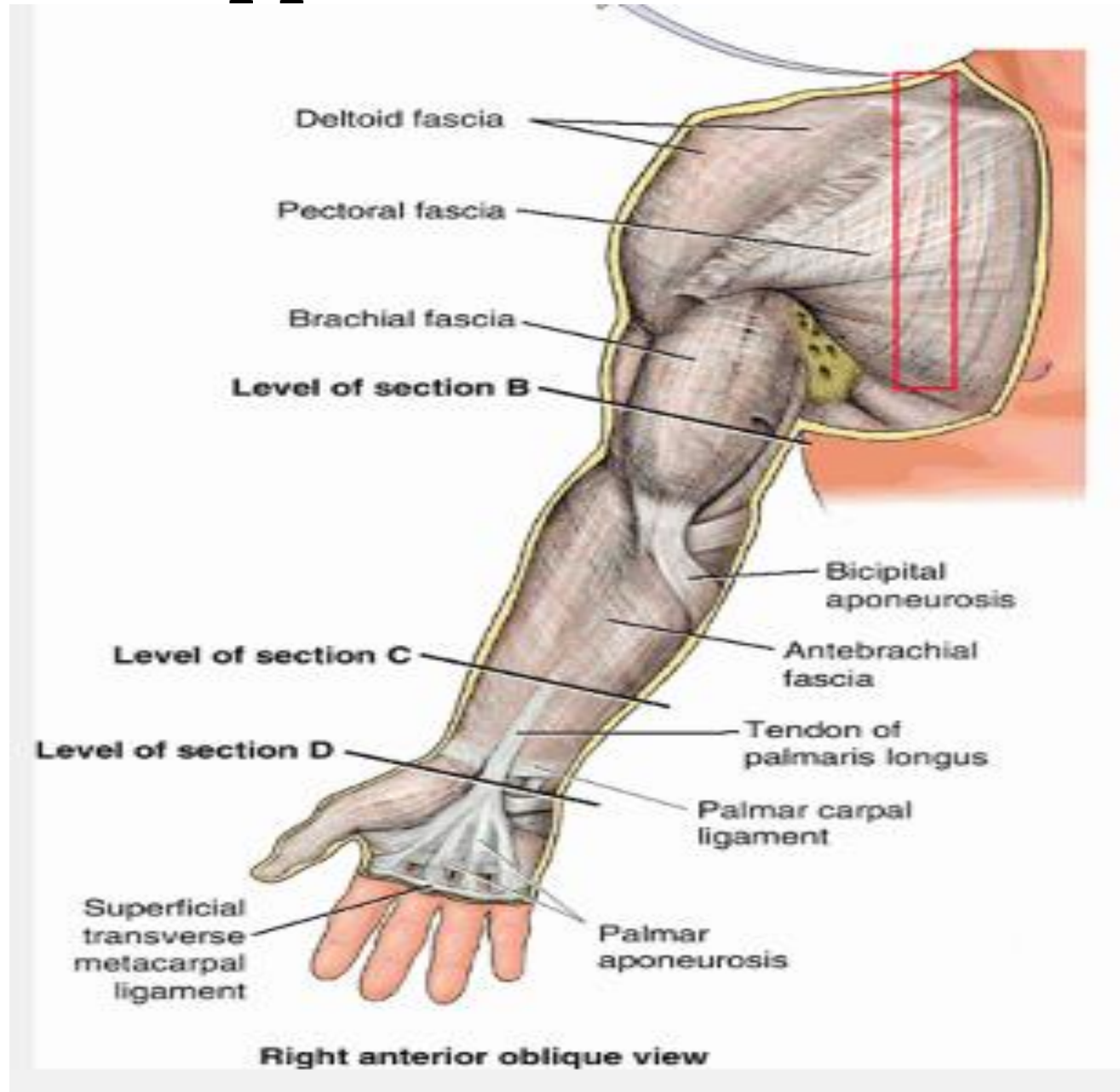
Fascia of the Upper Limb,,,

- Deep fascia of upper limb has different names in different regions of the upper limb.
- **Pectoral fascia:-** covers pectoralis major & attached to clavicle & sternum.
- Continuous with fascia of anterior abdominal wall inferiorly and axillary fascia laterally.
- **Clavipectoral fascia:-** lie under pectoralis minor muscles attached to clavicle
- **Deltoid fascia:-** covers deltoid muscle, attached to coracoid, acromion and spine of scapula.

Fascia of the Upper Limb,,

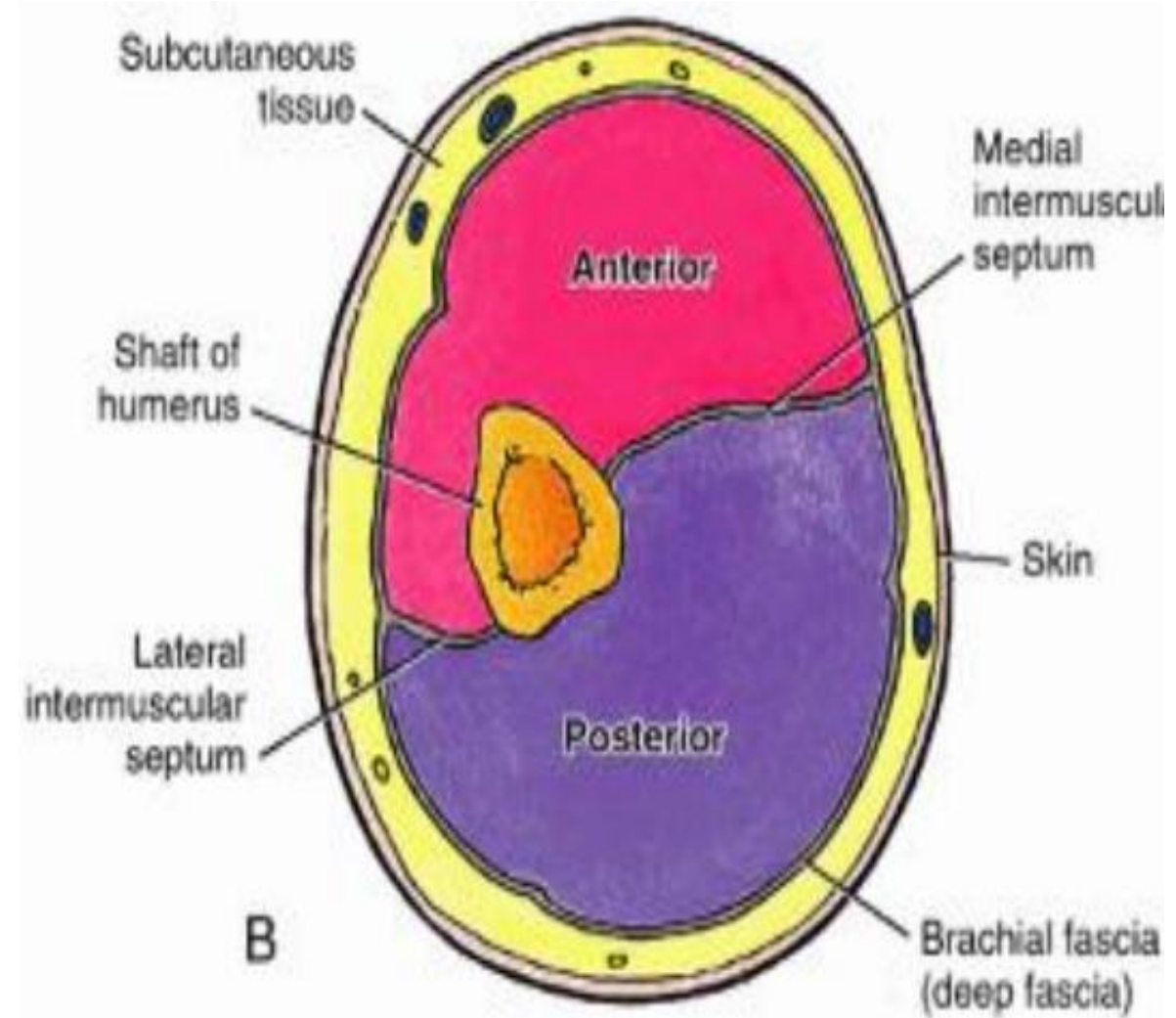


Fascia of the Upper Limb,,,

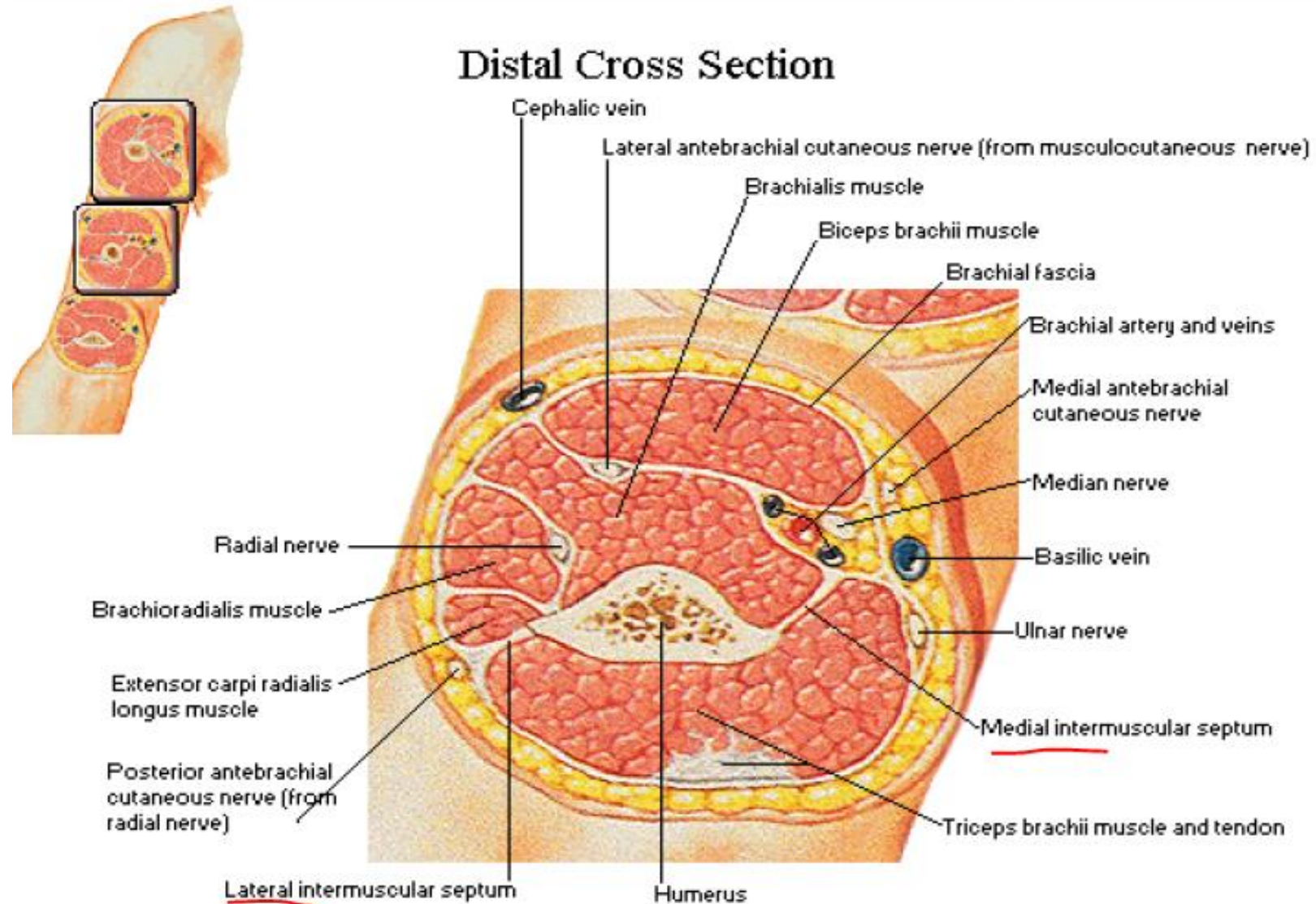


Fascia of the Upper Limb,,,

- Subscapular, Supraspinous & infraspinous similarly their name is based on the muscle they covered respectively.
- Brachial fascia:- enclosing arm, superiorly continuous with axillary, pectoral, deltoid and infraspinous fasciae
 - **Attached inferiorly with the epicondyles and olecranon process**
 - ✚ form two intermuscular septa: **medial and lateral** intermuscular septa



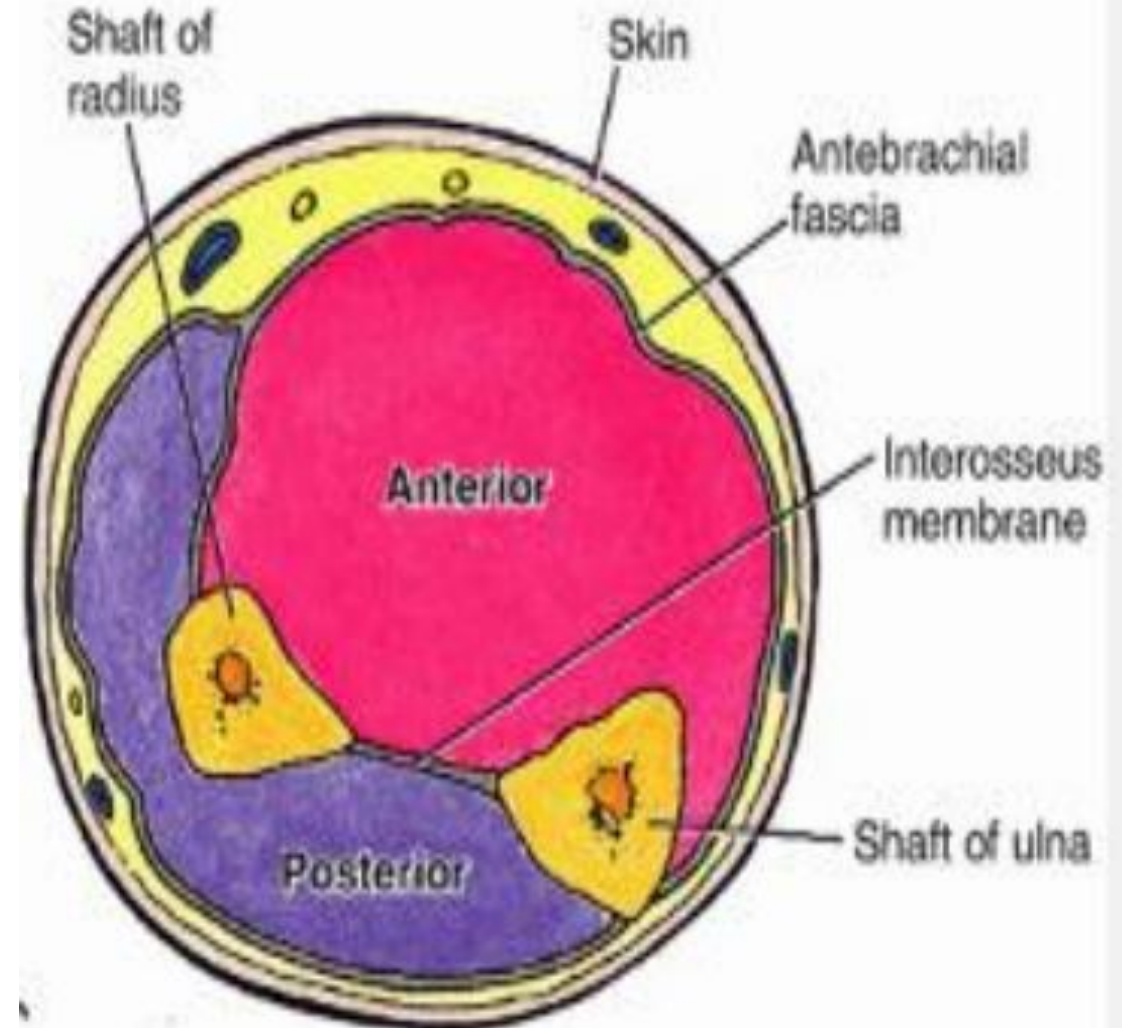
Fascia of the Upper Limb,,,



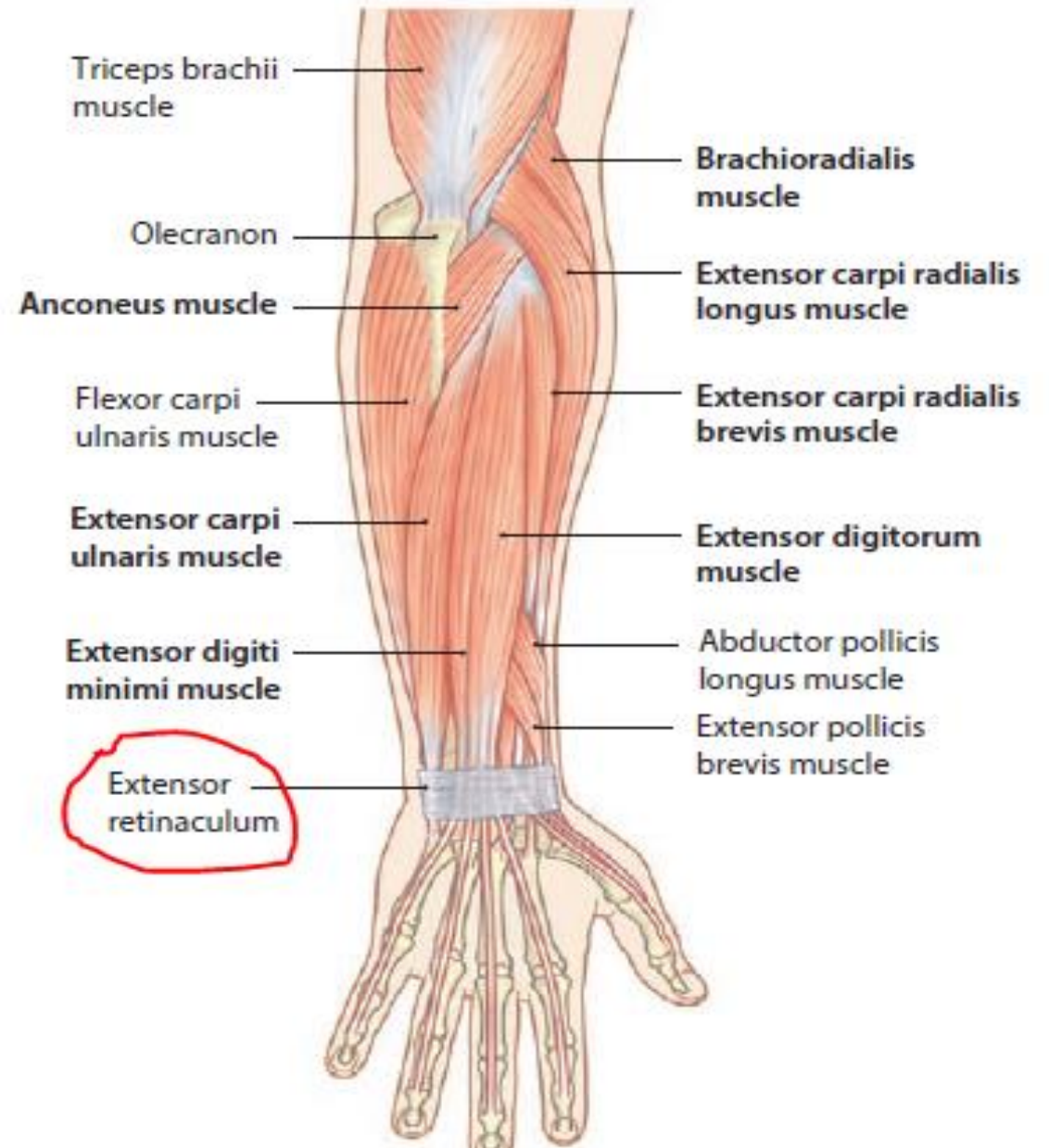
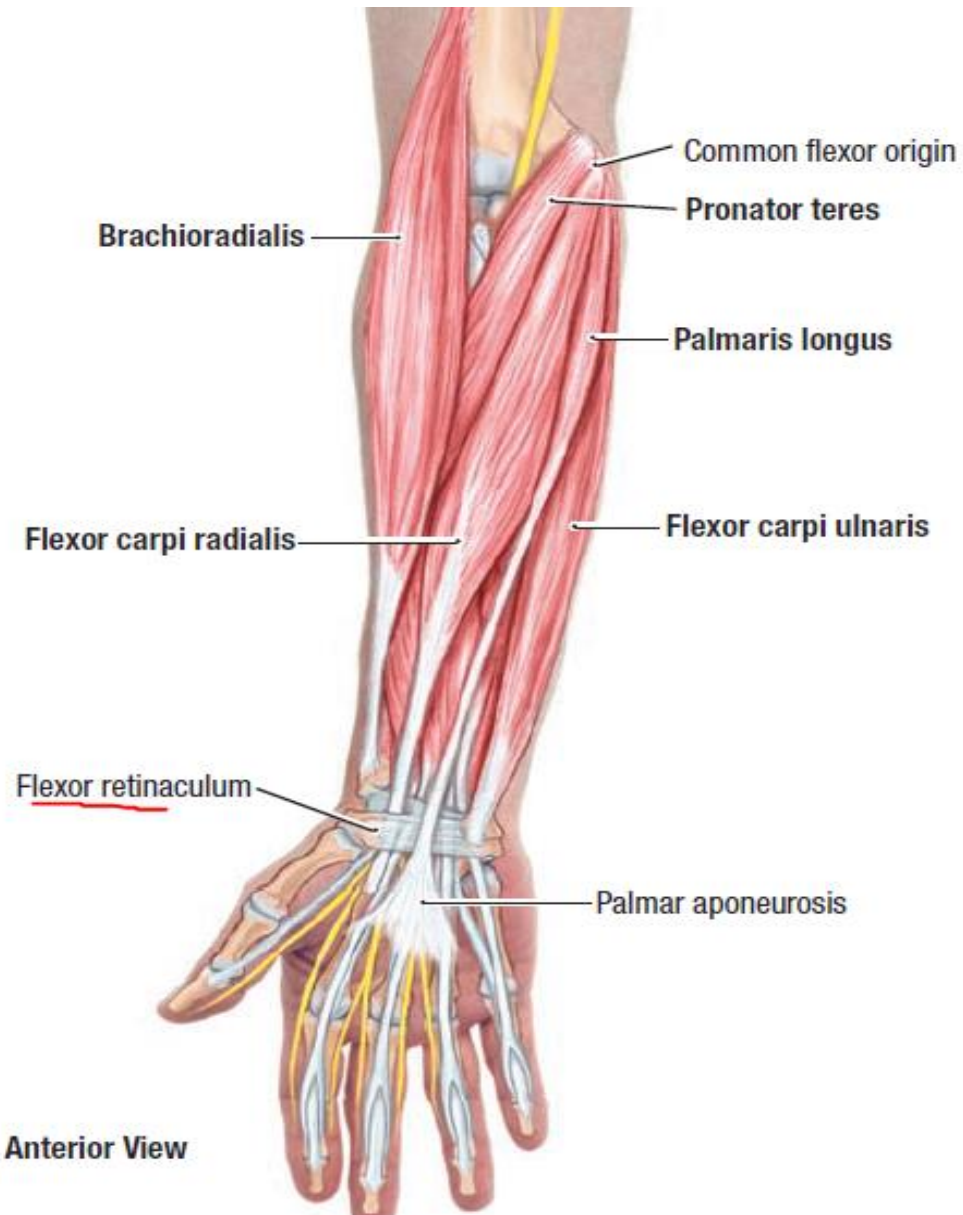
Fascia of the Upper Limb,,,

➤Antebrachial fascia:

- ✚ In the forearm
- ✚ Surrounds fascial compartments separated by interosseous membrane.
- ✚ Thickens posteriorly over the distal end to form a transverse band, **extensor retinaculum**, which holds the extensor tendons in position.
- ✚ Anteriorly at distal end forms an anterior thickenings, palmar carpal ligament and **flexor retinaculum**.



Fascia of the Upper Limb,,,



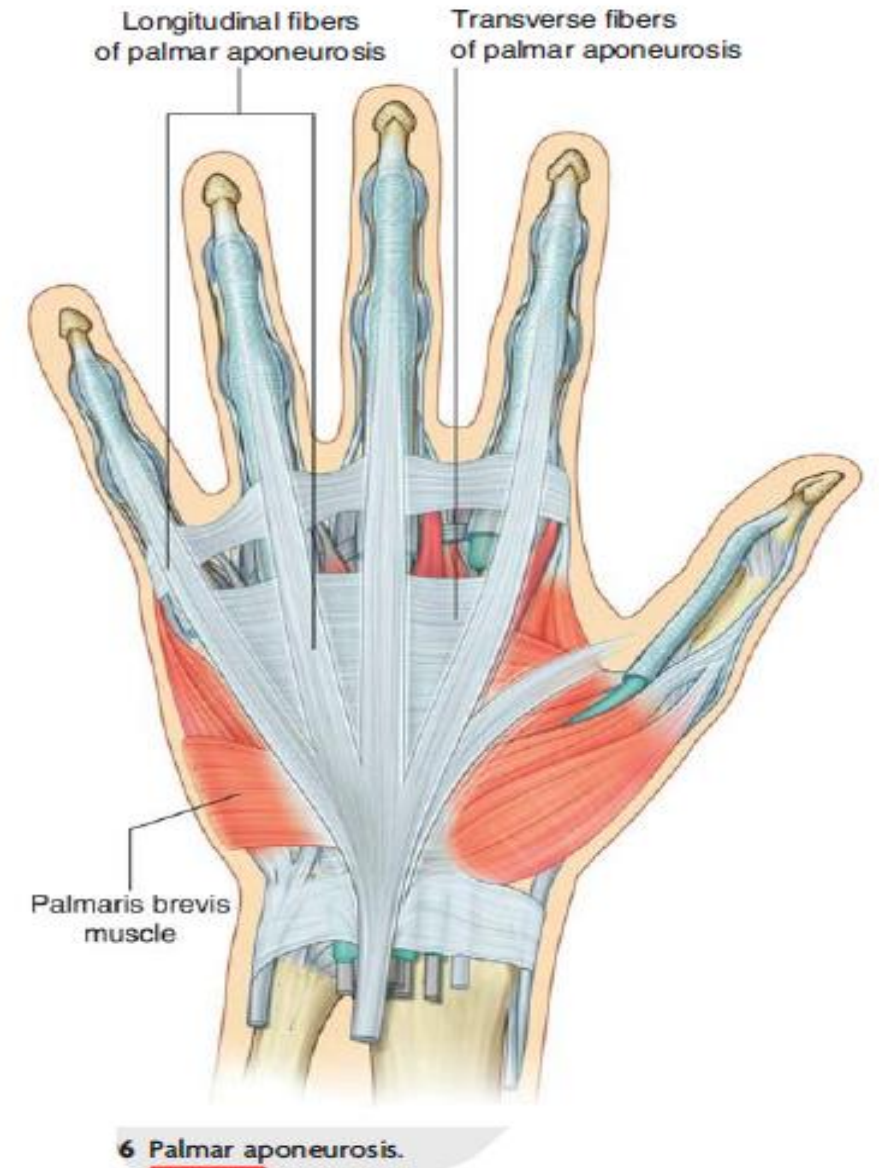
Fascia of the Upper Limb,,,

➤ Palmar fascia

✚ Extension of deep fascia beyond retinacula

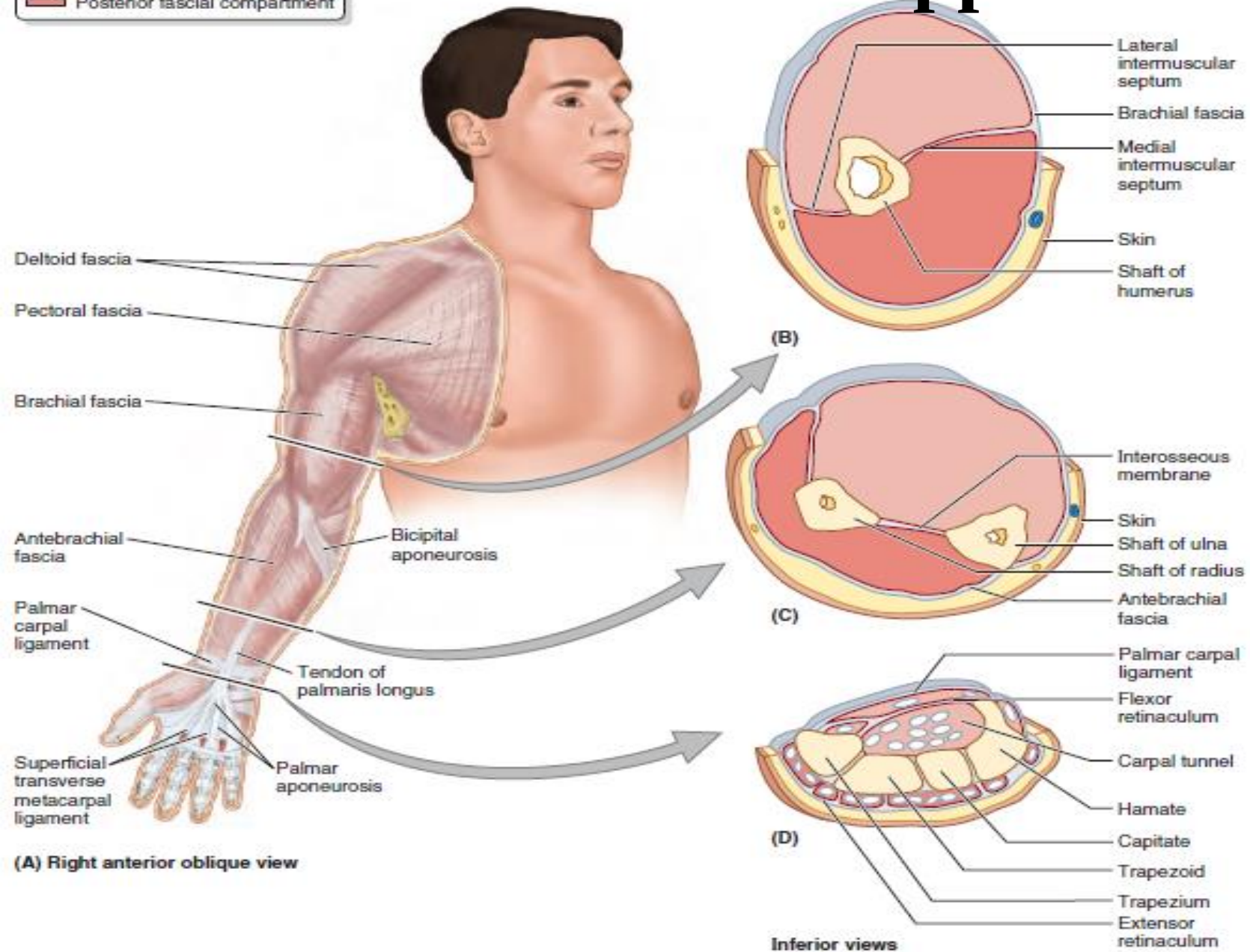
✚ Palmar aponeurosis:

- Central part, thick, triangular
- Forms four thickenings that radiate to the base of fingers.
- Superficial transverse metacarpal ligament: transverse bands distally and form the base of palmar aponeuroses.
- Ligaments extend from aponeurosis to skin, hold the skin closer.

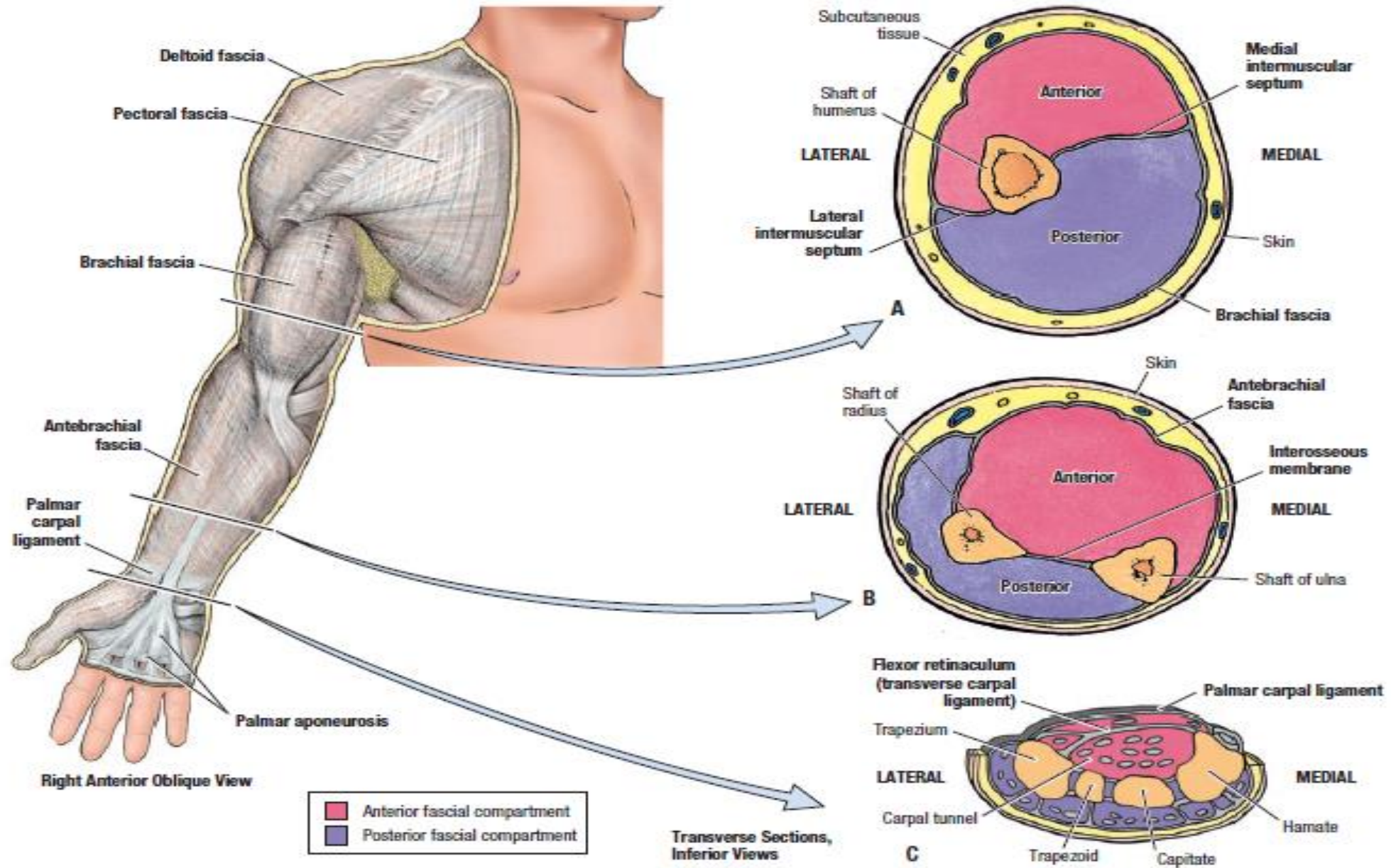


Fascia	
	Anterior fascial compartment
	Posterior fascial compartment

Fascia of the Upper Limb,,



Fascia of the Upper Limb,,,



Fascia of the Upper Limb,,,

❖The superficial fascia of the upper limb is crucial for:

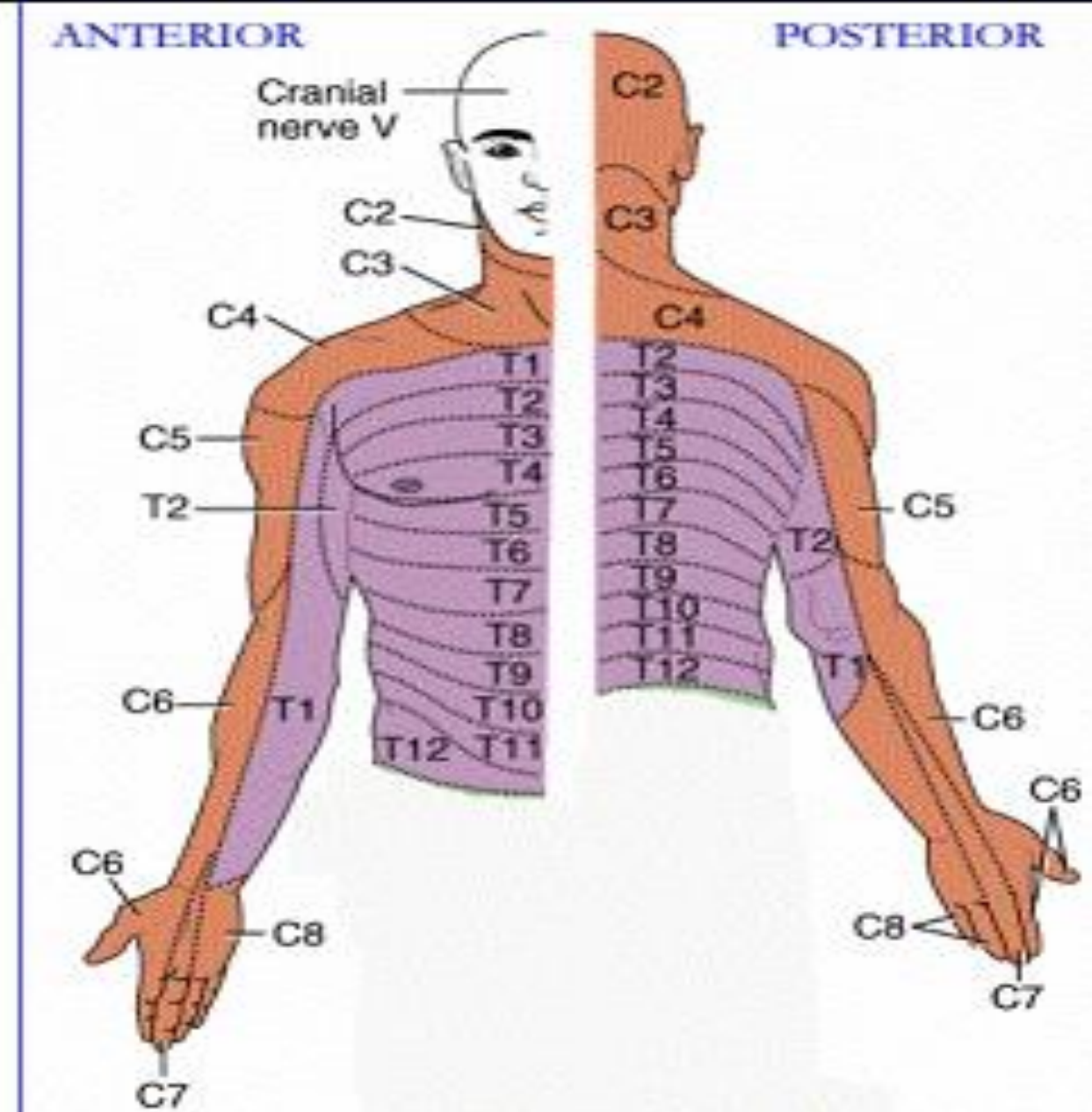
- ✓ **Movement and flexibility:** It allows for smooth gliding of the skin over the underlying muscles and bones.
- ✓ **Thermoregulation:** The fatty layer within the superficial fascia helps in heat insulation.
- ✓ **Sensory perception:** It houses nerve endings that contribute to tactile sensation and proprioception.
- ✓ **Support and protection:** It provides a cushioning effect, protecting the underlying structures from external trauma.

Cutaneous nerves of the upper limb

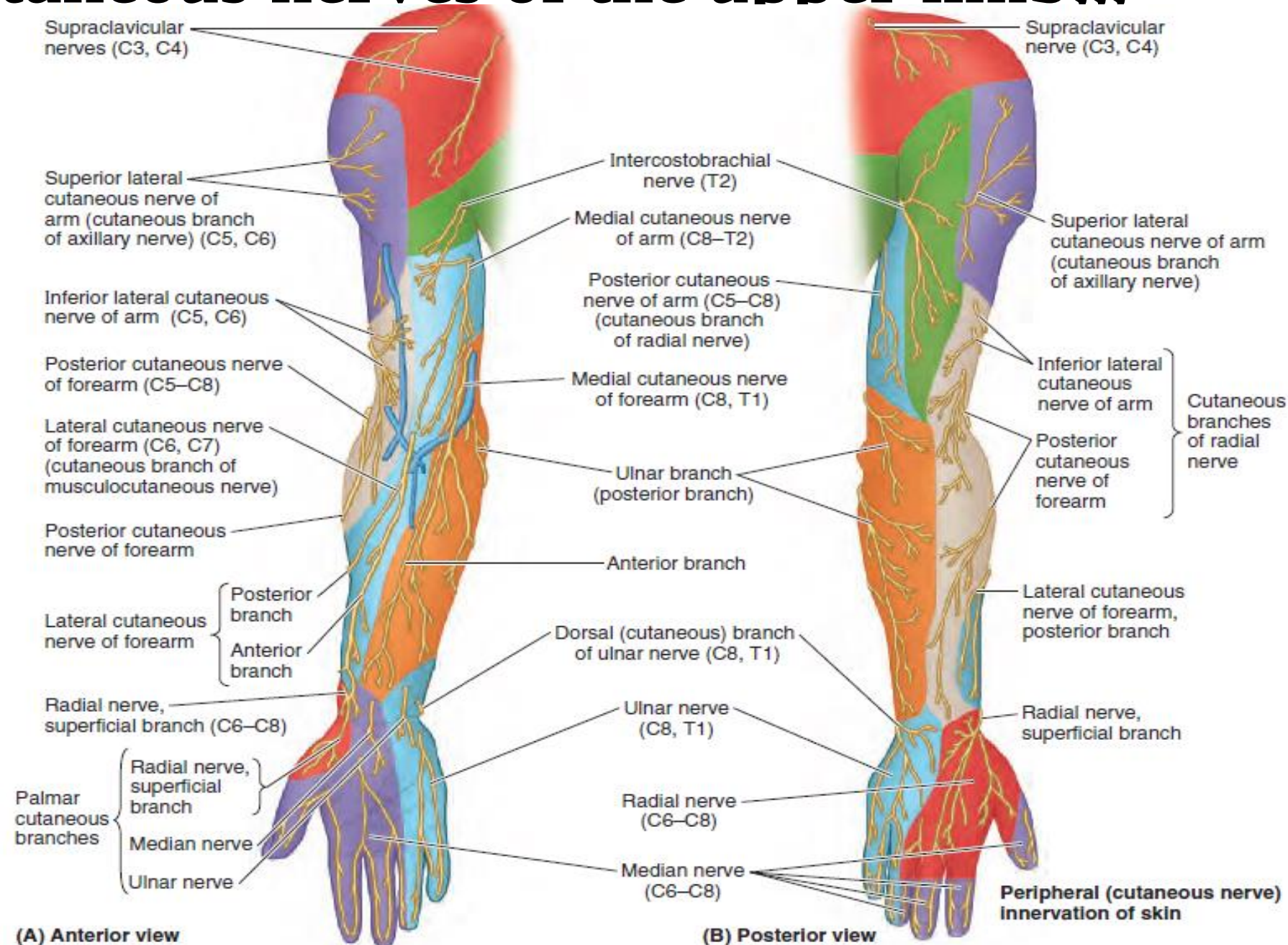
- ❑ **Most cutaneous nerves of upper limb are derived from brachial plexus, a major nerve network formed by the anterior rami of the C5–T1 spinal nerves.**
- ❑ Cutaneous nerves to shoulder are derived from cervical plexus
 - C4: base of the neck, shoulder
 - C5: arm laterally
 - C6: forearm laterally and thumb
 - C7: middle three fingers and middle of posterior surface
 - C8: little finger, the medial side
 - T1: middle of forearm to axilla
 - T2: small part of arm and axilla

Cutaneous nerves of the upper limb,,,

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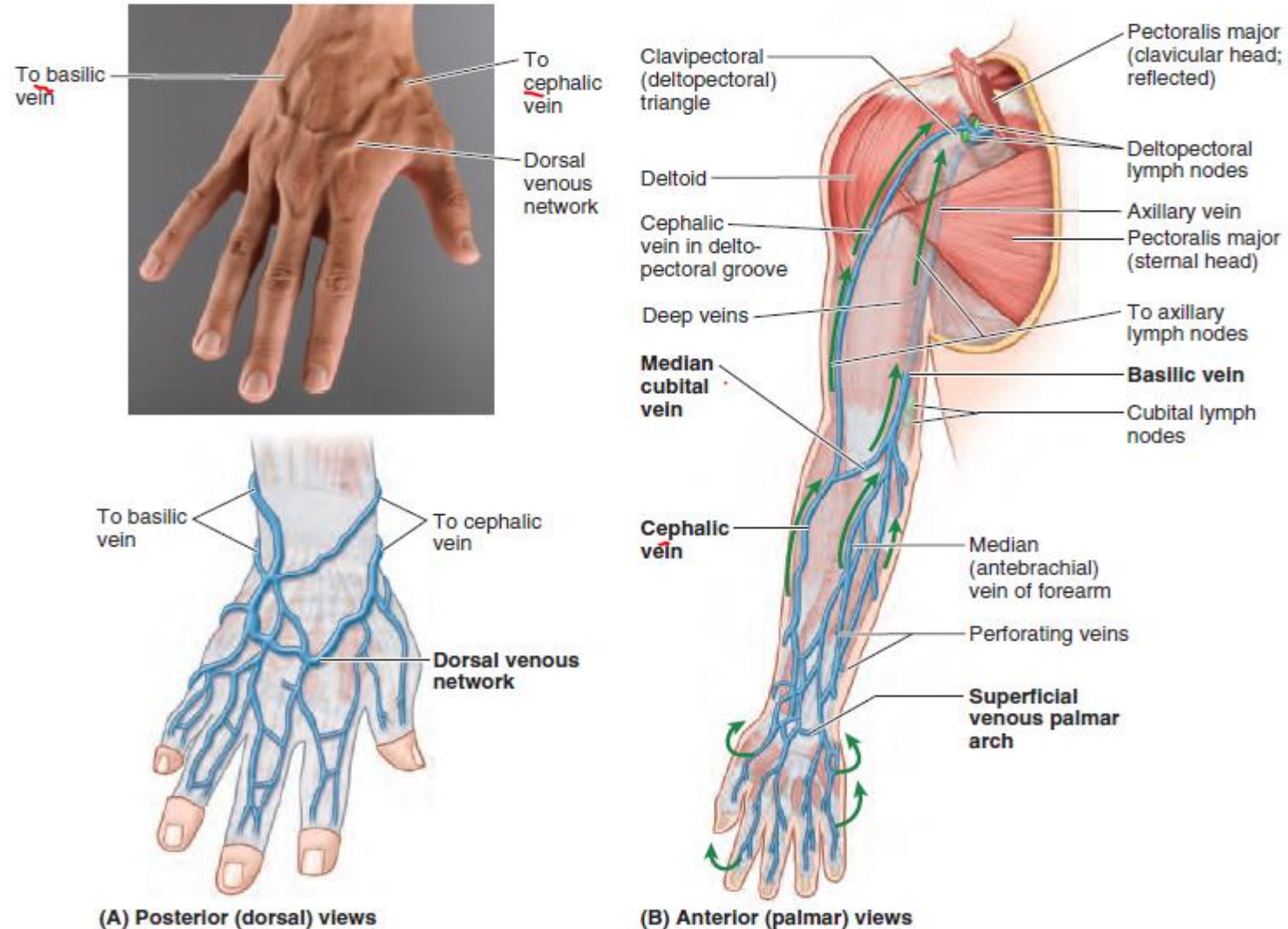


Cutaneous nerves of the upper limb...



Superficial veins of the upper limb

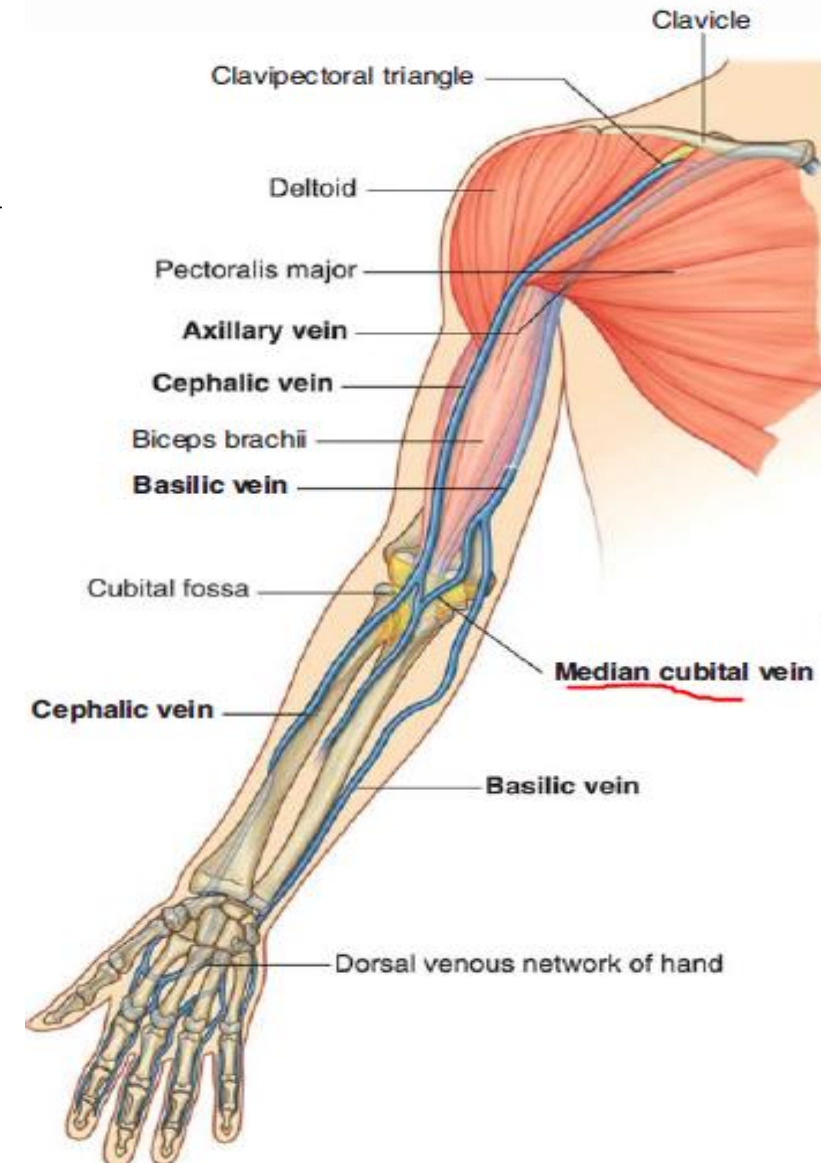
- The main superficial veins of the upper limb, the cephalic and basilic veins, originate in the subcutaneous tissue on the dorsum of the hand from the **dorsal venous network**.



Superficial veins of the upper limb,,

■ Cephalic vein

- Ascends from **lateral part** of dorsal venous network
- Proceed along anterolateral surface of forearm and arm
- Anterior to elbow **cephalic vein and basilic vein** communicate each other with **median cubital vein**.
- **Cephalic vein passes** superiorly between deltoid and pectoralis major muscles and enters clavipectoral triangle
- Pierces clavipectoral fascia and joins **axillary vein**.



Superficial veins of the upper limb,,

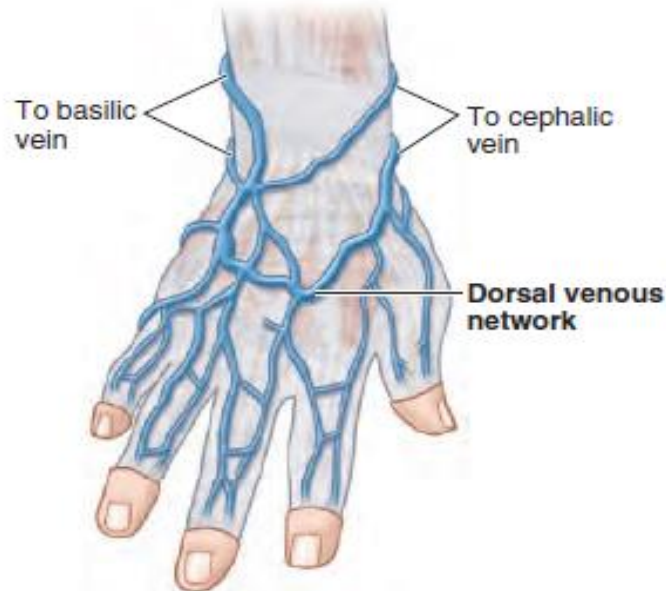
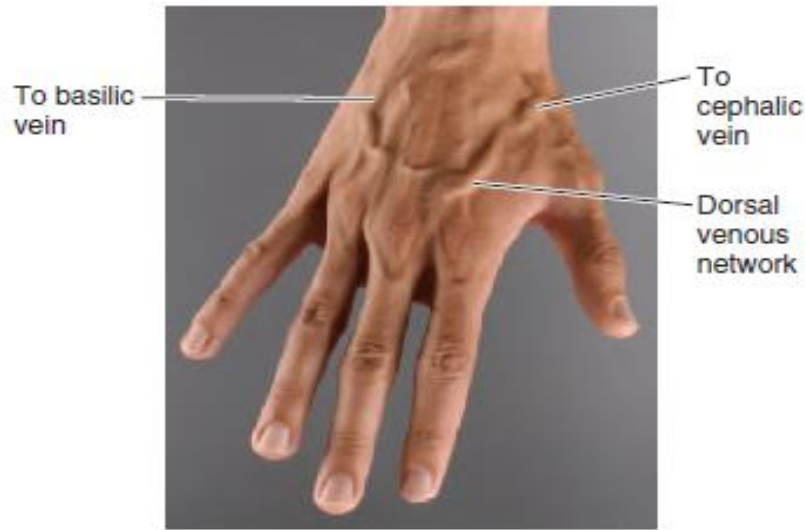
■ Basilic vein

- ✚ Ascends from **medial part** of dorsal venous network of the hand.
- ✚ Proceed along medial side of forearm and inferior part of arm
- ✚ Passes deeply near the junction of the middle and inferior thirds of the arm, piercing the brachial fascia and running superiorly parallel to the brachial artery.

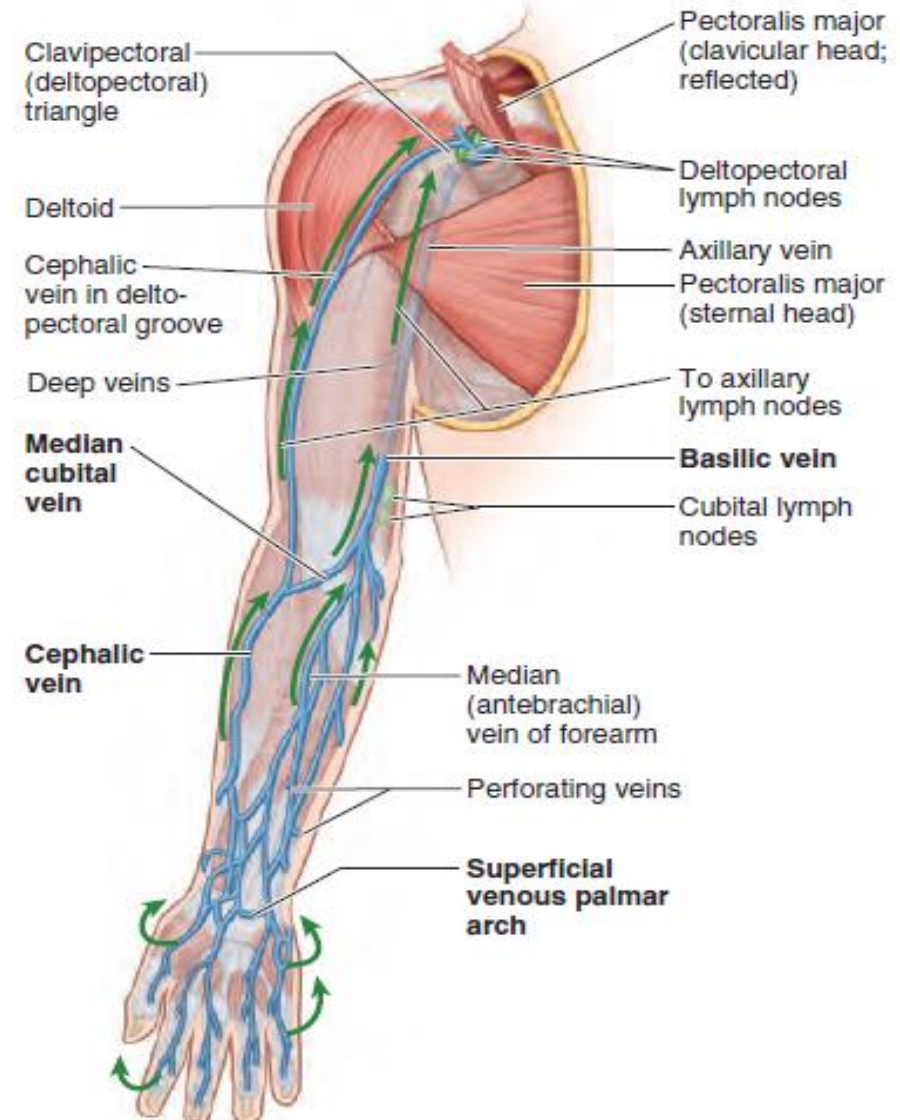
■ Median antebrachial vein (median vein of forearm)

- ✚ Ascends in the middle of anterior aspect of forearm between cephalic and basilic veins.

Superficial veins of the upper limb,,



(A) Posterior (dorsal) views



(B) Anterior (palmar) views

Superficial Lymphatic drainage of the upper limb

□ Lymphatic Drainage of the Upper Limb

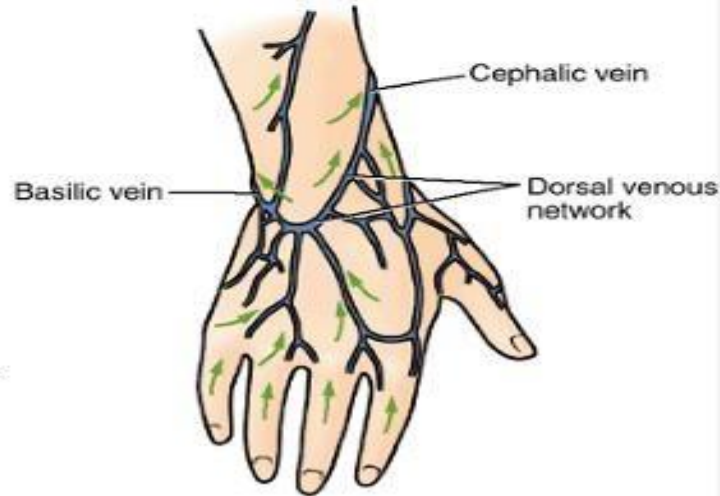
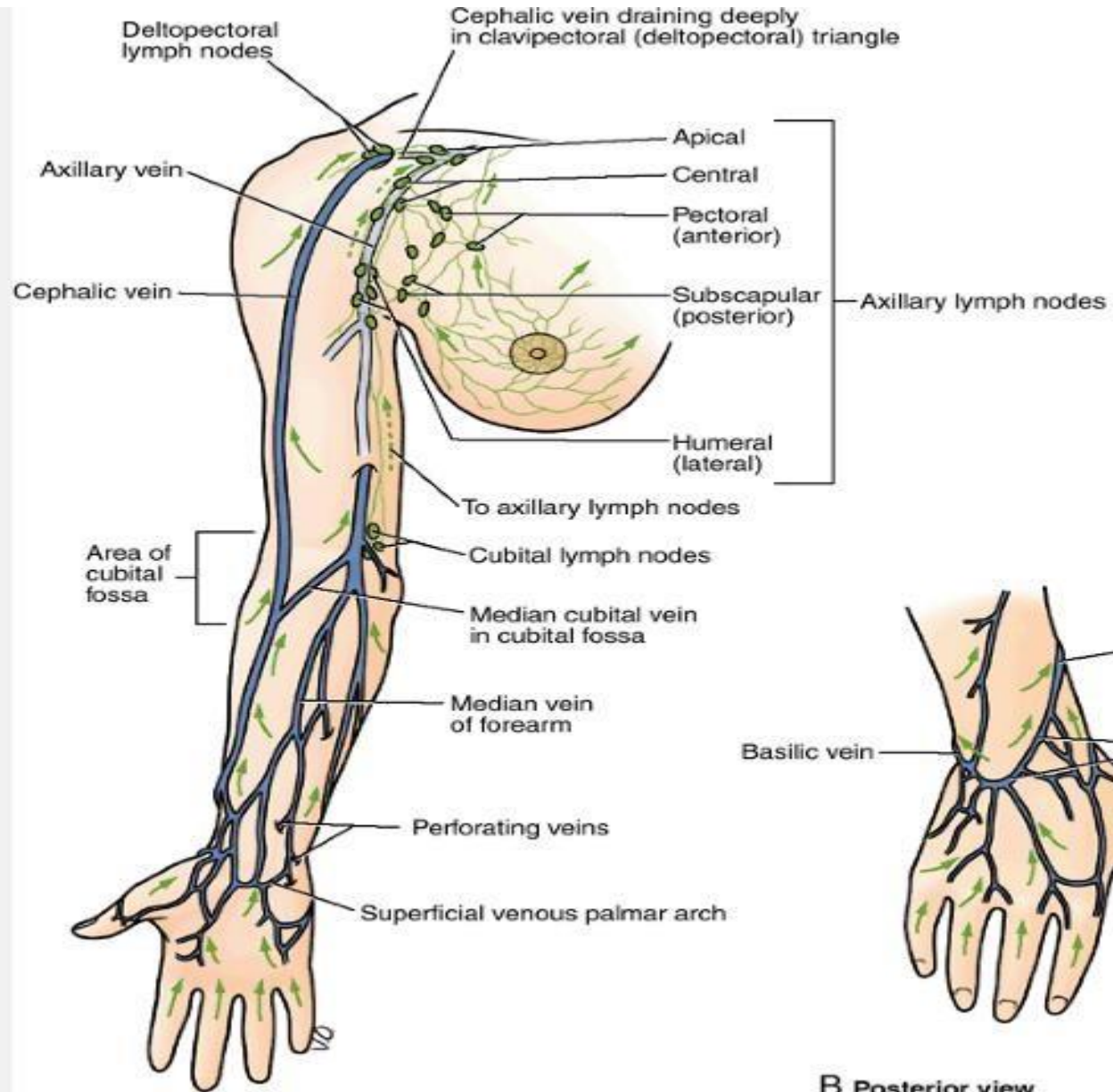
1. Superficial:- are more significant

- Originate from lymphatic plexuses in the hand (palm, dorsum)
- Accompany superficial veins
 - ✓ Most accompany cephalic veins & terminate in the **apical axillary LN (most) & deltopectoral LN**
- Some with basilic terminate in **cubital lymph nodes** near the medial epicondyle
- Efferent lymph vessels ascend up to end in lateral (humeral) axillary lymph nodes

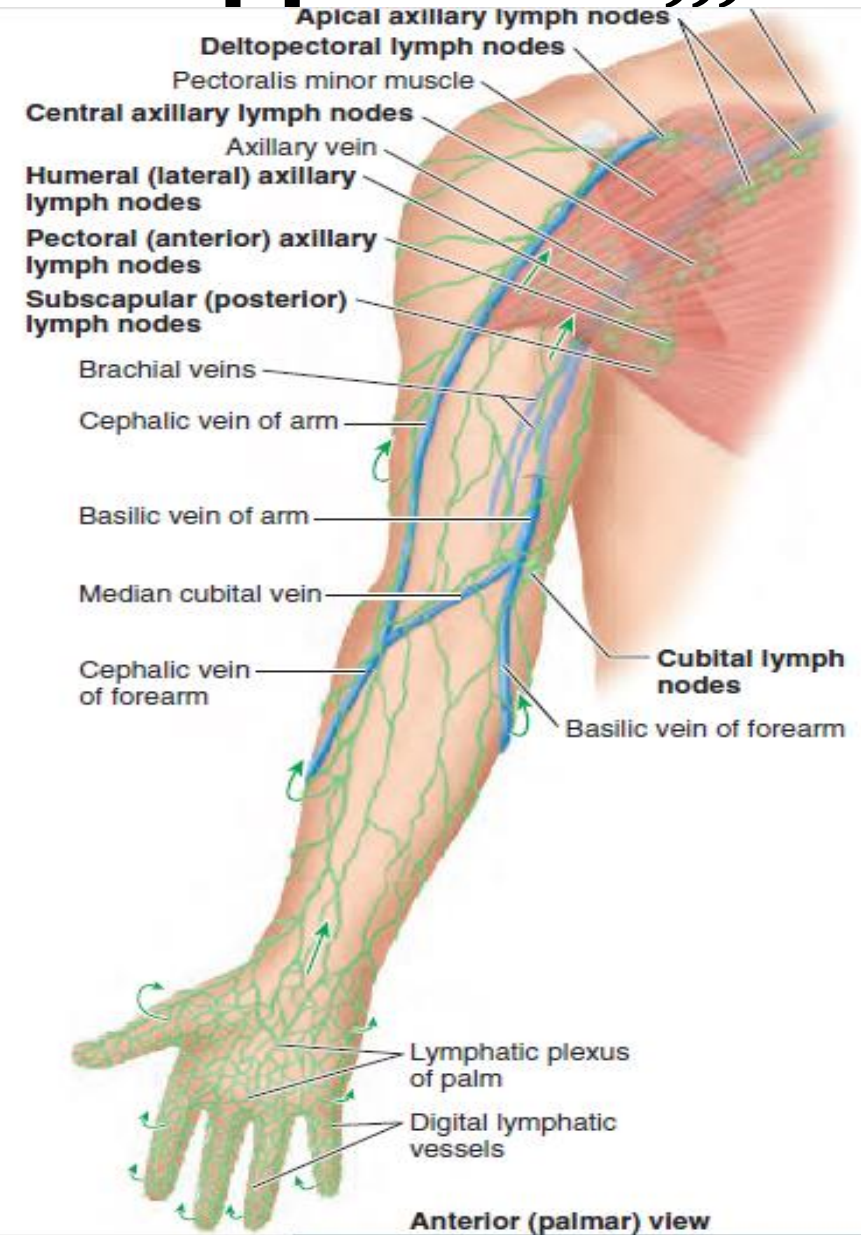
2. Deep:- less significant & course with deep veins

- Drains most of joints, bones nerves and muscles
- Terminate in the **lateral axillary lymph nodes**

Superficial Lymphatic drainage of the upper limb,,



B Posterior view



Anterior (palmar) view

Axioappendicular Muscles

❑ Axioappendicular muscles are Muscles of shoulder, scapular and pectoral region

- They are intermediates b/n axial and appendicular muscles

❑ Divided in to anterior, posterior and intrinsic groups

❖ Anterior axioappendicular muscles (pectoral muscles)

- ✓ Pectoralis major

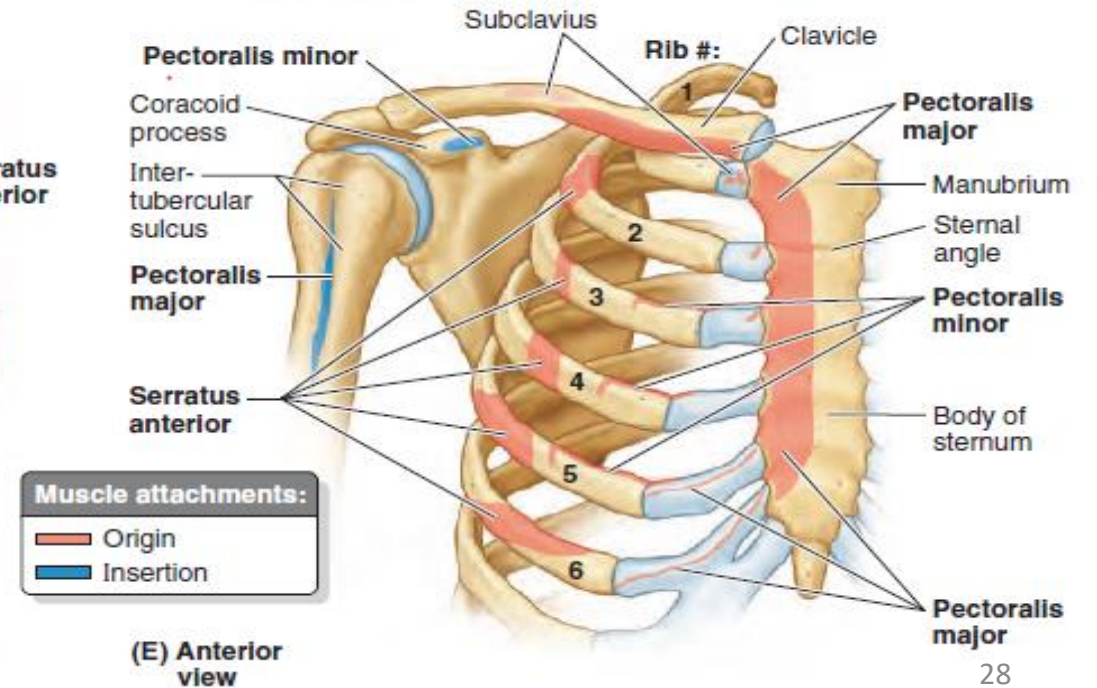
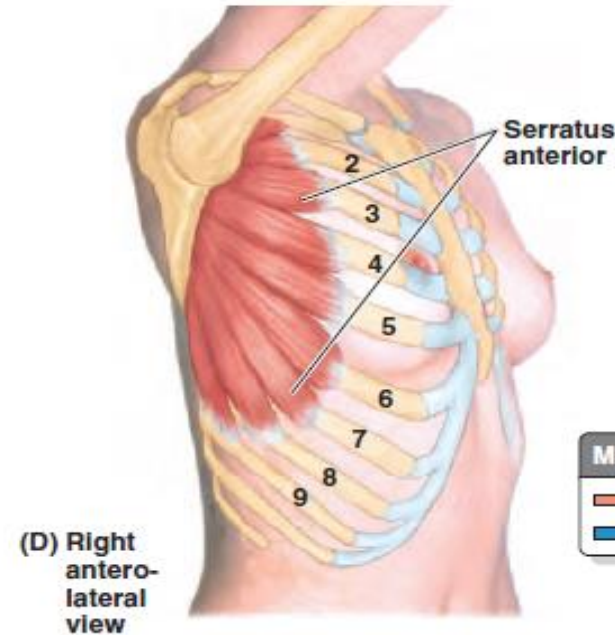
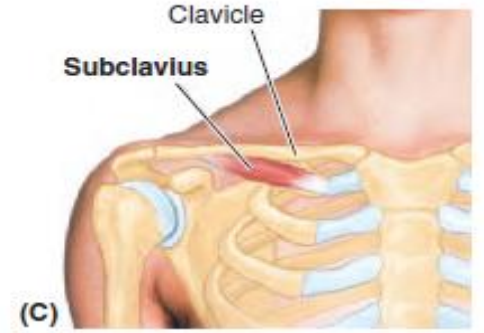
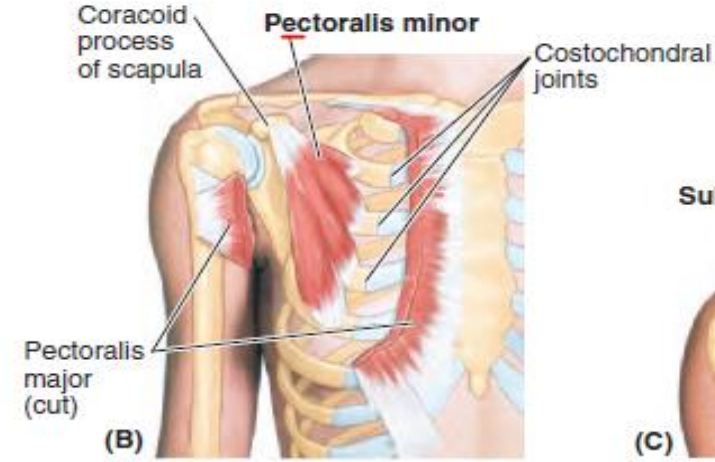
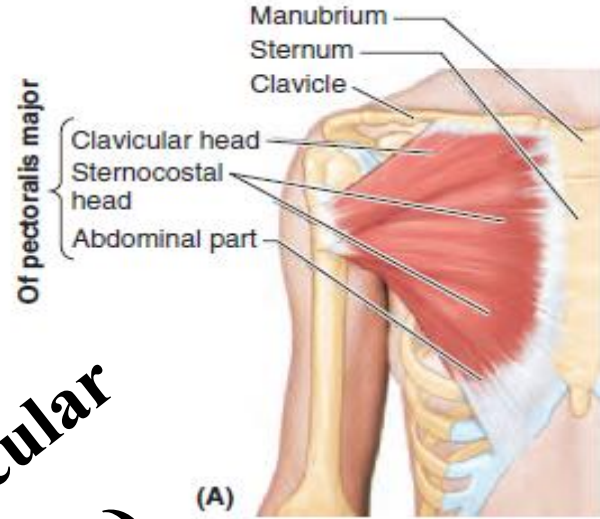
- ✓ Pectoralis minor

- ✓ Subclavius

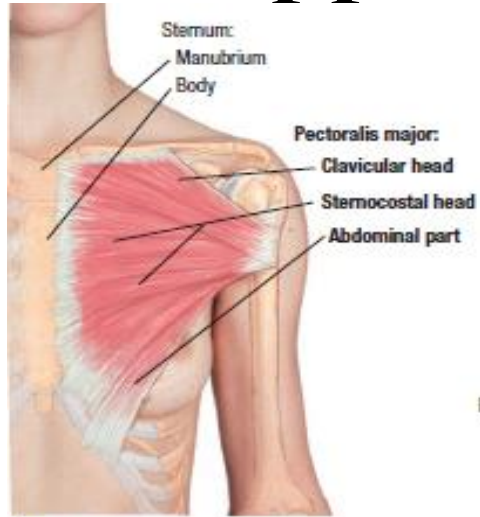
- ✓ Serratus anterior

Axioappendicular Muscles,,,

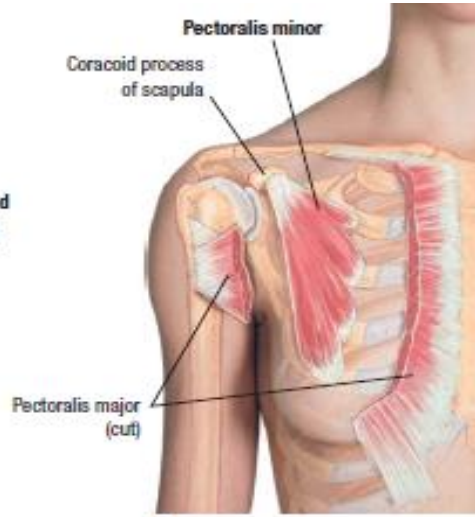
➤ Anterior axioappendicular muscles (pectoral muscles)



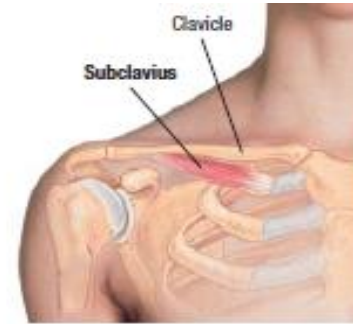
Axioappendicular Muscles,,,



A. Anterior View

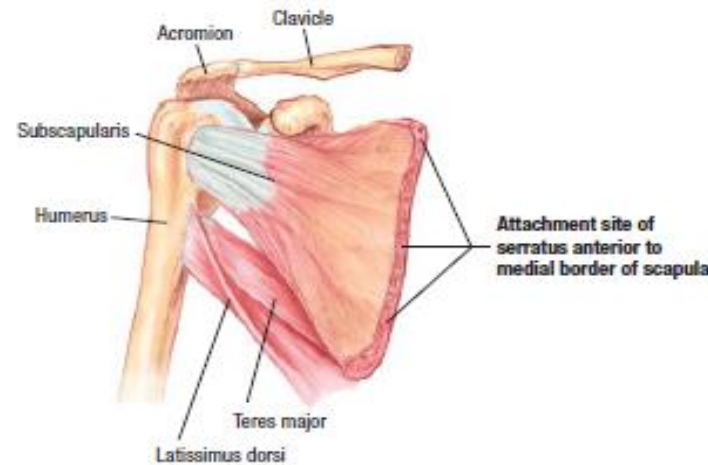
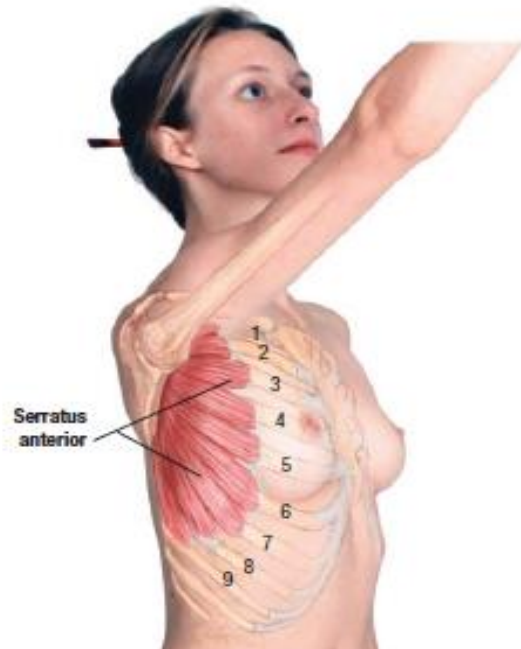


B. Anterior View



C. Anterior View

➤ Anterior axioappendicular muscles (pectoral muscles)



E. Anterior View

Axioappendicular Muscles,,,

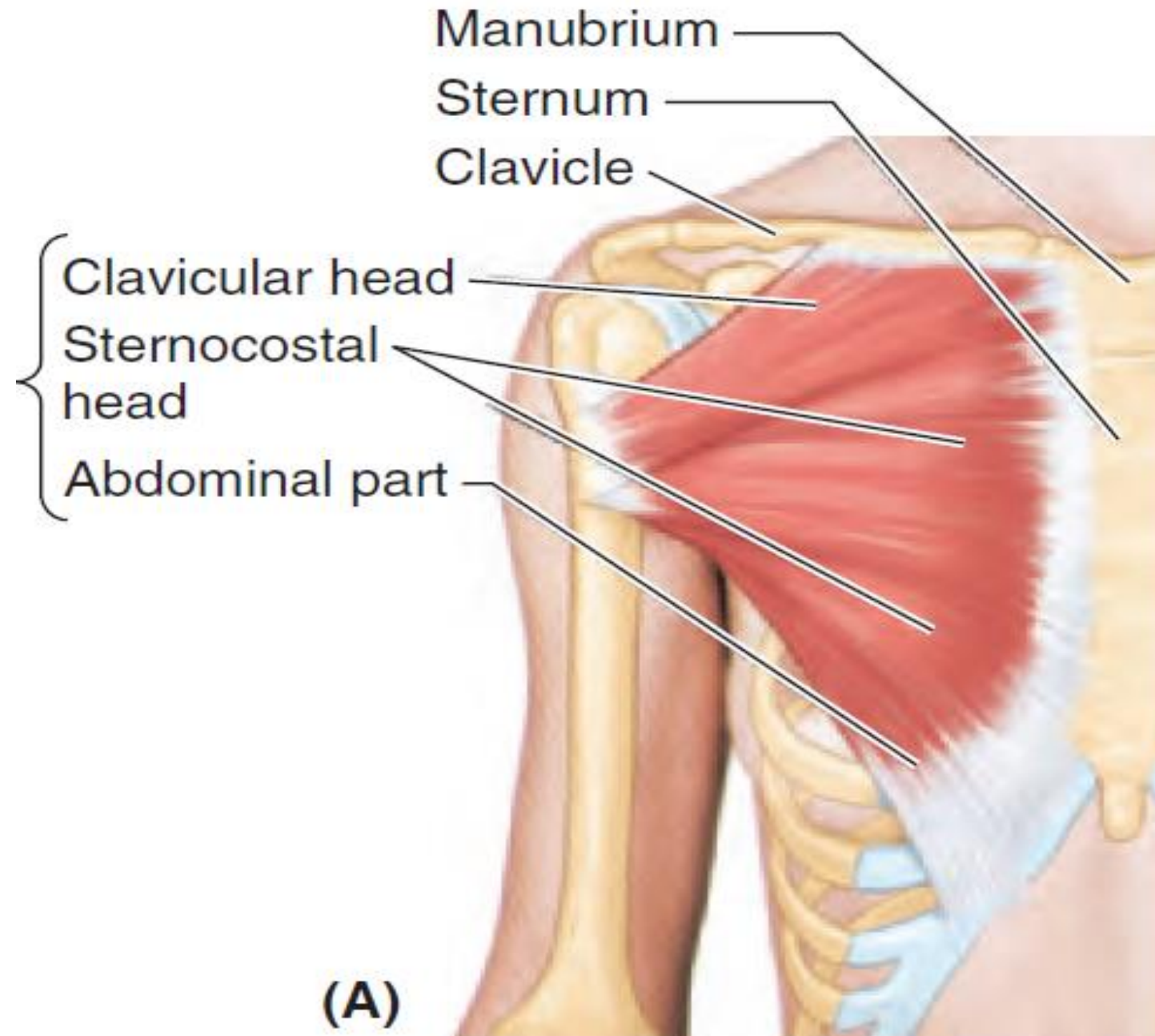
Table for anterior axio-appendicular muscles

Muscle	Proximal Attachment	Distal Attachment	Innervation ^a	Main Action
Pectoralis major	Clavicular head: anterior surface of medial half of clavicle Sternocostal head: anterior surface of sternum, superior six costal cartilages, aponeurosis of external oblique muscle	Lateral lip of intertubercular sulcus of humerus	Lateral and medial pectoral nerves; clavicular head (C5, C6), sternocostal head (C7, C8, T1)	Adducts and medially rotates humerus; draws scapula anteriorly and inferiorly Acting alone, clavicular head flexes humerus and sternocostal head extends it from the flexed position
Pectoralis minor	3rd–5th ribs near their costal cartilages	Medial border and superior surface of coracoid process of scapula	Medial pectoral nerve (C8, T1)	Stabilizes scapula by drawing it inferiorly and anteriorly against thoracic wall
Sub-clavius	Junction of 1st rib and its costal cartilage	Inferior surface of middle third of clavicle	Nerve to subclavius (C5, C6)	Anchors and depresses clavicle
Serratus anterior	External surfaces of lateral parts of 1st–8th ribs	Anterior surface of medial border of scapula	Long thoracic nerve (C5, C6, C7)	Protracts scapula and holds it against thoracic wall; rotates scapula

Pectoralis major

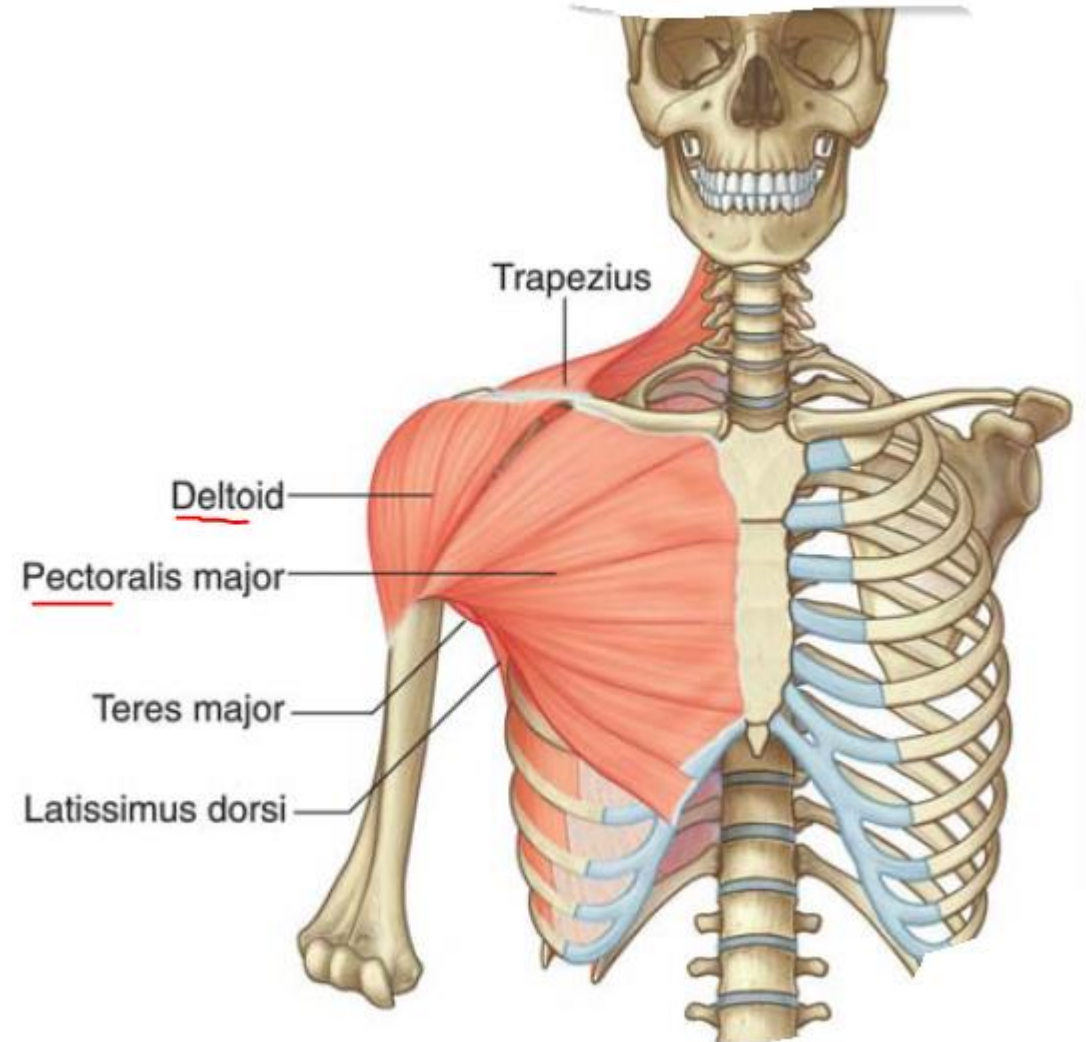
- The **pectoralis major** is a large, fan-shaped muscle that covers the superior part of the thorax .
- *It has clavicular and sternocostal heads.*
- The sternocostal head is much larger, and its lateral border forms the muscular mass that makes up most of the anterior wall of the axilla.
- Producing powerful adduction and medial rotation of the arm when acting together, the two parts of the pectoralis major can also act independently:
 - the clavicular head flexing the humerus, and
 - the sternocostal head extending it back from the flexed position.

Pectoralis major,,



Pectoralis major,,

- ✚ With deltoid form deltopectoral groove
- ✚ Superiorly separate with deltoid and with clavicle form **deltopectoral triangle**
- ✚ Origin
 - **Clavicular head:** medial half of clavicle
 - **Sternocostal head:** sternum, superior six costal cartilages, aponeurosis of external oblique muscle



Pectoralis major,,

❑ Insertion

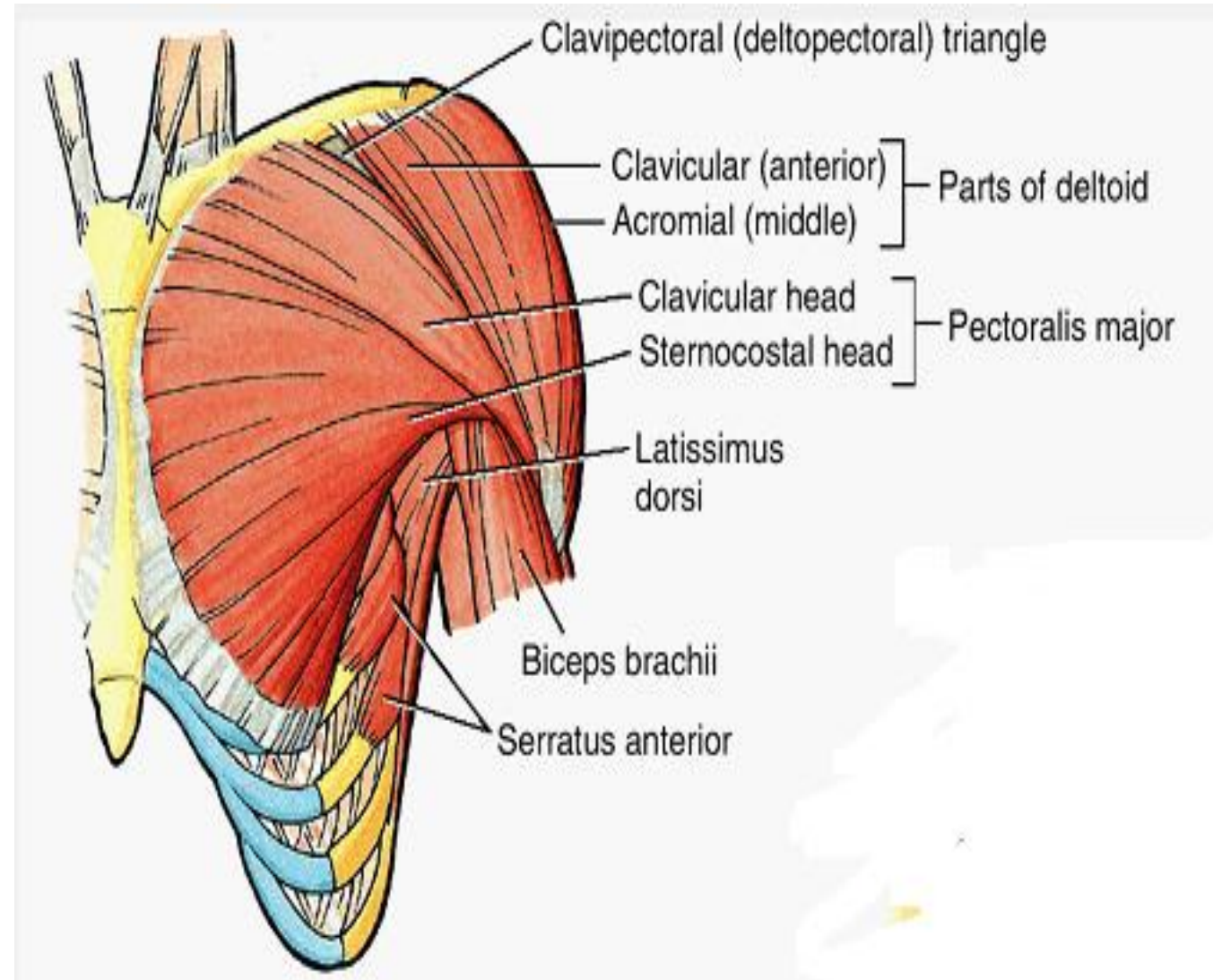
- Intertubercular groove of humerus

❑ Innervation

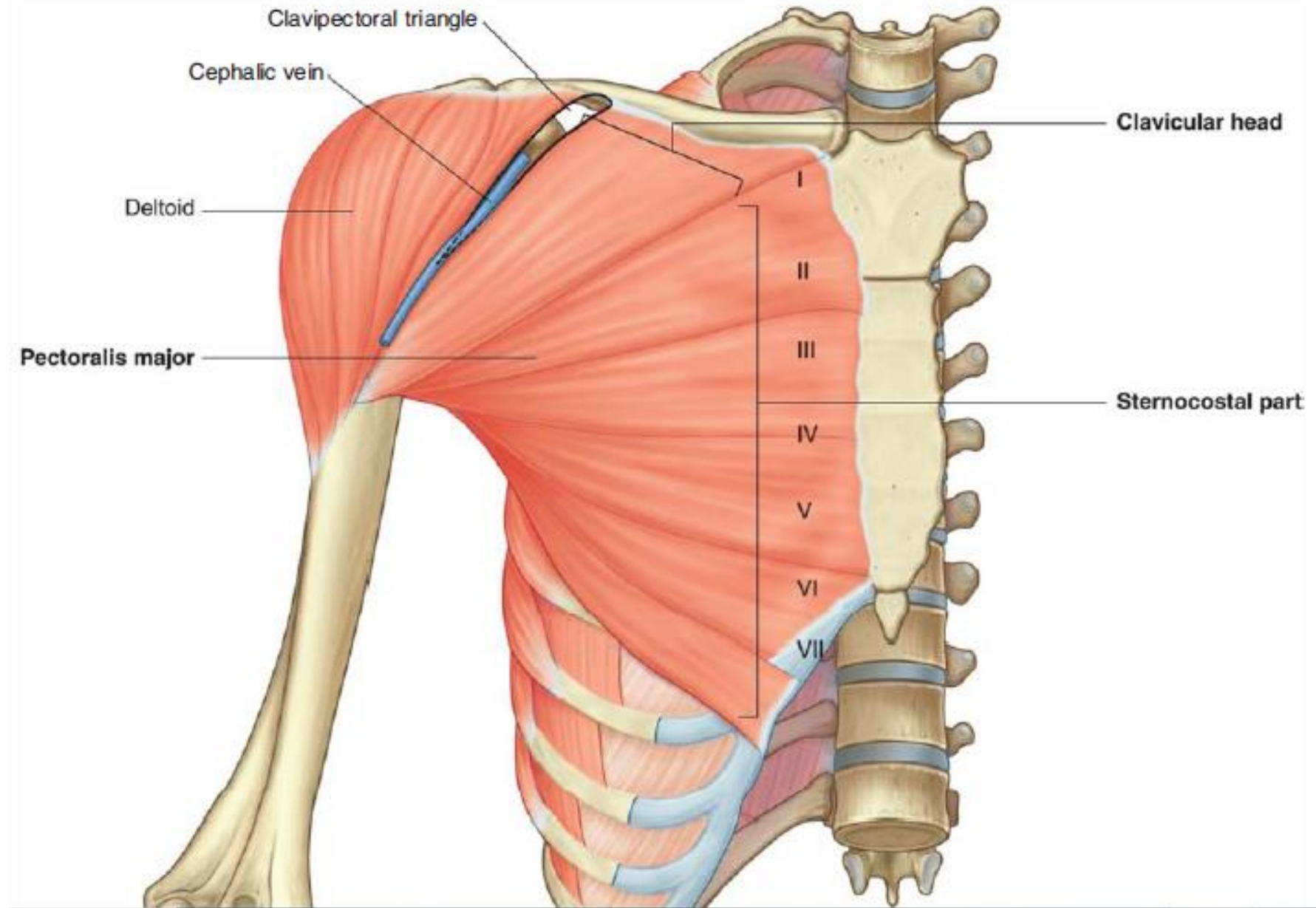
- **Lateral and medial pectoral nerves**

❑ Action

- Adducts and medially rotates humerus; draws scapula anteriorly and inferiorly



Pectoralis major,,



Pectoralis minor

❖ The **pectoralis minor** lies in the anterior wall of the axilla where it is almost completely covered by the much larger pectoralis major

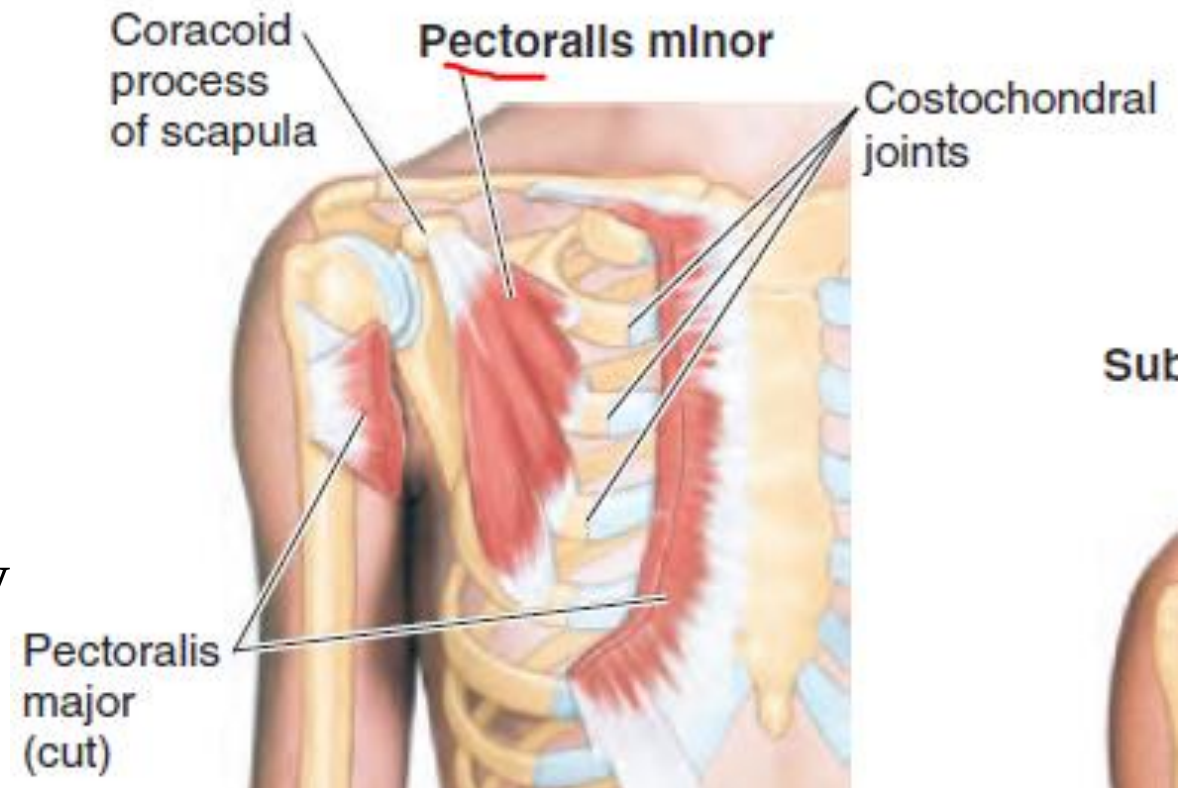
❑ **Origin:** 3rd to 5th ribs near their costal cartilages

❑ **Insertion:** coracoid process of scapula

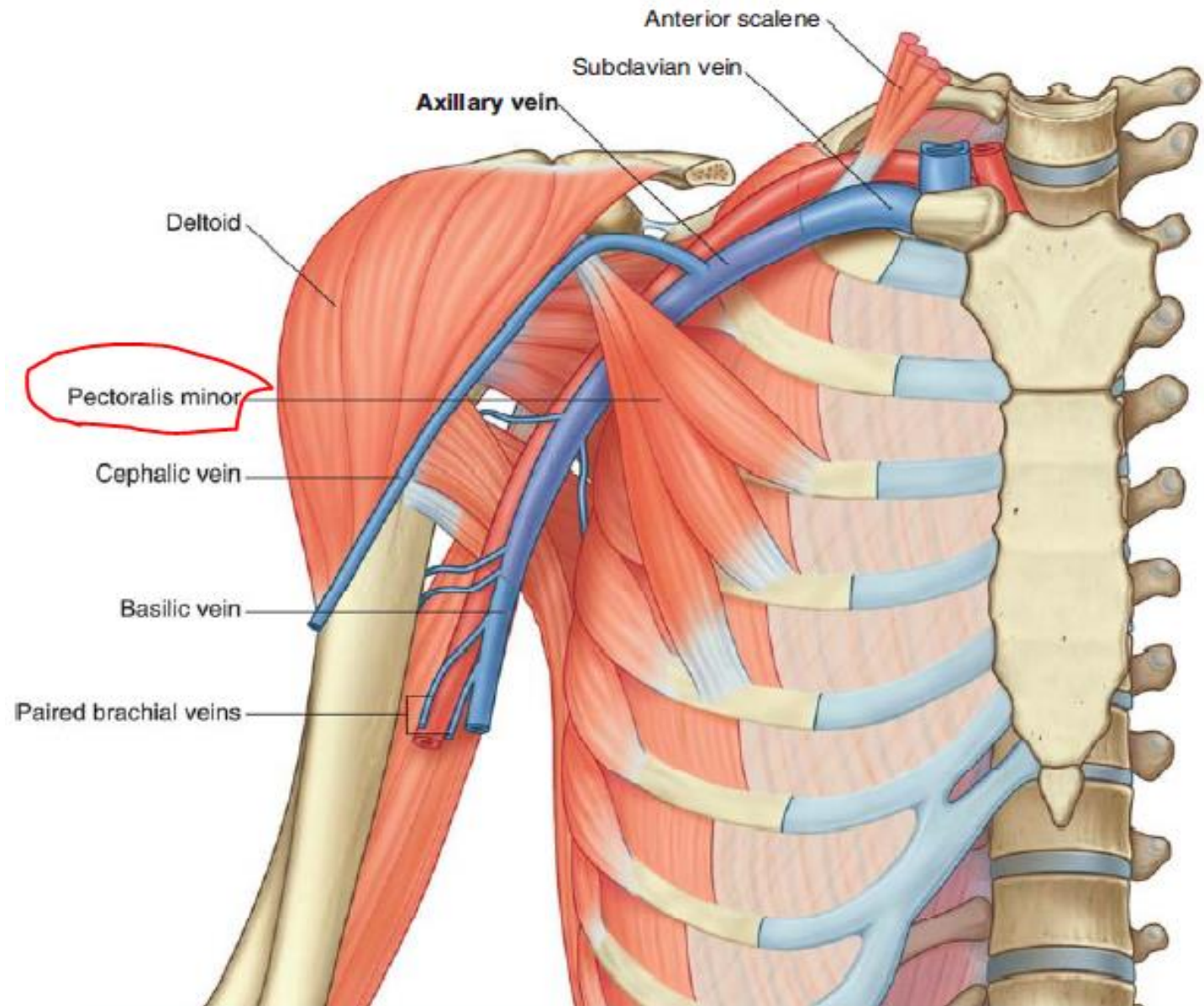
❑ **Innervation:** Medial pectoral nerve

❑ **Action:**

- Stabilizes scapula by drawing it inferiorly and anteriorly against thoracic wall
- Elevates ribs in forced inspiration

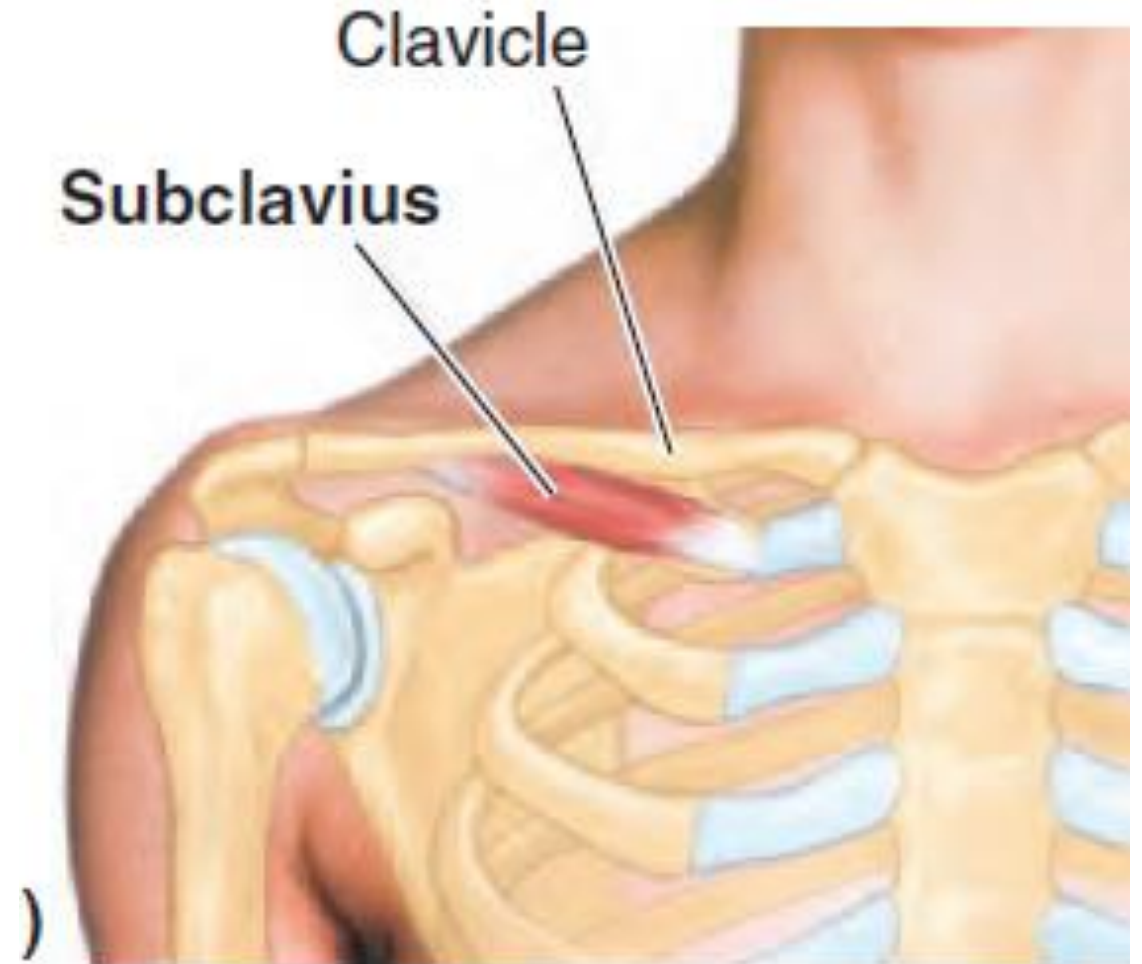


Pectoralis minor,,



Subclavius

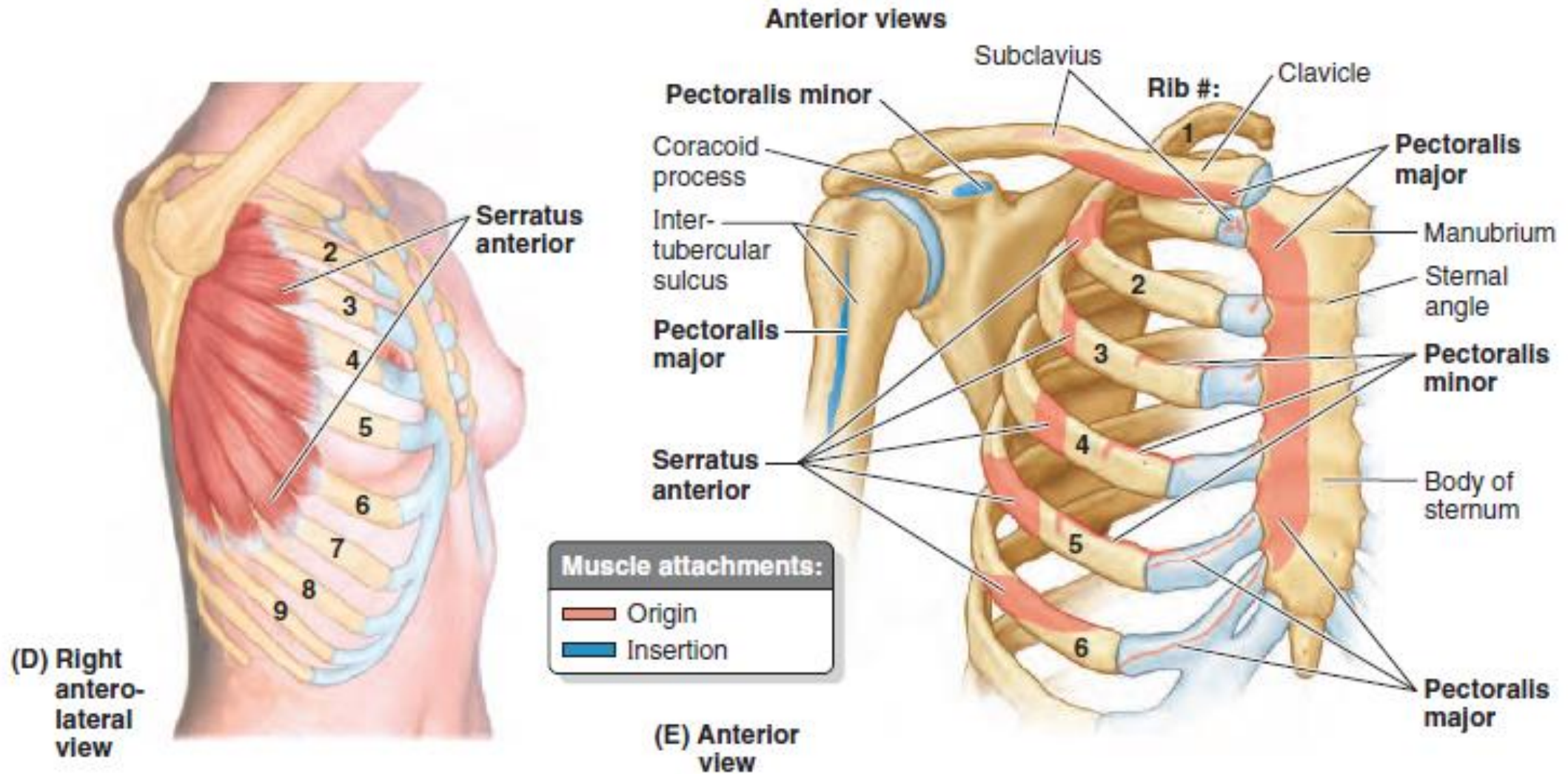
- ✦ Lies horizontally inferior to clavicle
- ✦ Small, round
- ✦ **Origin:** Junction of 1st rib and its costal cartilage
- ✦ **Insertion:** Inferior surface of middle third of clavicle
- ✦ **Innervation:** **Nerve to subclavius**
- ✦ **Action:** Anchors and depresses clavicle



Serratus anterior

- This muscle is named because of the sawtoothed appearance of its fleshy slips.
- **Serratus anterior** the most powerful muscles of the pectoral girdle.
- ▶ It overlies lateral part of thorax and forms medial wall of axilla
- ▶ Anchors scapula, enabling other muscles to use it as a fixed bone for movement of humerus
- ▶ **Origin:** External surfaces of lateral parts of 1st to 8th ribs
- ▶ **Insertion:** Anterior surface of medial border of scapula
- ▶ **Innervation:** Long thoracic nerve
- ▶ **Action:** Protracts scapula and holds it against thoracic wall; rotates scapula

Serratus anterior,,



Serratus anterior,,

❖ Paralysis of the Serratus Anterior

- serratus anterior is paralyzed owing to injury to the **long thoracic nerve**
- the medial border of the scapula moves laterally and posteriorly away from the thoracic wall, giving the scapula the appearance of a wing
- When the arm is raised, the medial border and inferior angle of the scapula pull markedly away from the posterior thoracic wall, a deformation known as **a winged scapula**
- **Arm cannot be abducted above horizontal position**

Serratus anterior,,



Right winged scapula

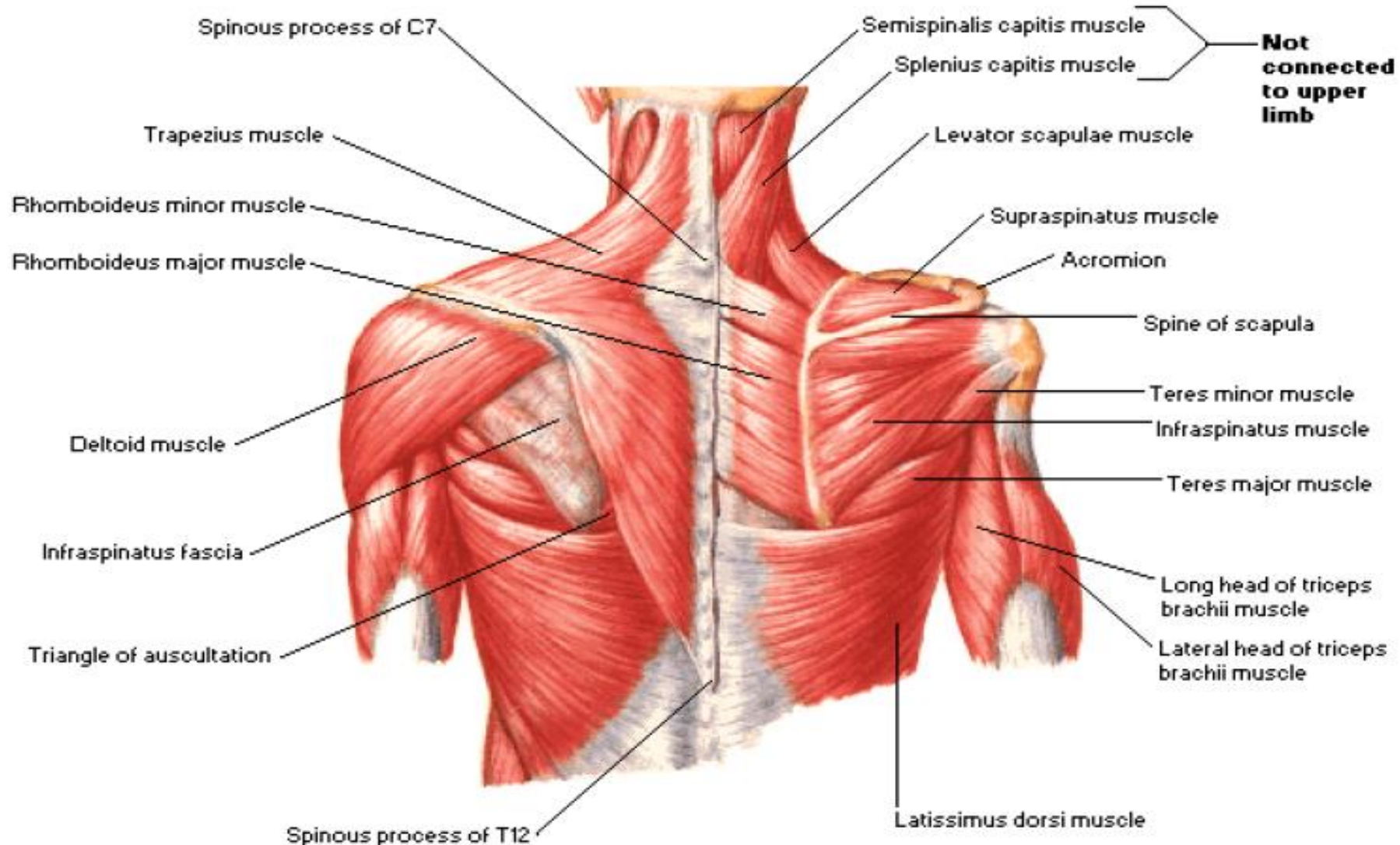
Posterior axio-appendicular muscles

- ❖ Posterior axioappendicular (extrinsic shoulder **muscles**) connect axial skeleton with appendicular skeleton
- ❑ **The posterior shoulder muscles are divided into three groups:**
 - ✓ ***Superficial** posterior axio-appendicular (extrinsic shoulder) muscles:* trapezius and latissimus dorsi.
 - ✓ ***Deep** posterior axio-appendicular (extrinsic shoulder) muscles:* levator scapulae and rhomboids.
 - ✓ ***Scapulohumeral** (intrinsic shoulder) muscles:* deltoid, teres major, and the four rotator cuff muscles (supraspinatus, infraspinatus, teres minor, and subscapularis).

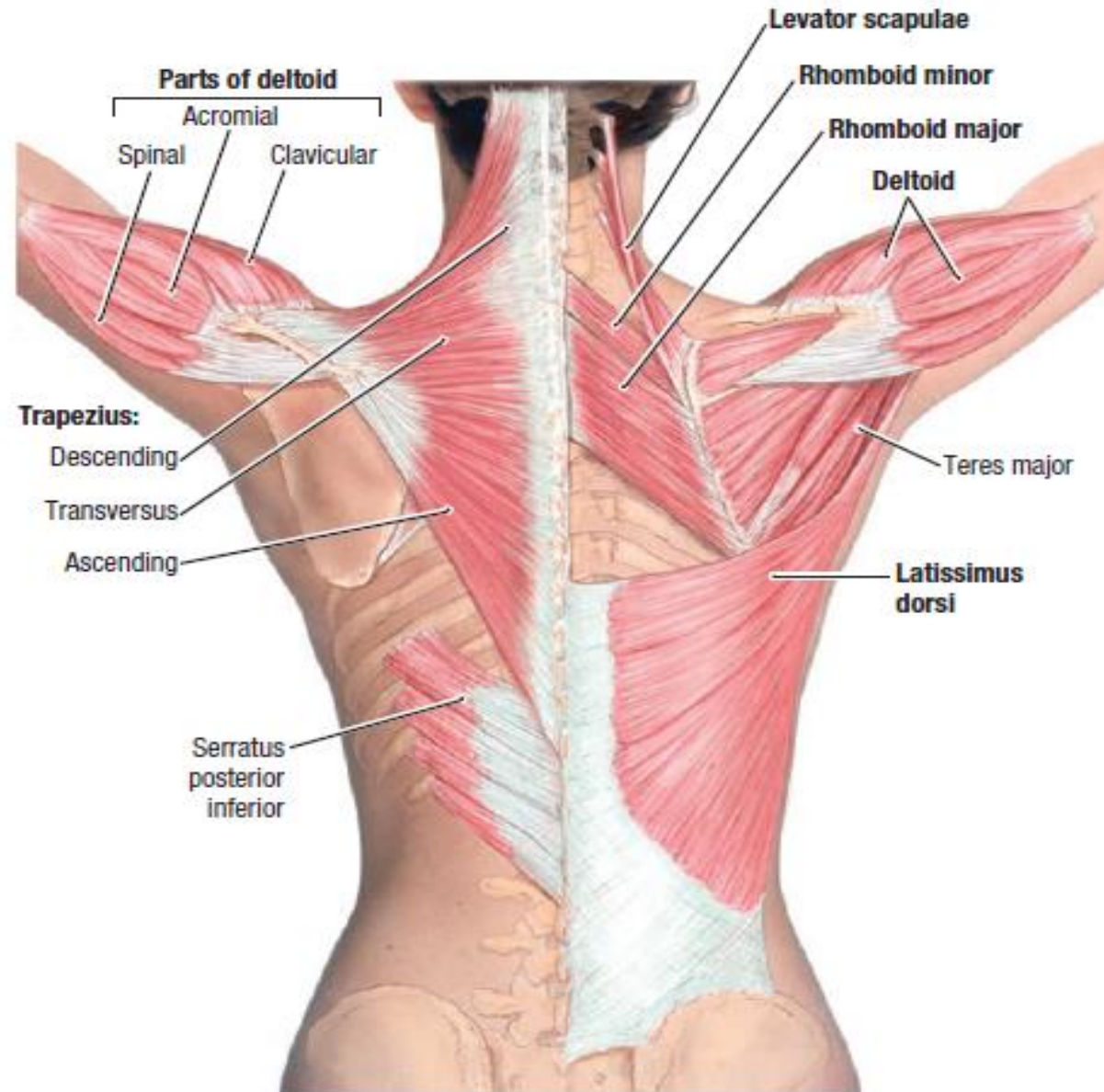
Posterior axio-appendicular muscles,,,

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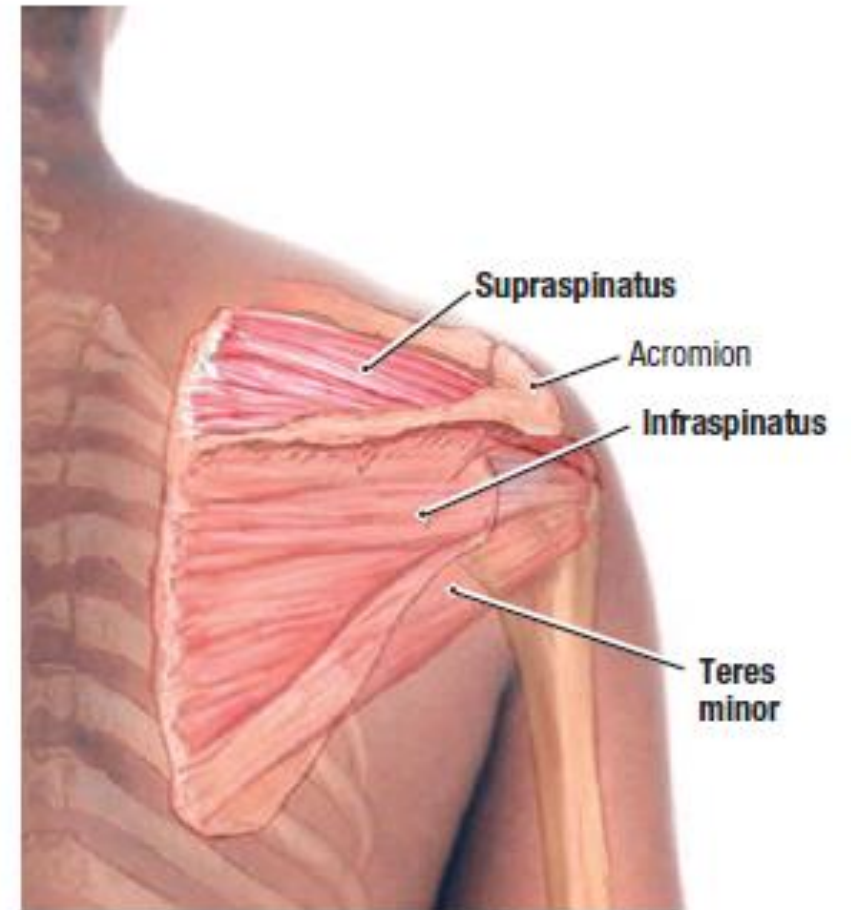
Posterior View



Posterior axioappendicular muscles,,,

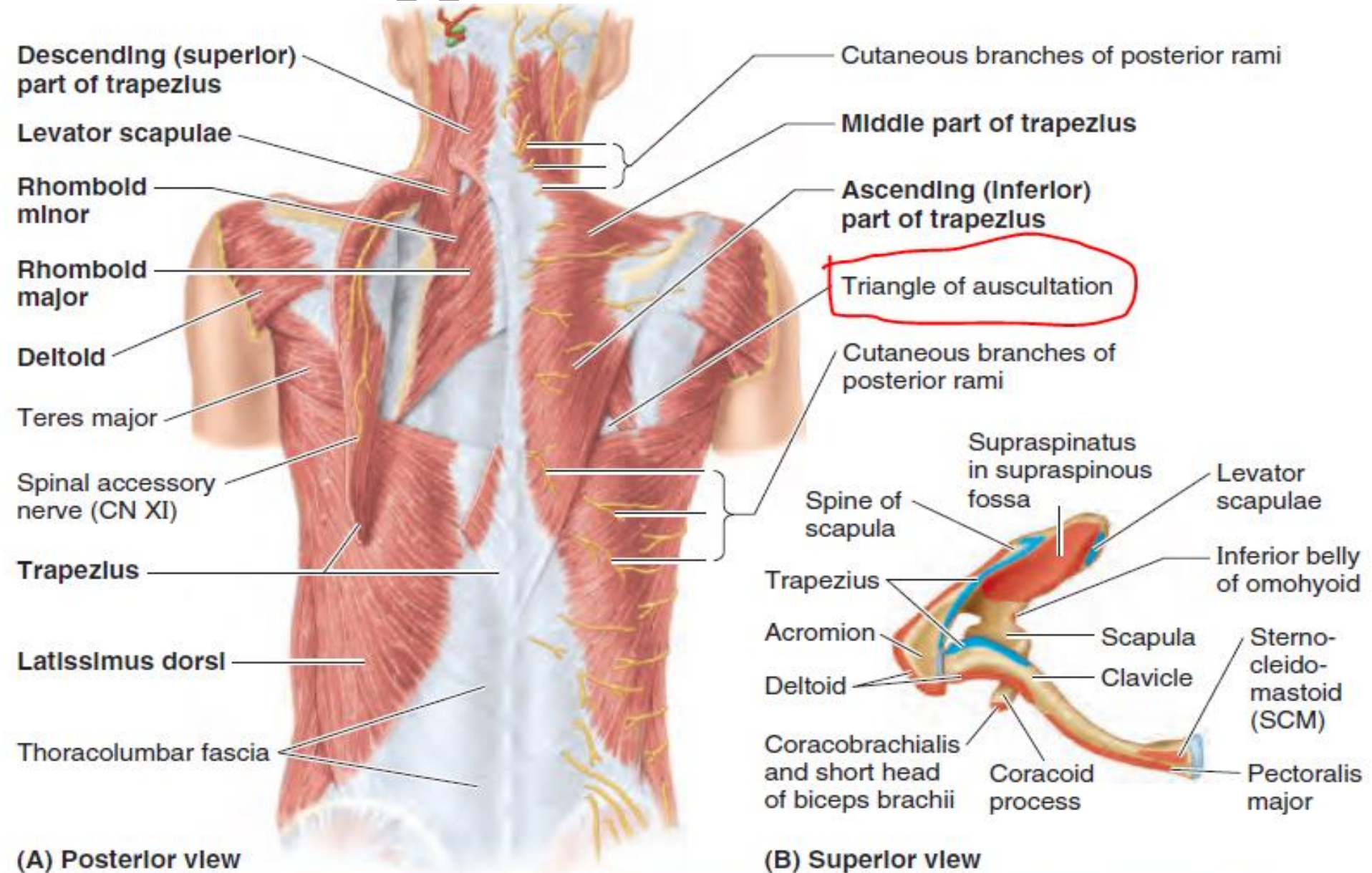


A. Posterior View

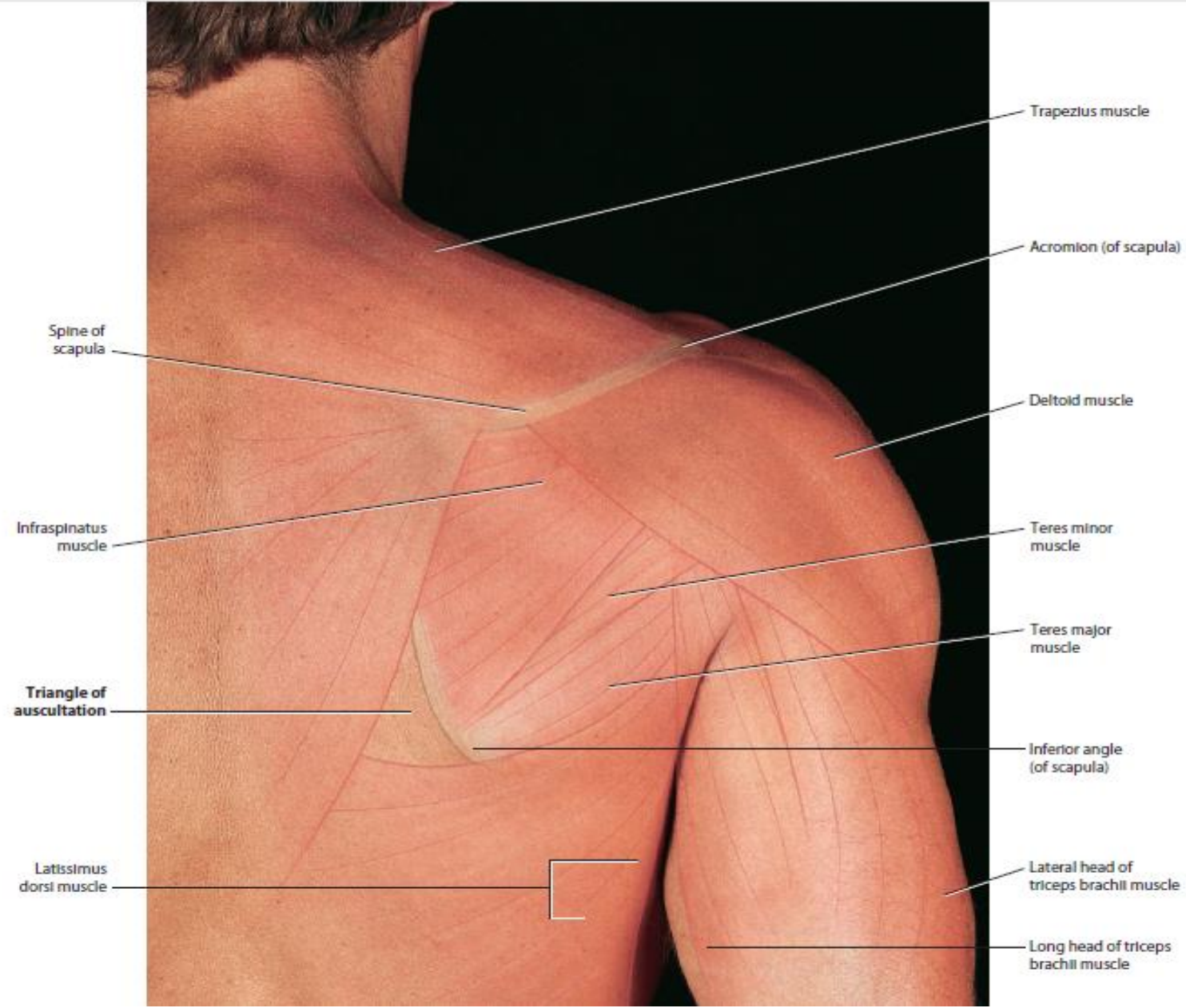


D

Posterior axio-appendicular muscles,,,



Posterior axio-appendicular muscles,,,



Trapezius muscle

❑ **Trapezius** provides direct attachment of pectoral girdle to trunk.

- It was given its name because the muscles of the two sides form a *trapezium*

- + Covers posterior part of neck and superior half of trunk

- + Attach pectoral girdle to cranium and vertebral column

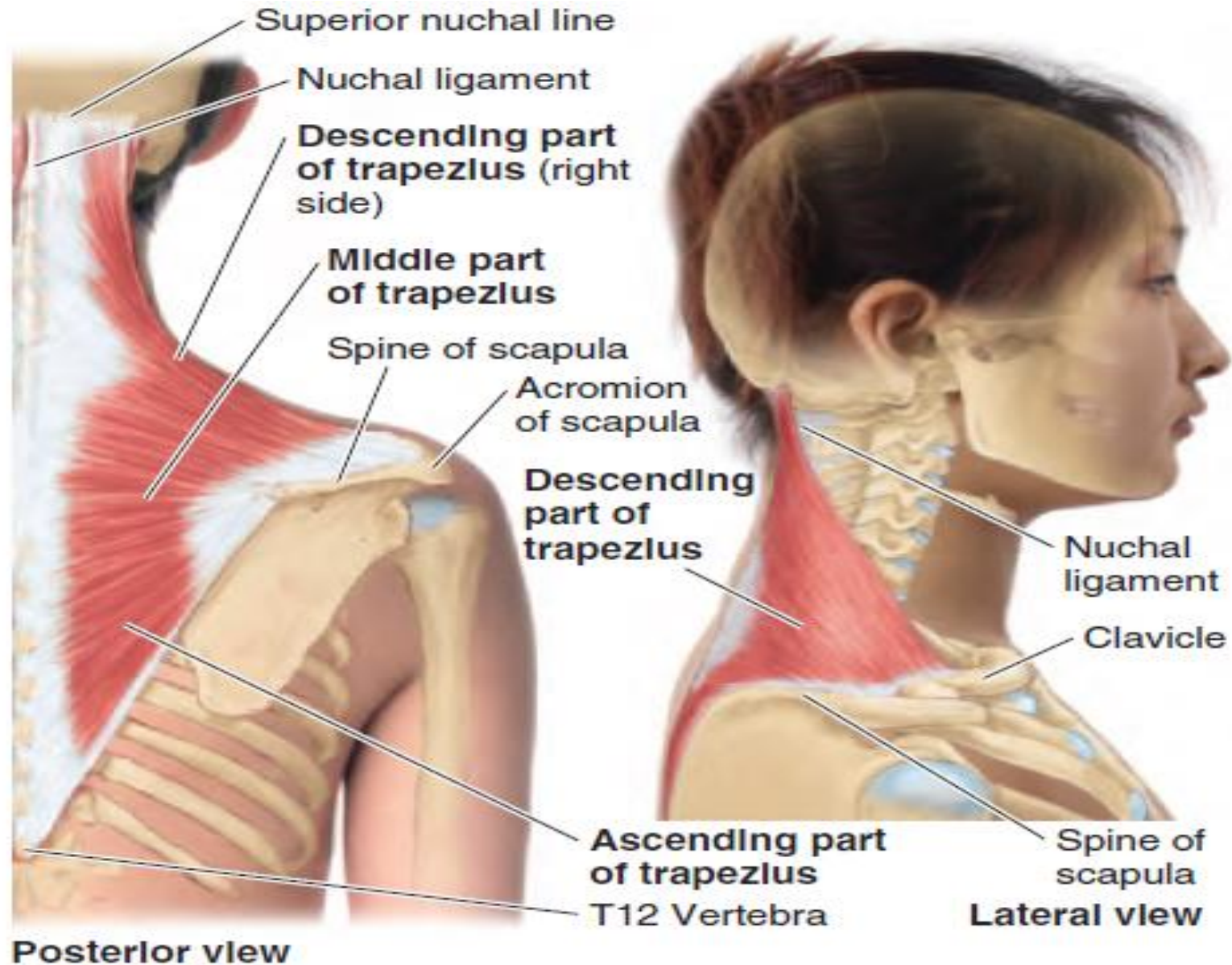
❑ **The fibers of the trapezius are divided into three parts,**

- ✓ **Descending** (superior) fibers elevate the scapula (e.g., when squaring the shoulders).

- ✓ **Middle** fibers retract the scapula (i.e., pull it posteriorly).

- ✓ **Ascending** (inferior) fibers depress the scapula and lower the shoulder.

Trapezius muscle,,



Trapezius muscle,,,

- **Origin:** Medial third of superior nuchal line; external occipital protuberance; nuchal ligament; spinous processes of C7 to T12 vertebrae
- **Insertion:** Lateral third of clavicle; acromion and spine of scapula
- **Innervation:** Accessory nerve (CN XI) (motor fibers) and C3, C4 spinal nerves (pain and proprioceptive fibers)
- **Action:**
 - descending part elevates scapula
 - ascending part depresses scapula
 - middle part (or all parts together) retracts scapula
 - descending and ascending parts act together to rotate glenoid cavity superiorly

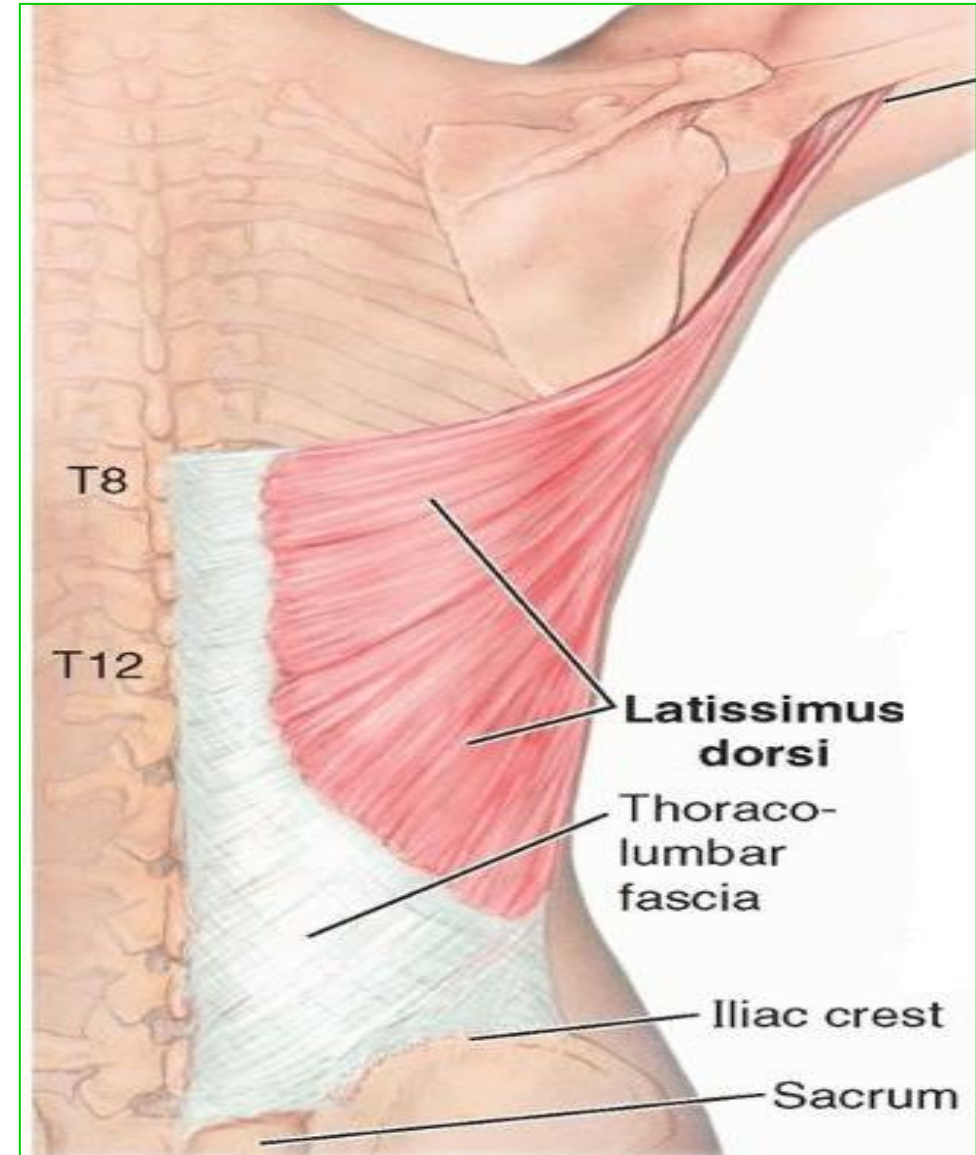
Latissimus dorsi muscle

- ❖ The name latissimus dorsi (L. widest of the back) was well chosen because the muscle covers a wide area of the back.
- Large, fan-shaped muscle passing from the trunk to the humerus.
- Extends, retracts, and rotates the humerus medially.
- **In combination with the pectoralis major:** -is a powerful adductor of the humerus
 - is also useful in climbing a tree, as it raises the trunk to the arm (thus Climbing muscle).
- It is also useful in restoring the upper limb from abduction superior to the shoulder; hence the latissimus dorsi is important in climbing.

Latissimus dorsi muscle,,,

❖ Latissimus dorsi

- **Origin:** Spinous processes of inferior 6 thoracic vertebrae, thoracolumbar fascia, iliac crest & inferior 3/4 ribs
- **Insertion:** Floor of intertubercular groove of humerus
- **Nerve Supply:** Thoraco dorsal nerve
- **Main Action:** Extends, adducts, and medially rotates humerus
 - Raises body toward arms during climbing

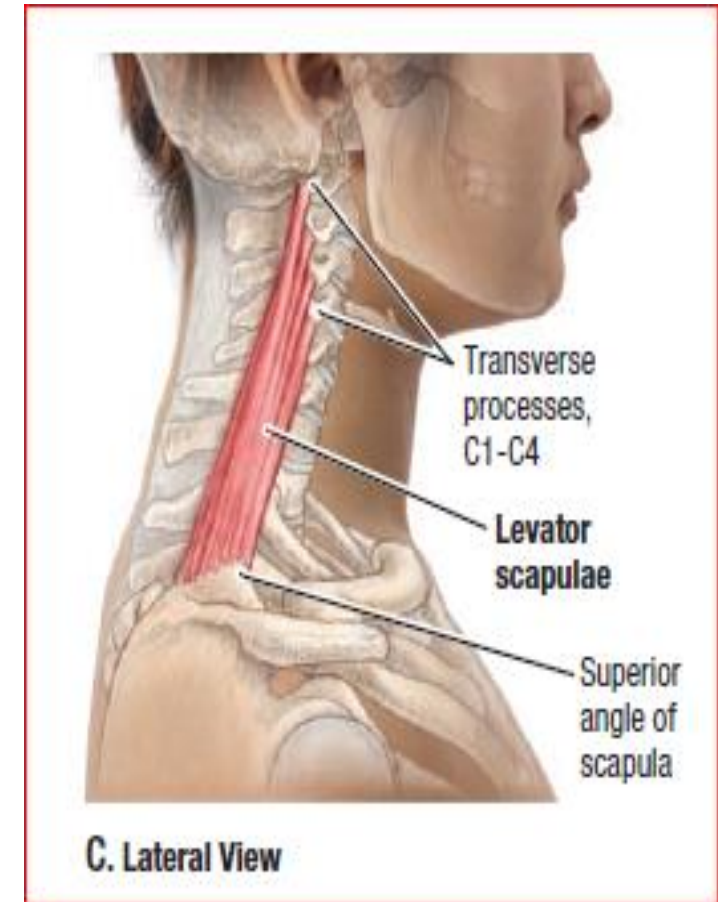


Deep posterior axioappendicular (extrinsic shoulder) muscles

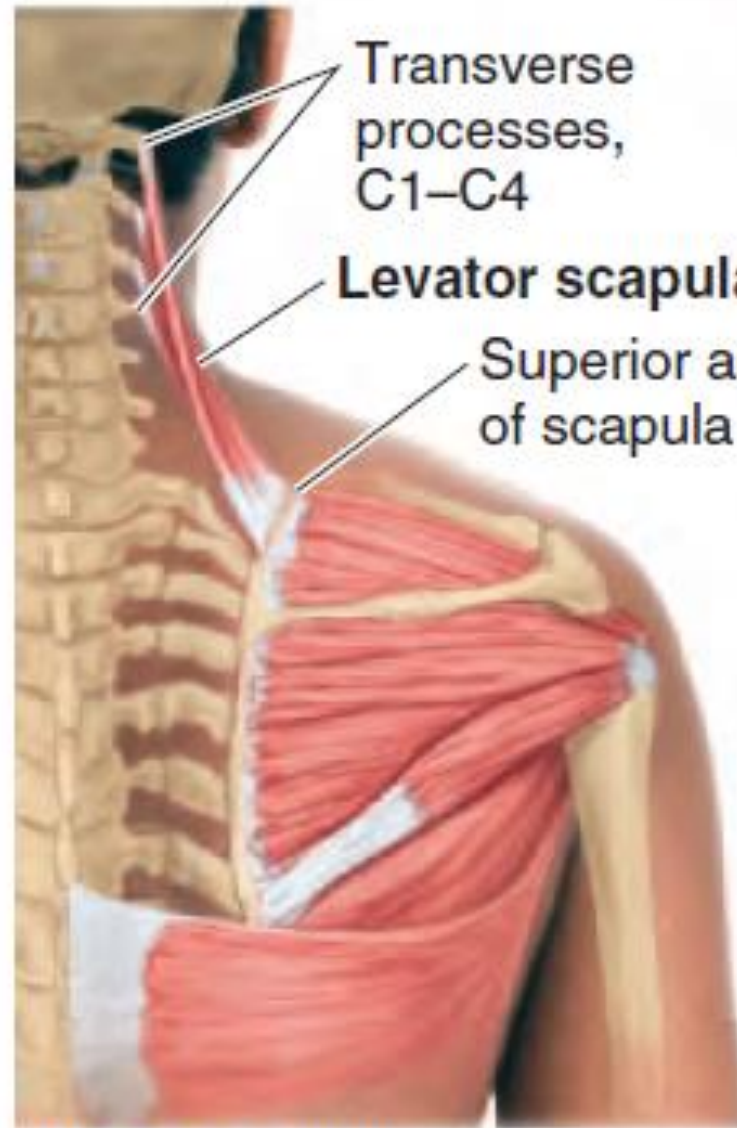
✓ *Deep posterior axio-appendicular (extrinsic shoulder) muscles:*
levator scapulae and rhomboids.

❖ **Levator scapulae:** lies deep to the SCM & Trapezius

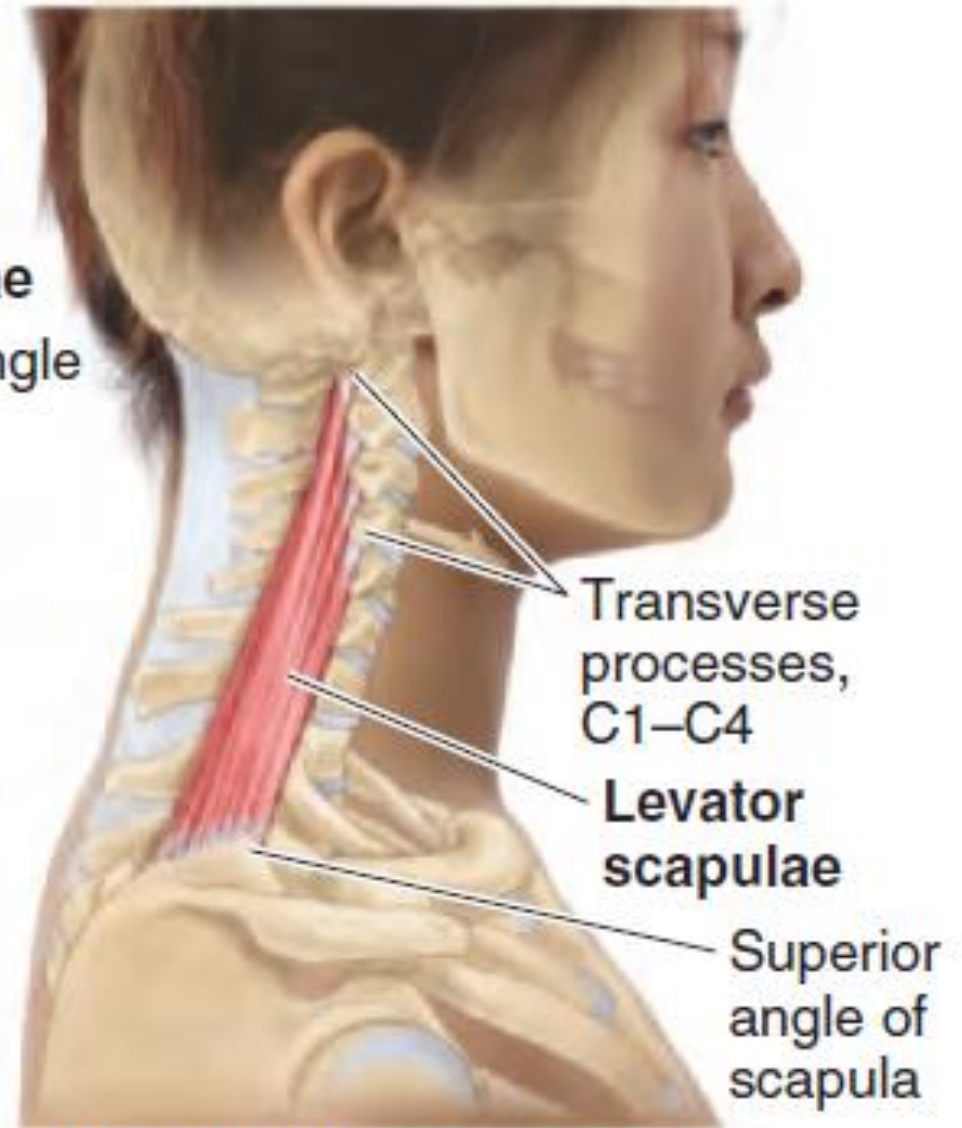
- True to its name, it acts to elevate the scapula
- **Origin:** Posterior tubercles of transverse processes of C1 - C4 vertebrae
- **Insertion:** Medial border of scapula superior to root of spine
- **Nerve Supply:** **Dorsal scapular** (C5) and **cervical** (C3, C4) nerves
- **Main Action:** Elevates scapula and tilts its glenoid cavity inferiorly by rotating scapula



Deep posterior axioappendicular (extrinsic shoulder) muscles,,,



Posterior view



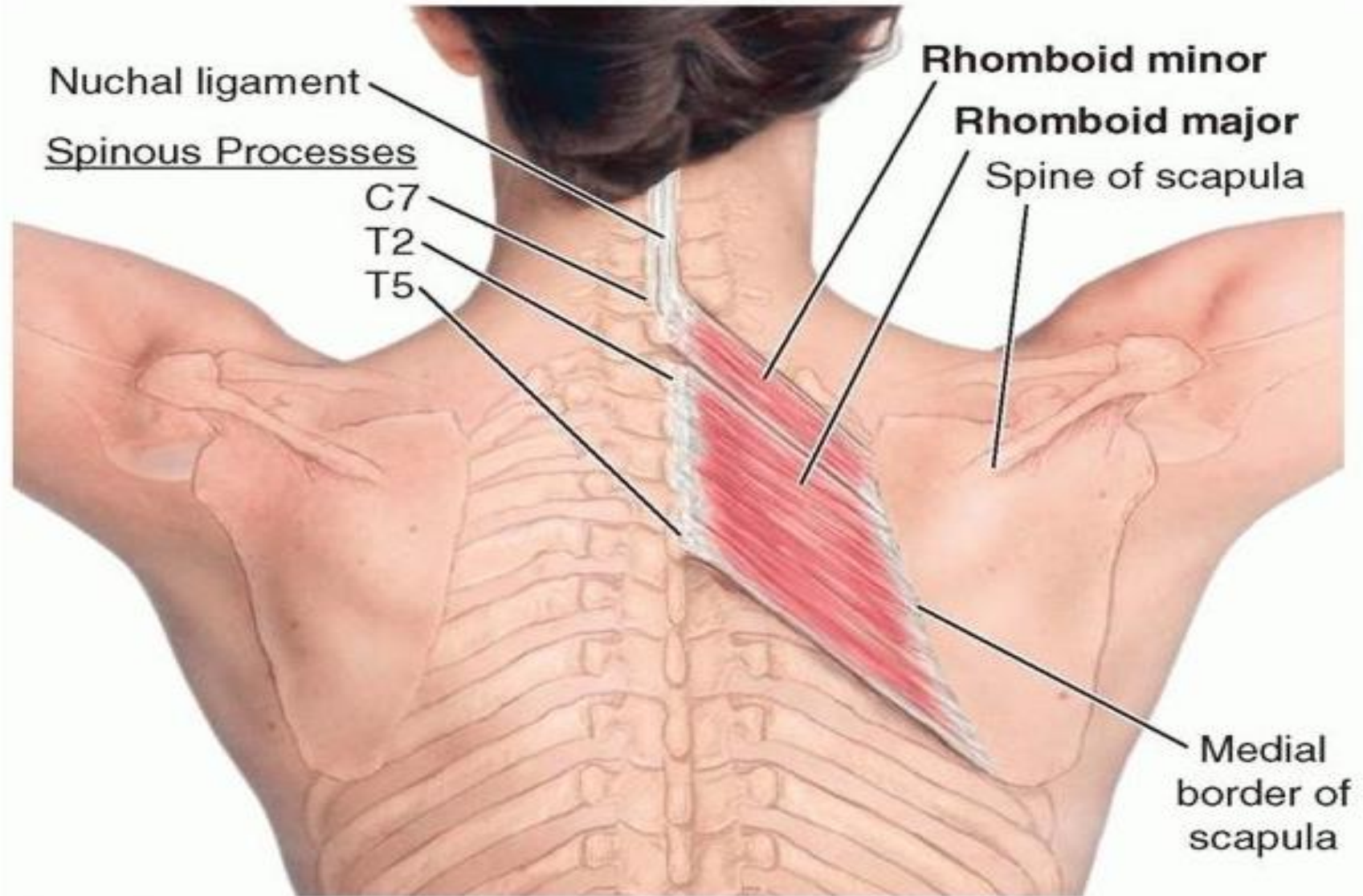
Lateral view

Deep posterior axioappendicular (extrinsic shoulder) muscles,,,-

❖ **Rhomboids (major and minor:** Not always clearly separated from each other

- Have a rhomboid appearance
- They lie deep to the trapezius and form broad parallel bands that pass inferolaterally from the vertebrae to the medial border of the scapula
- **Rhomboid Major: THINNER**, ~2x **wider** than **rhomboid minor**
- **Rhomboid minor-THICKER**, superior in position
- The rhomboids retract and rotate the scapula, depressing its glenoid cavity

Deep posterior axioappendicular (extrinsic shoulder) muscles,,,



Posterior view

Deep posterior axioappendicular (extrinsic shoulder) muscles,,,

□ Rhomboid **Major** : **Origin**: spinous processes of T2 - T5 vertebrae

- **Insertion**: medial border of scapula from level of spine to inferior angle
- **Nerve Supply**: Dorsal scapular nerve (C4, C5)
- **Main Action**: Retract scapula and rotate it to depress glenoid cavity; fix scapula to thoracic wall
- Rhomboid **minor** : **Origin**: Nuchal ligament, spinous processes of C7 and T1 vertebrae
- **Insertion**: Smooth triangular area at medial end of the spine of the scapula
- **Nerve Supply**: Dorsal scapular nerve (C4, C5)
- **Main Action**: Retract scapula and rotate it to depress glenoid cavity; fix scapula to thoracic wall.

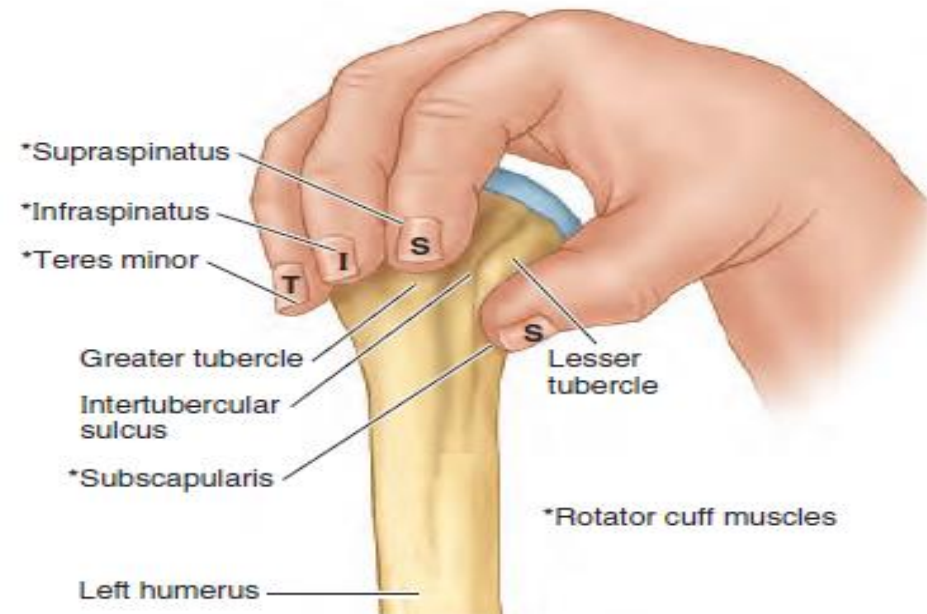
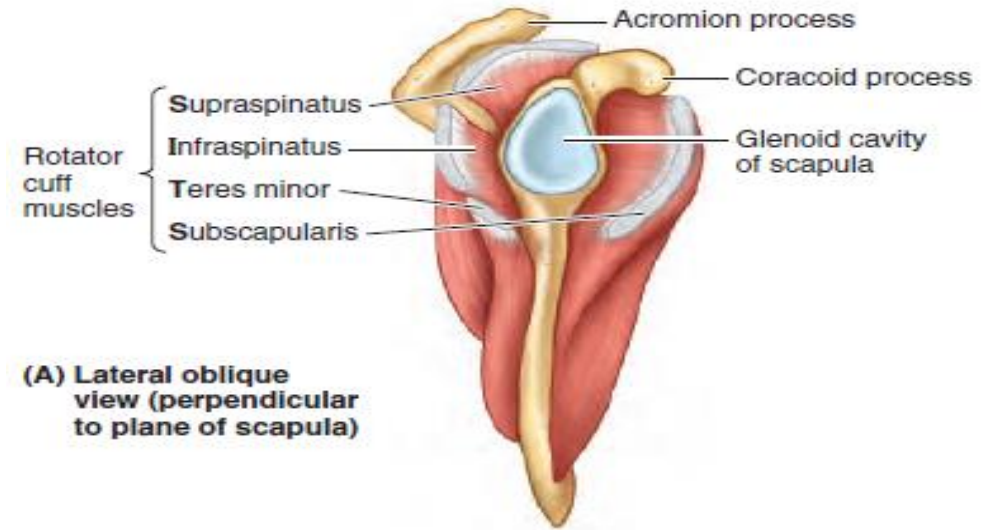
Scapulohumeral (intrinsic shoulder) muscles

- The **six scapulohumeral muscles** are relatively short muscles that pass from the scapula to the humerus and act on the **glenohumeral** joint.

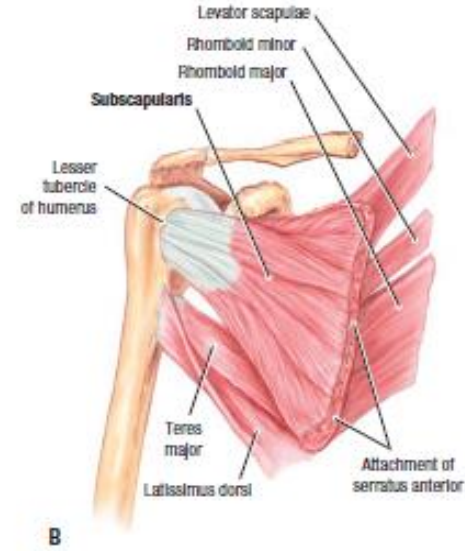
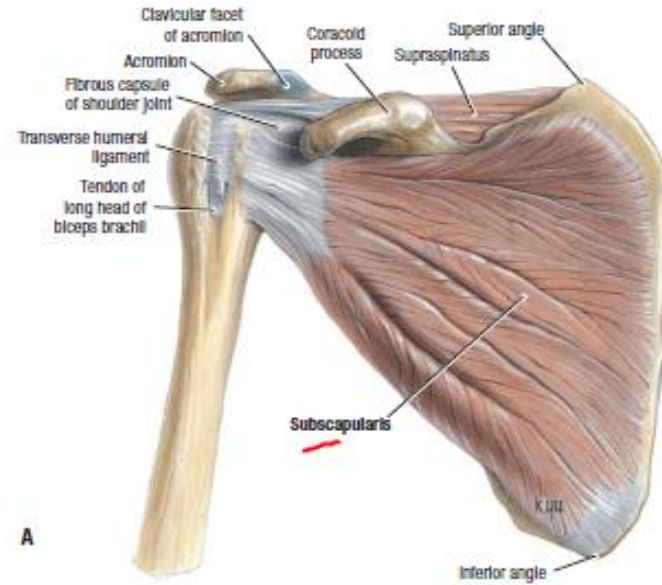
✓ **Deltoid**,

✓ **Teres major**, and The four **Rotator cuff muscles** which includes

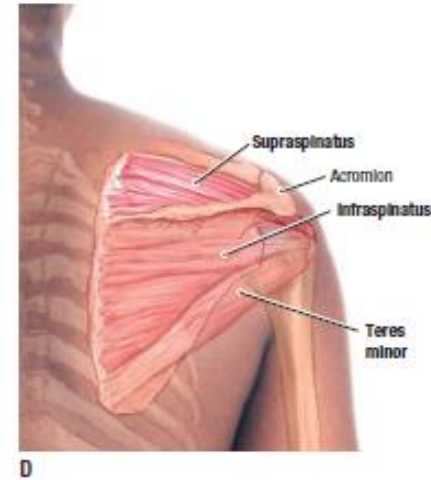
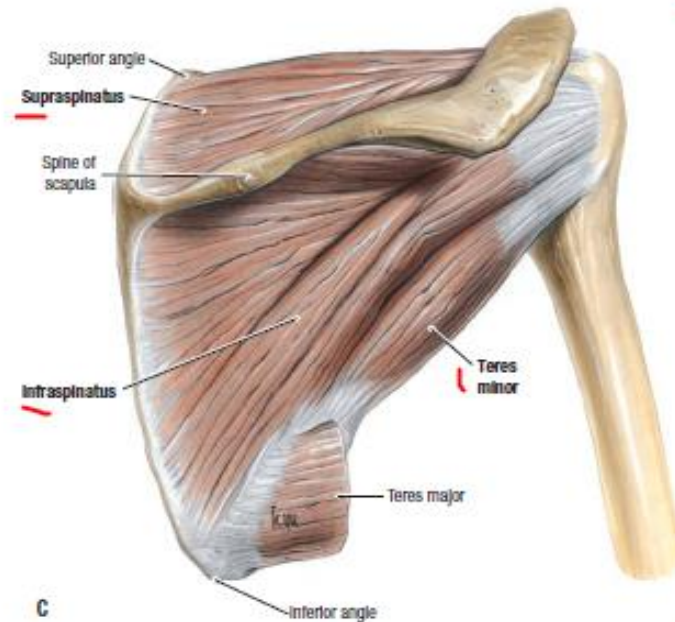
- **Supraspinatus**,
- **Infraspinatus**,
- **Teres minor**, and
- **Subscapularis**



Scapulohumeral (intrinsic shoulder) muscles,,,



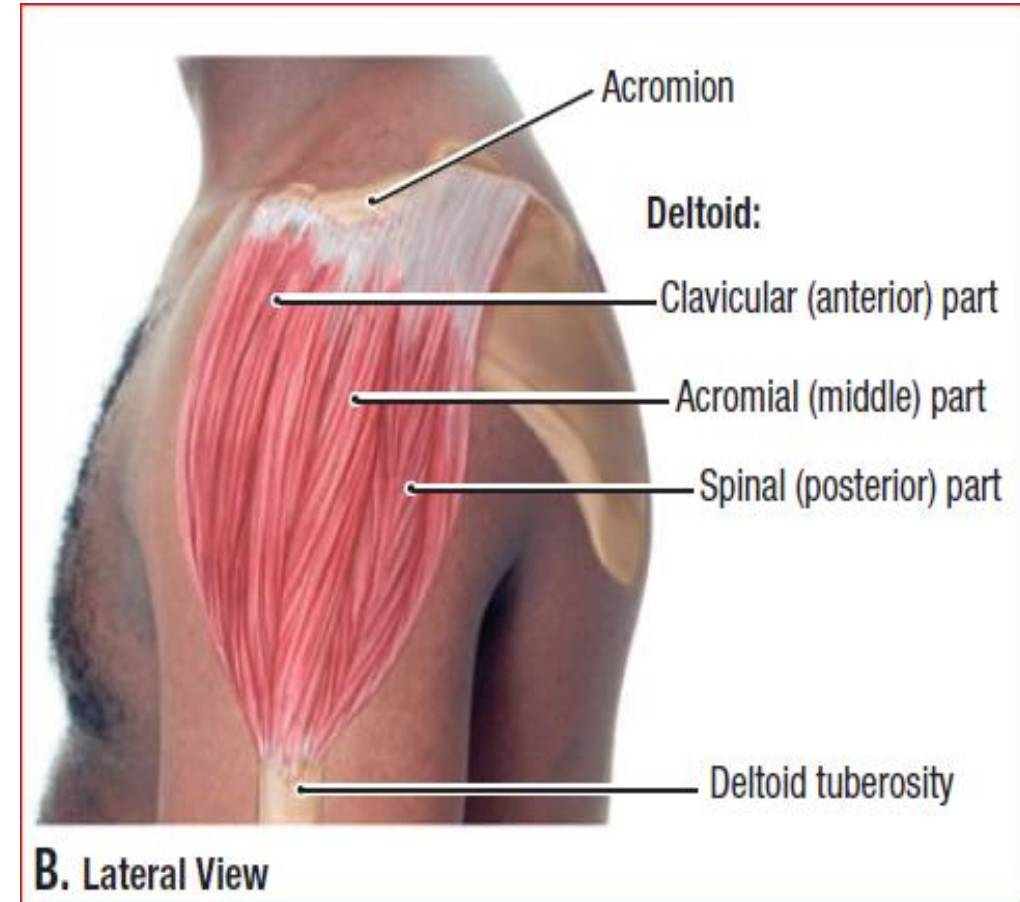
Anterior Views



Posterior Views

Scapulohumeral (intrinsic shoulder) muscles,,,

- ❖ **The deltoid muscle:** Is a thick, powerful muscle covering the shoulder and forming its rounded contour
- Is shaped like the inverted **Greek letter delta**
- Has anterior, middle & posterior parts
- The parts of the deltoid can act separately or as a whole
- **The anterior and posterior** parts of the deltoids are used to swing the limbs during walking
- The deltoid also helps **stabilize the glenohumeral joint.**



Scapulohumeral (intrinsic shoulder) muscles,,,

❖ **The deltoid muscle:,,** To initiate movement during the first 15° of abduction, the deltoid is assisted by the supraspinatus

- **Origin:** it has three origins:
- **Anterior fibers:** from the anterior border of the lateral third of clavicle
- **Middle fibers:** from the lateral border of acromion
- **Posterior fibers:** from the lower lip of the crest of the spine of scapula
- **Insertion:** into the deltoid tuberosity of the humerus

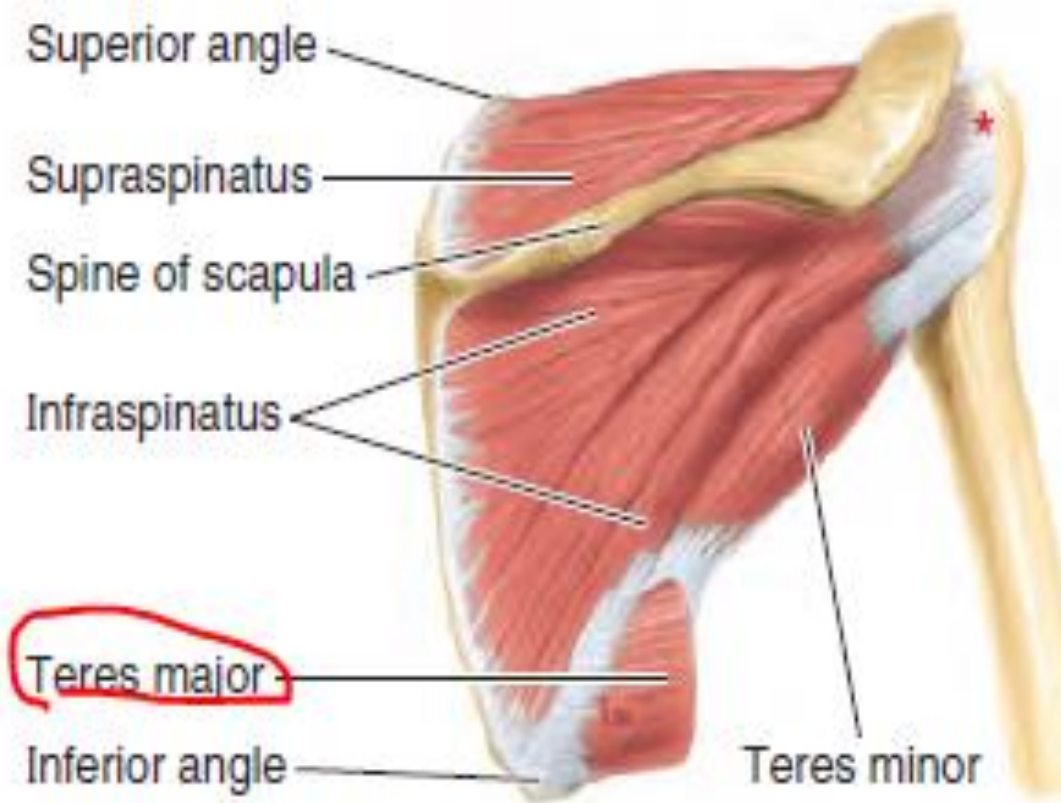
ACTION:

- **Anterior part:** flexes and medially rotates arm
- **Middle part:** abducts arm
- **Posterior part:** extends and laterally rotates arm
- **Nerve Supply:** Axillary nerve

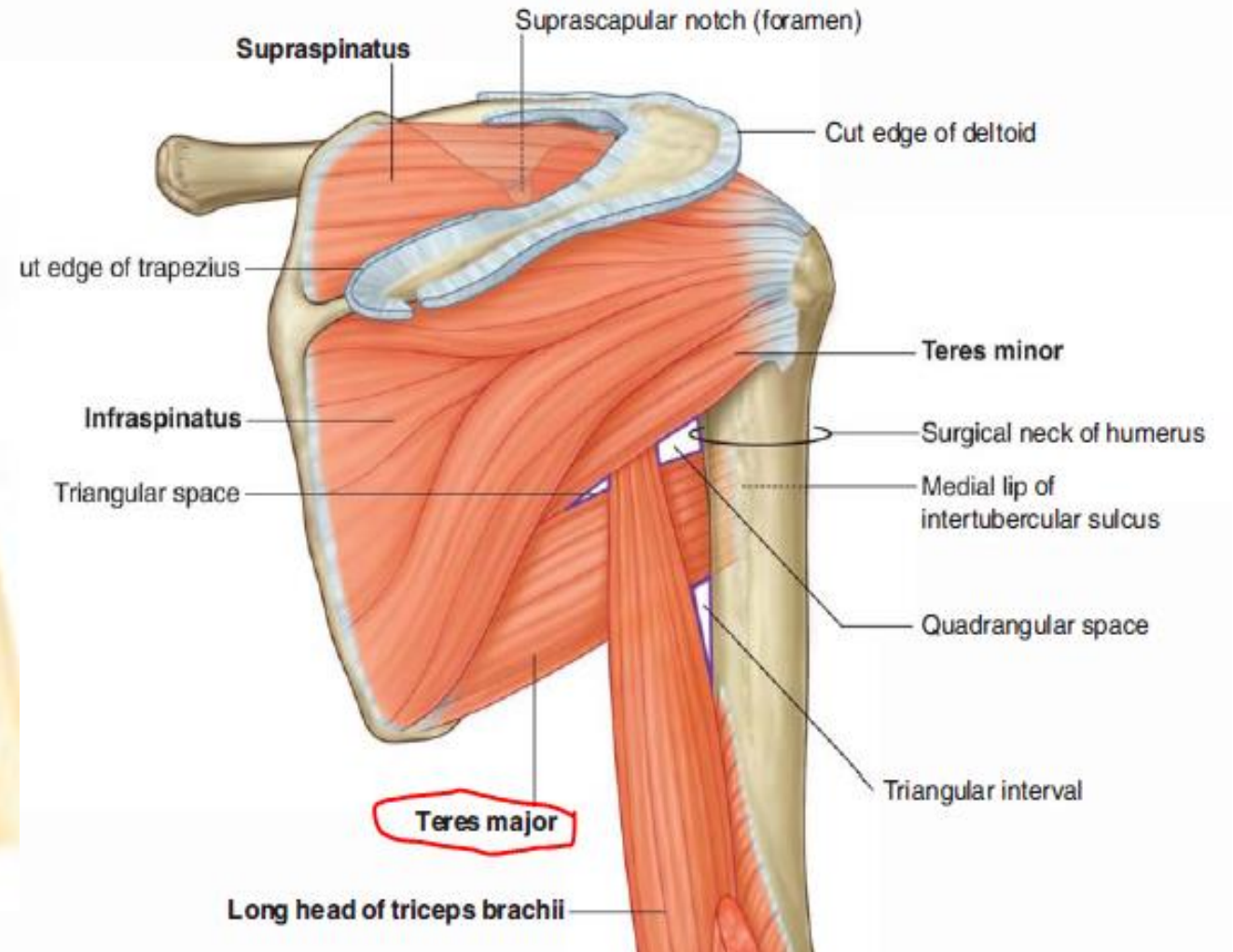
Scapulohumeral (intrinsic shoulder) muscles,,

- ❖ **Teres major:** the teres major is a thick, rounded muscle that forms a raised oval area on the inferolateral third of the scapula.
- ❖ *To test the teres major*, the abducted arm is adducted against resistance.
- ❖ It is also stabilizer of the humeral head in the glenoid cavity
- ✓ **Origin:** Posterior surface of inferior angle of scapula
- ✓ **Insertion:** Medial lip of intertubercular groove of humerus
- ✓ **Nerve Supply:** Lower subscapular nerve (C5, C6)
- ✓ **Main Action:** Adducts and medially rotates arm and assists in arm extension

Scapulohumeral (intrinsic shoulder) muscles,,,



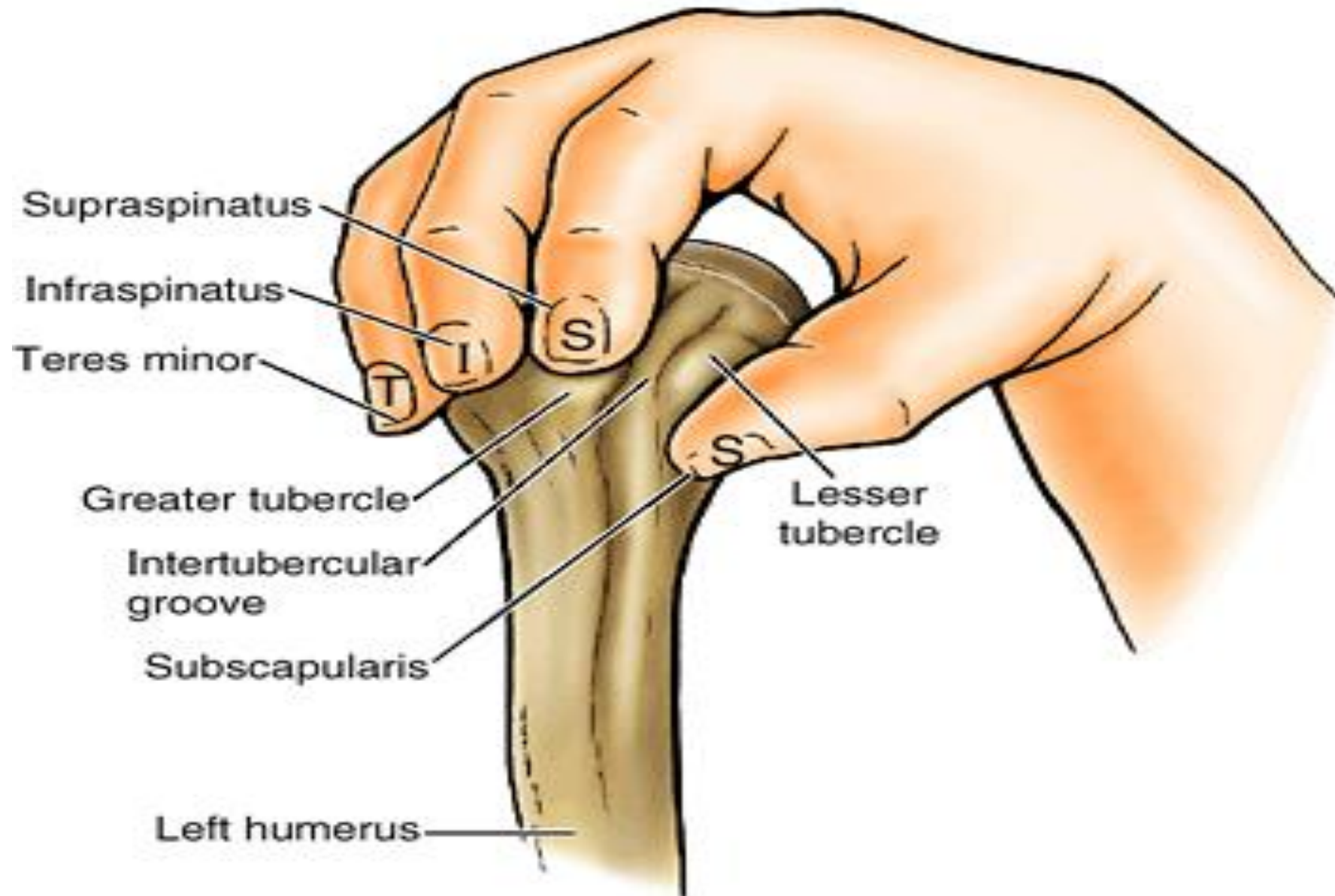
(B) Posterior view



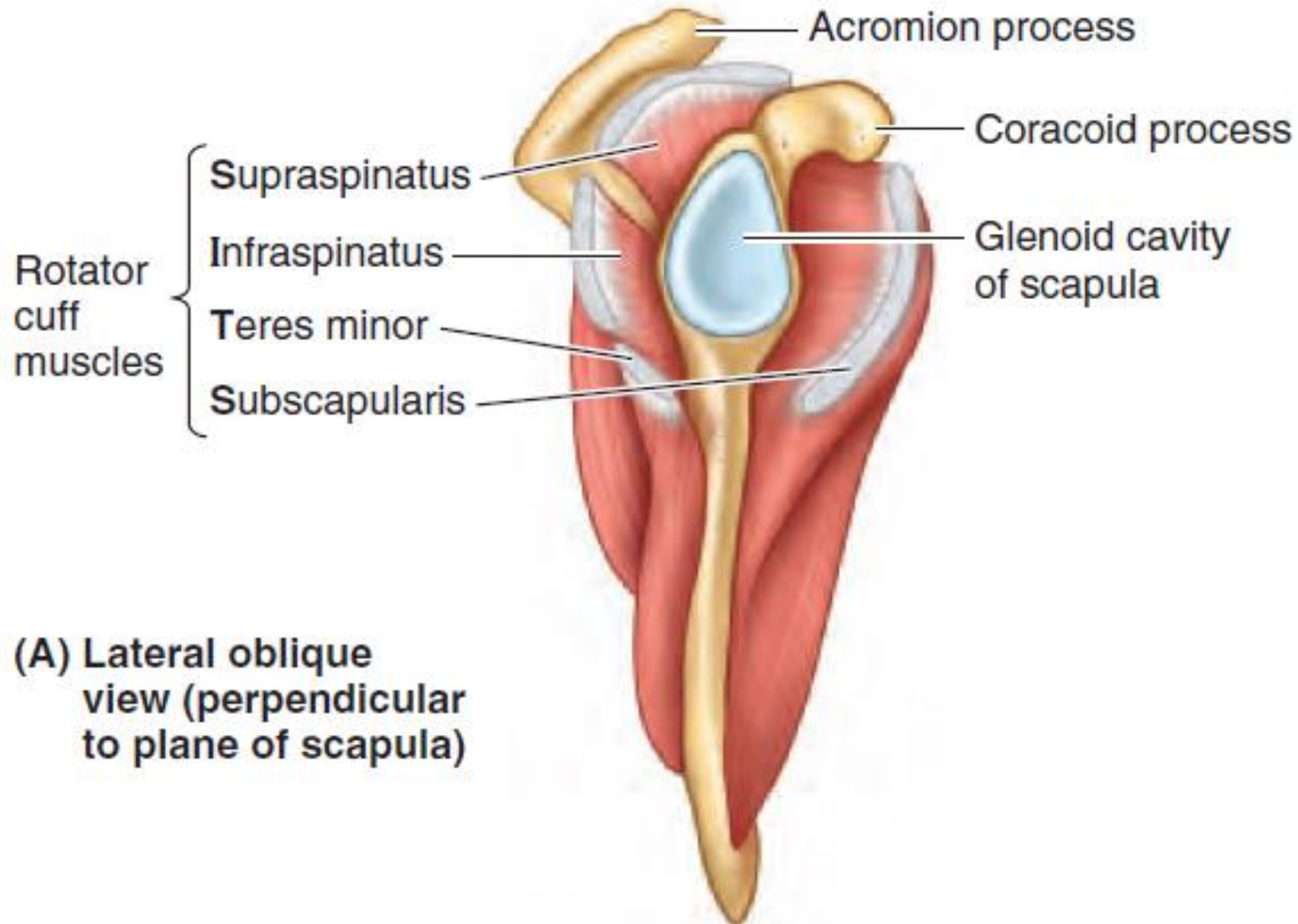
Rotator cuff muscle

- **Four of the scapulohumeral muscles** (intrinsic shoulder muscles) supraspinatus, infraspinatus, teres minor, and subscapularis (referred to as the SITS muscles) are called **rotator cuff muscles**
- These muscles form a **musculotendinous rotator cuff around the glenohumeral joint**
- The **SITS** muscles blend with and reinforce the fibrous layer of the joint capsule of the glenohumeral joint, thus forming the rotator cuff that protects the joint and gives it stability
- The tonic contraction of **SITS** holds the relatively large head of the humerus in the small, shallow glenoid cavity of the scapula during arm movements.

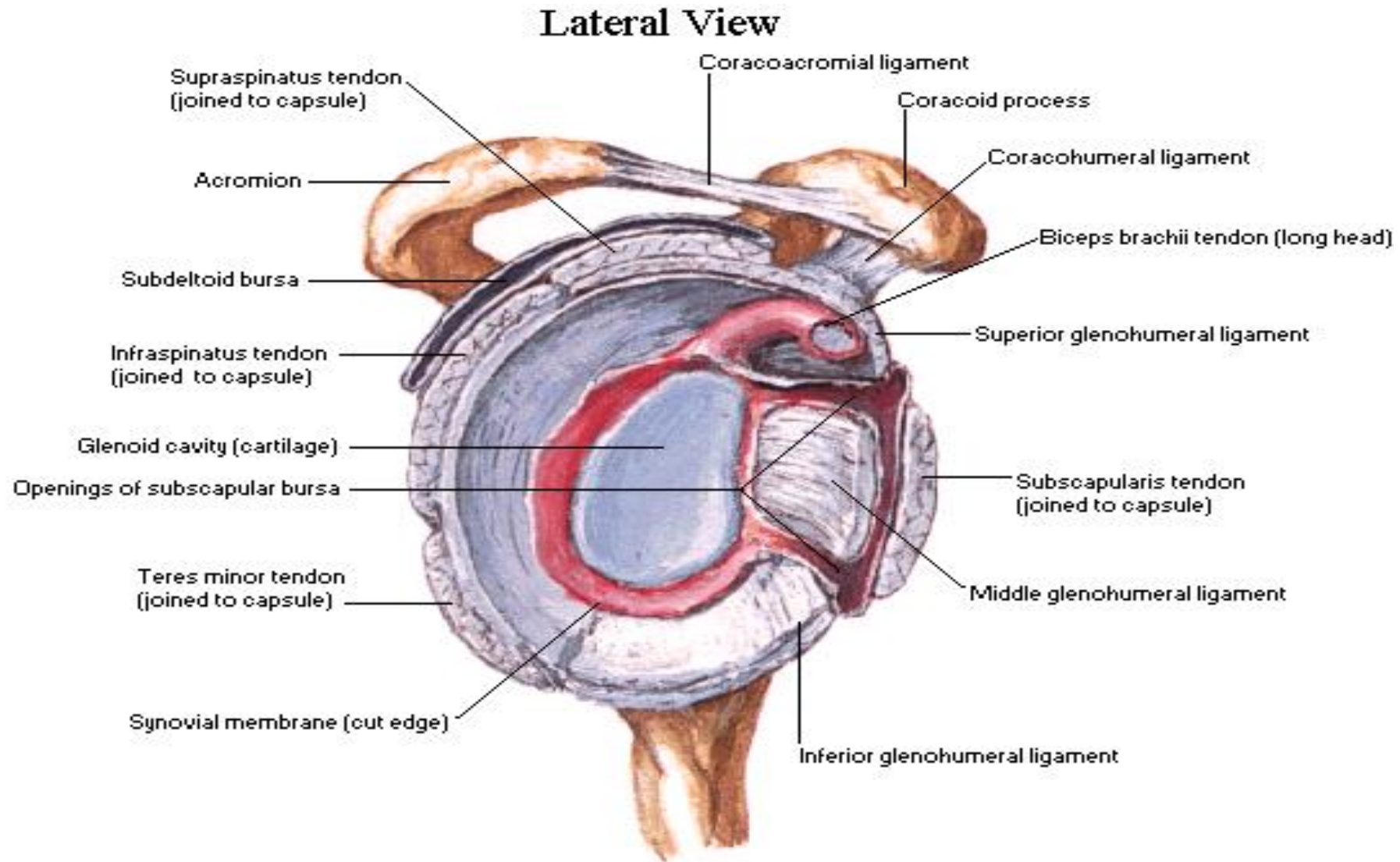
Rotator cuff muscle,,



Rotator cuff muscle,,

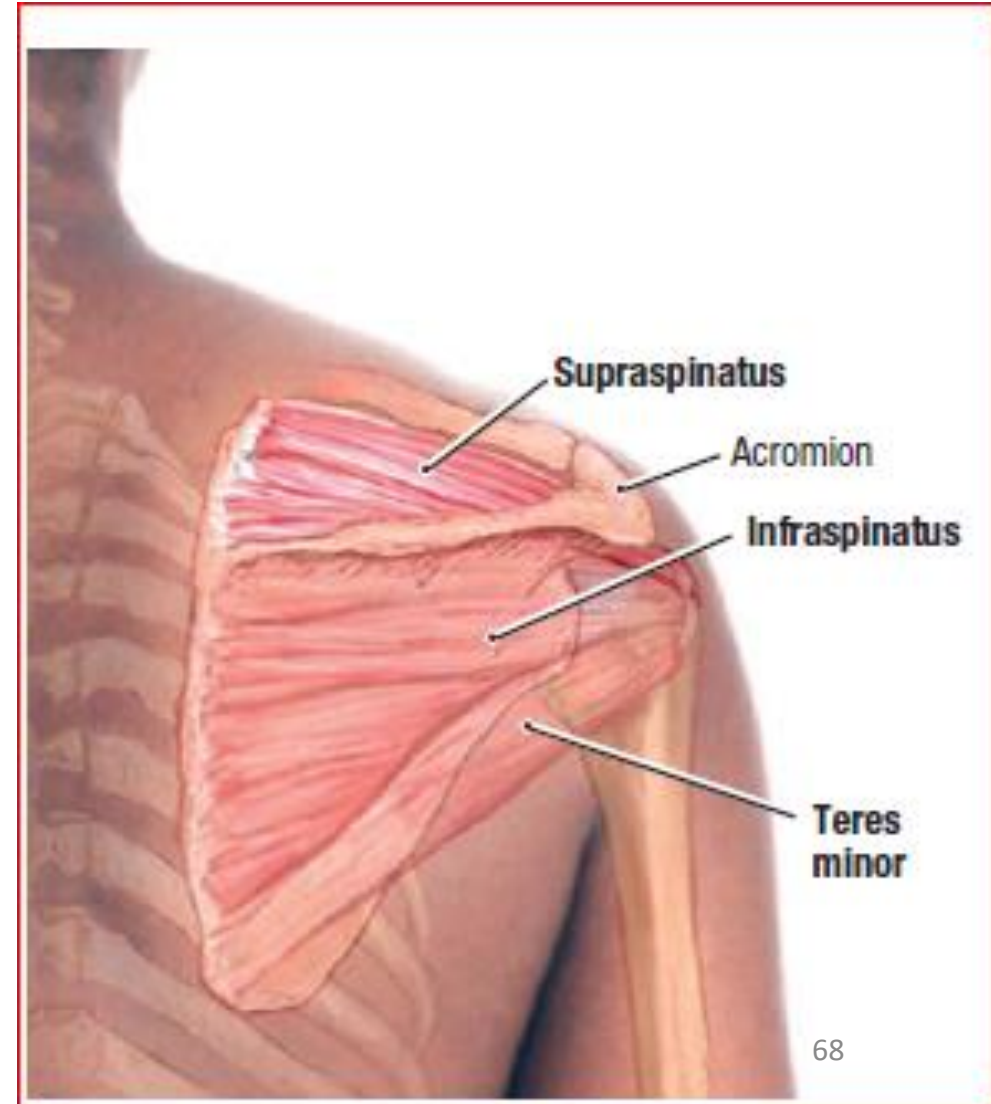


Rotator cuff muscle,,,

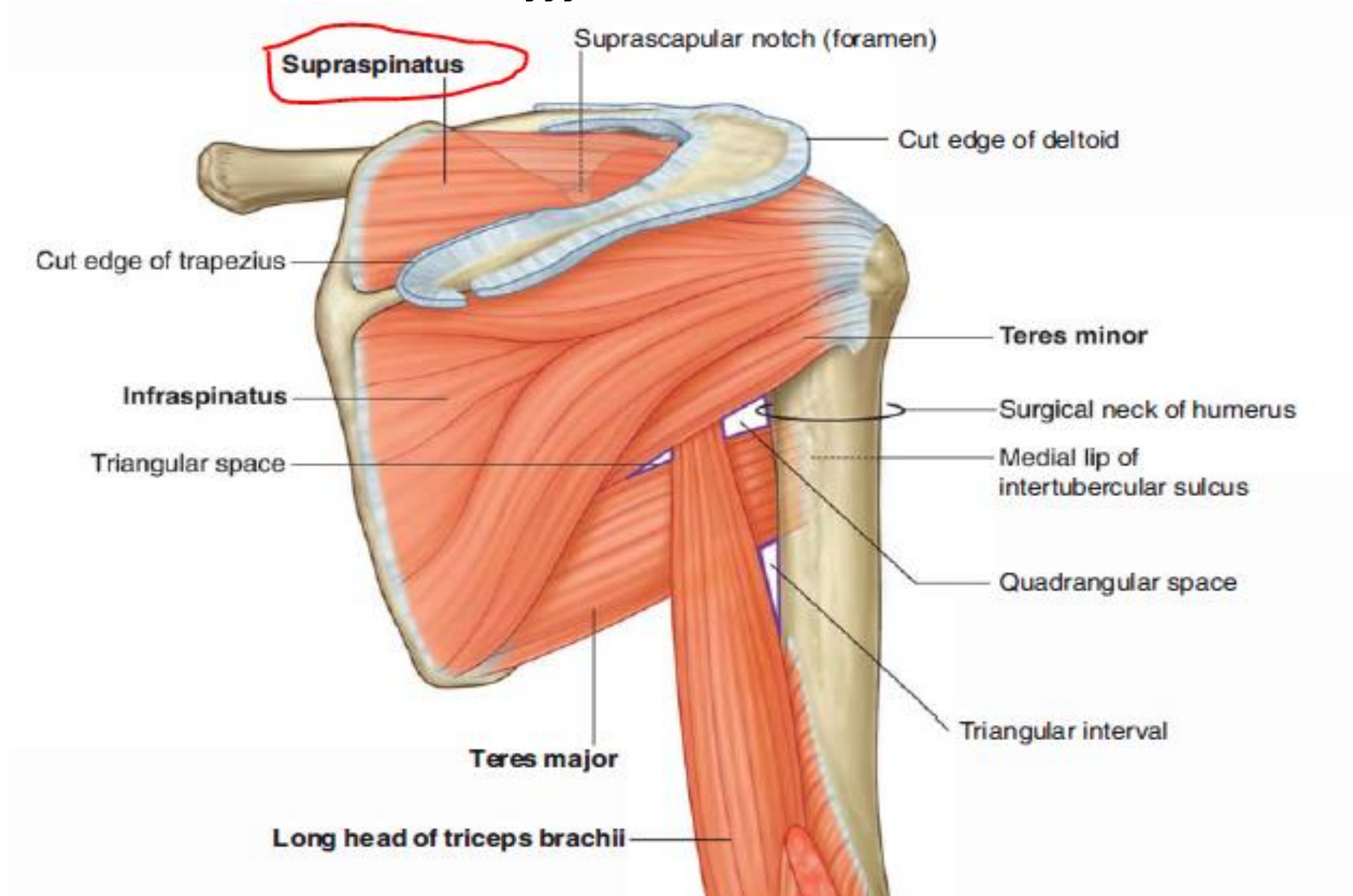


Rotator cuff muscle,,

- ❖ **The supraspinatus;** occupies the **supraspinous fossa** of the scapula
- **Origin:** Supraspinous fossa of scapula
- **Insertion:** Superior facet of greater tubercle of humerus
- **Nerve Supply:** Suprascapular nerve
- **Main Action:** Initiates and assists deltoid in abduction of arm and acts with rotator cuff muscles



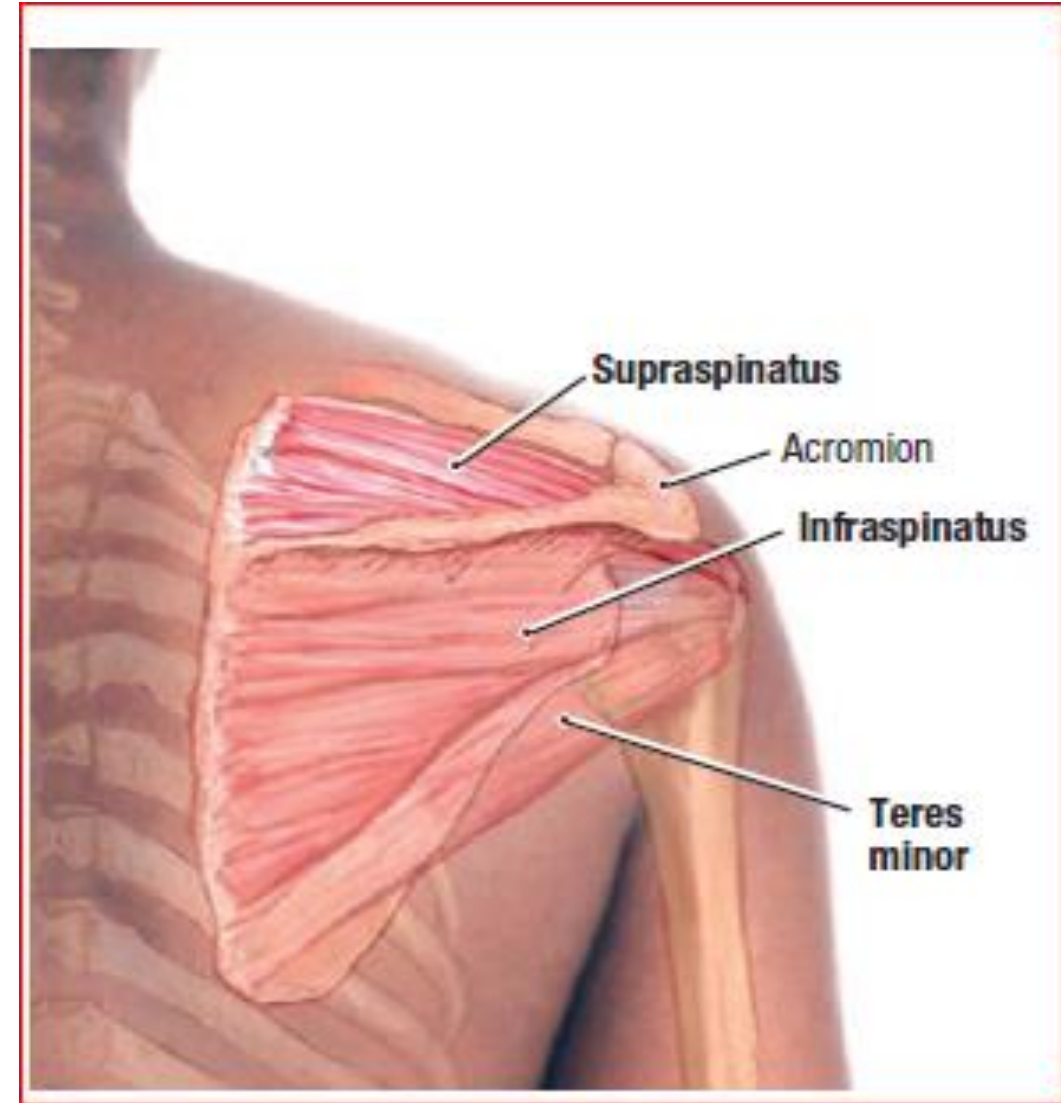
Rotator cuff muscle,,



Rotator cuff muscle,,

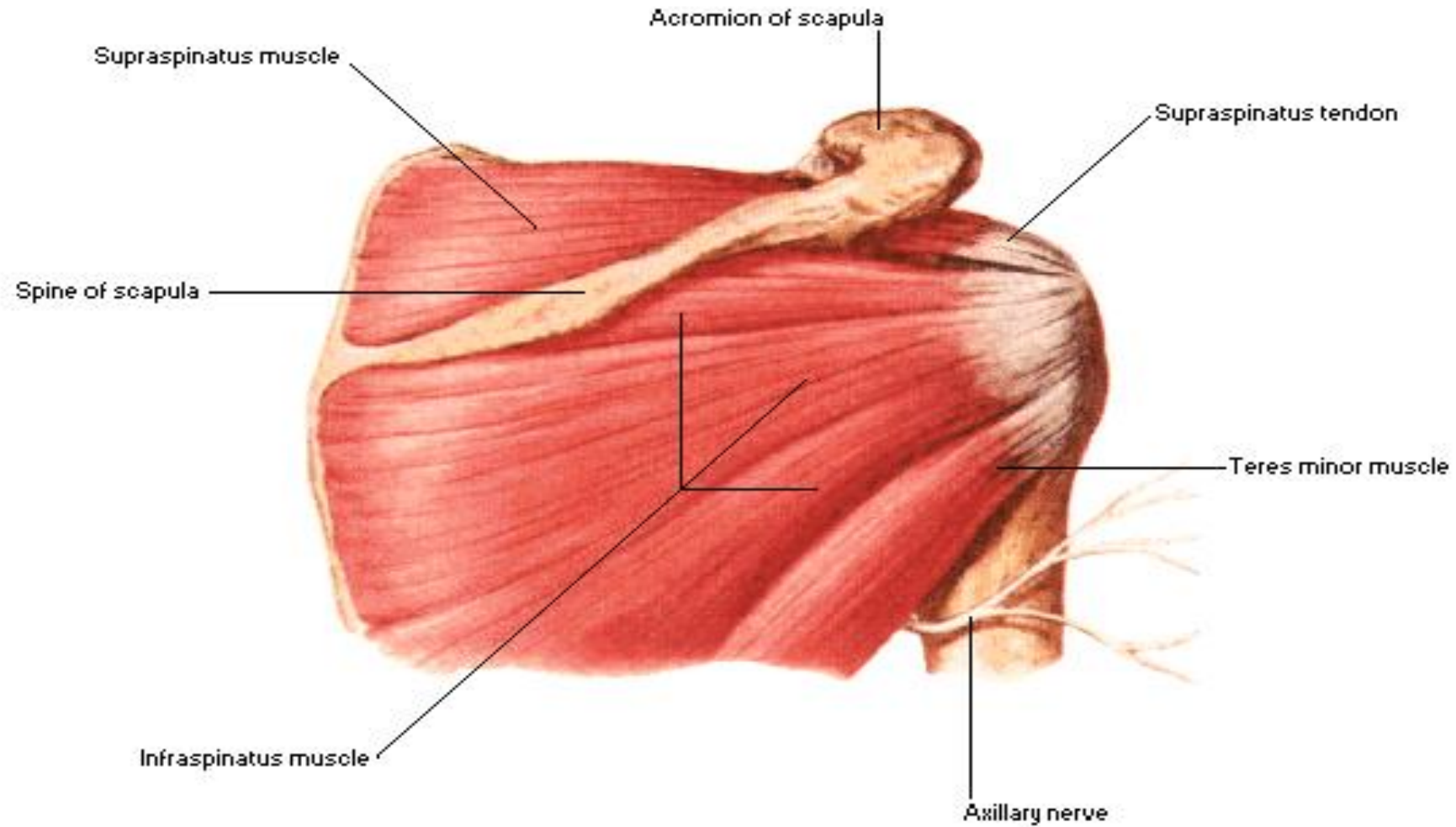
❖ Infraspinatus:

- **Origin:** Infraspinous fossa of scapula
- **Insertion:** Middle facet of greater tubercle of humerus
- **Nerve Supply:** Suprascapular nerve
- **Main Action:** Laterally rotate arm; help hold humeral head in glenoid cavity of scapula



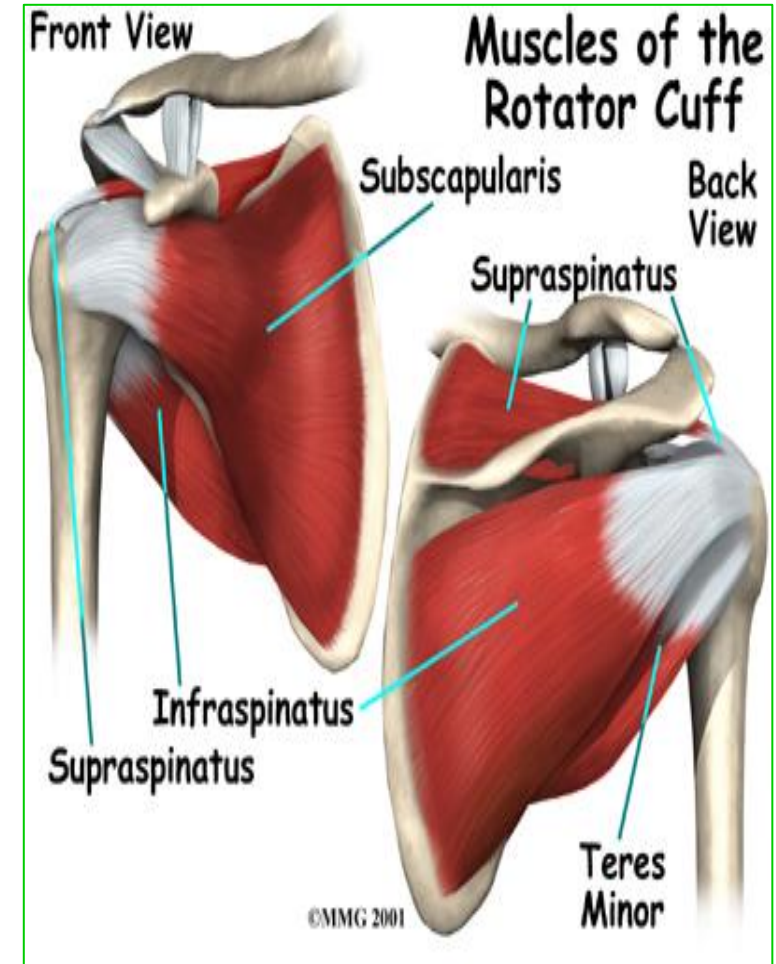
Rotator cuff muscle,,

Posterior View



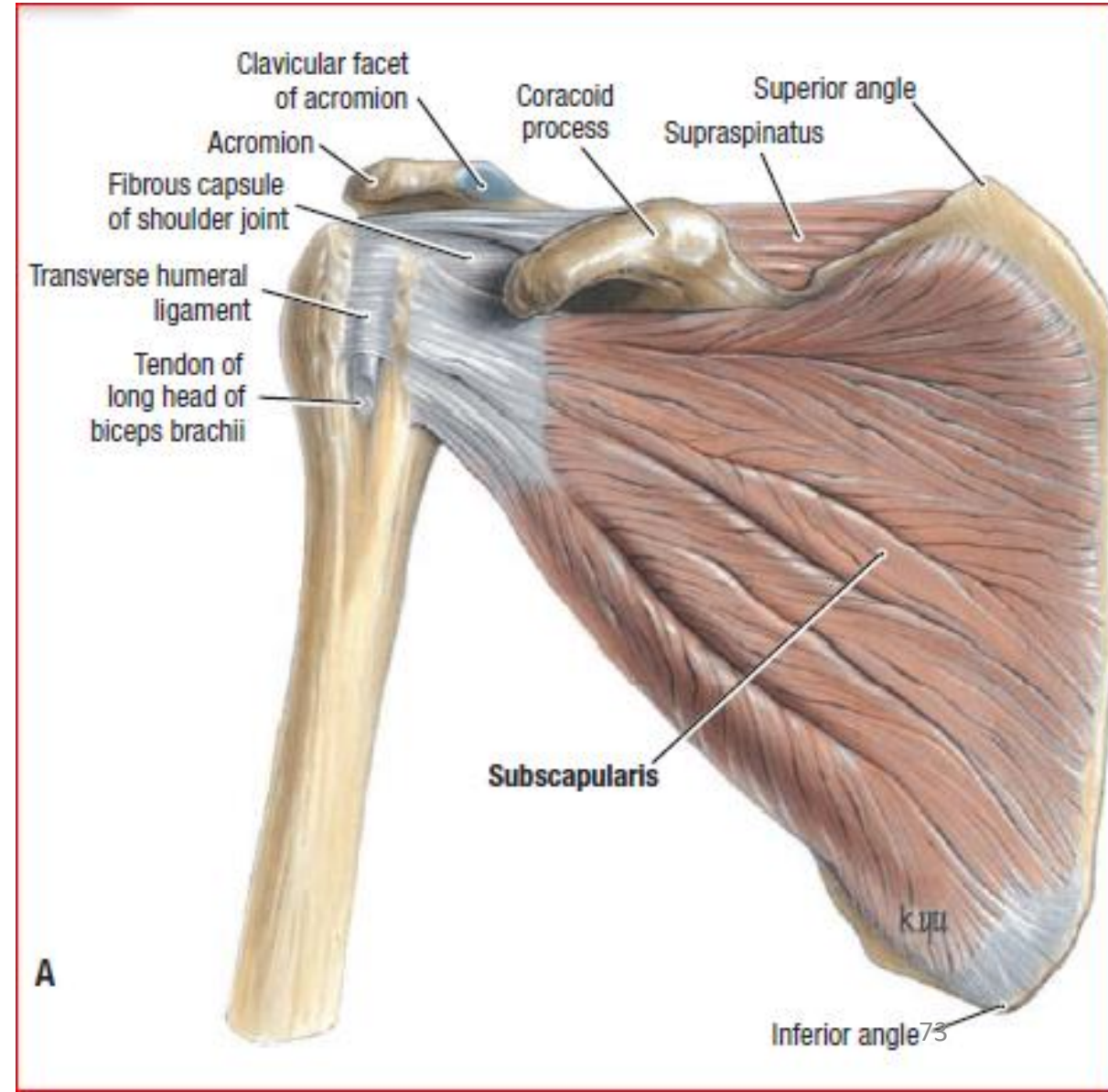
Rotator cuff muscle,,,

- ❖ **Teres minor:** Is a narrow, elongate muscle that is completely hidden by the deltoid and is often not clearly delineated from the infraspinatus
- Works with the infraspinatus to rotate the arm laterally & assist its adduction
- **Origin:** Middle part of lateral border of scapula
- **Insertion:** Inferior facet of greater tubercle of humerus
- **Nerve Supply:** Axillary nerve
- **Main Action:** Laterally rotate arm; help hold humeral head in glenoid cavity of scapula



Rotator cuff muscle,,

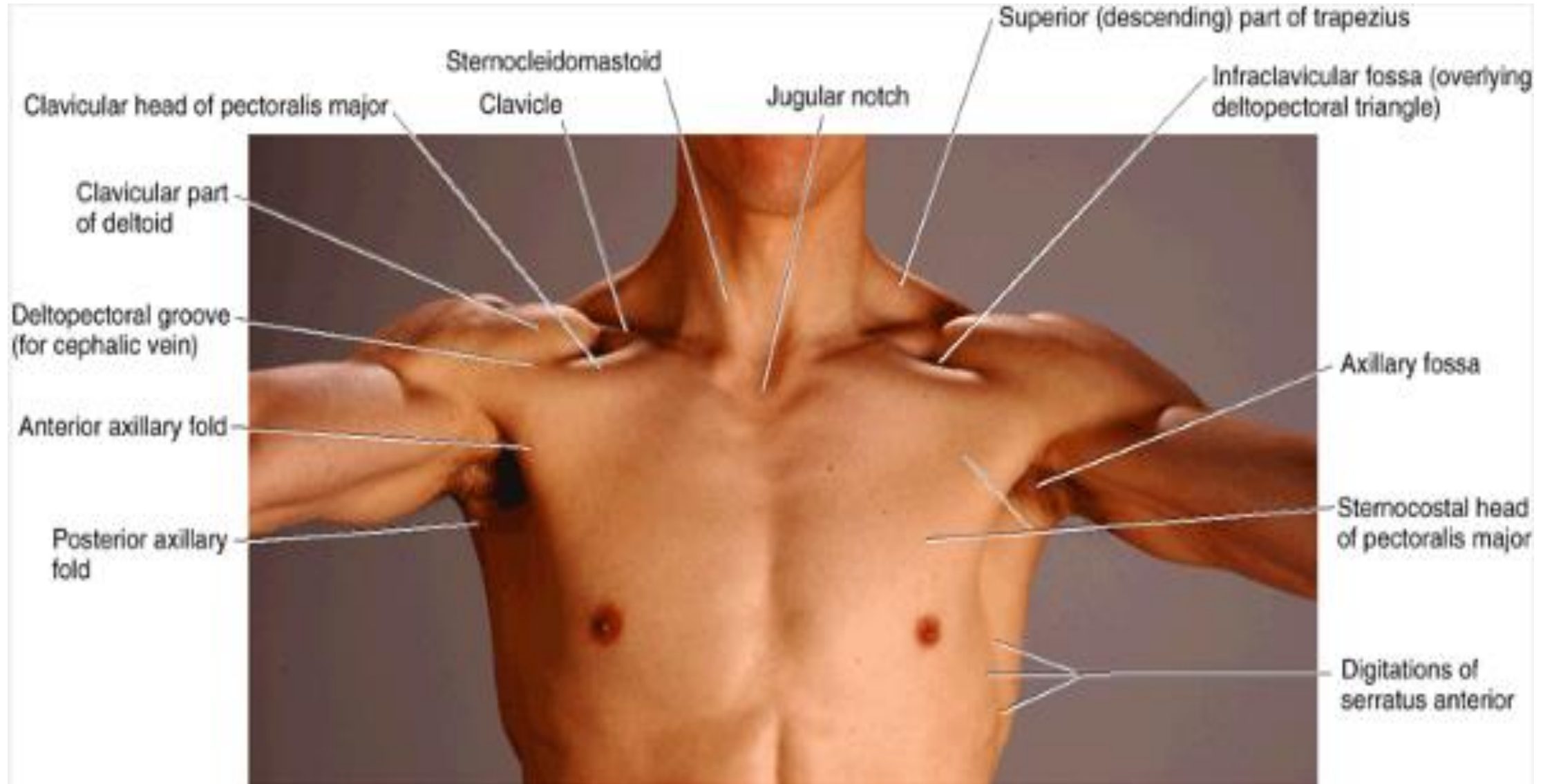
- ❖ **Subscapularis;** Is a thick, triangular muscle that lies on the costal surface of the scapula
- **Origin:** Subscapular fossa
- **Insertion:** Lesser tubercle of humerus
- **Nerve Supply:** Upper and lower subscapular nerves
- **Main Action:** Medially rotates and adduct arm
 - Helps hold humeral head in glenoid cavity



Axilla

- The **axilla** (armpit) is the pyramidal space inferior to glenohumeral joint and superior to skin and axillary fascia at junction of arm and thorax.
- The axilla provides a passageway, or “distribution center,” for vessels and nerves to and from upper limb.
- It is area of transition between the arm the neck.

Axilla,,,



Axilla,,,

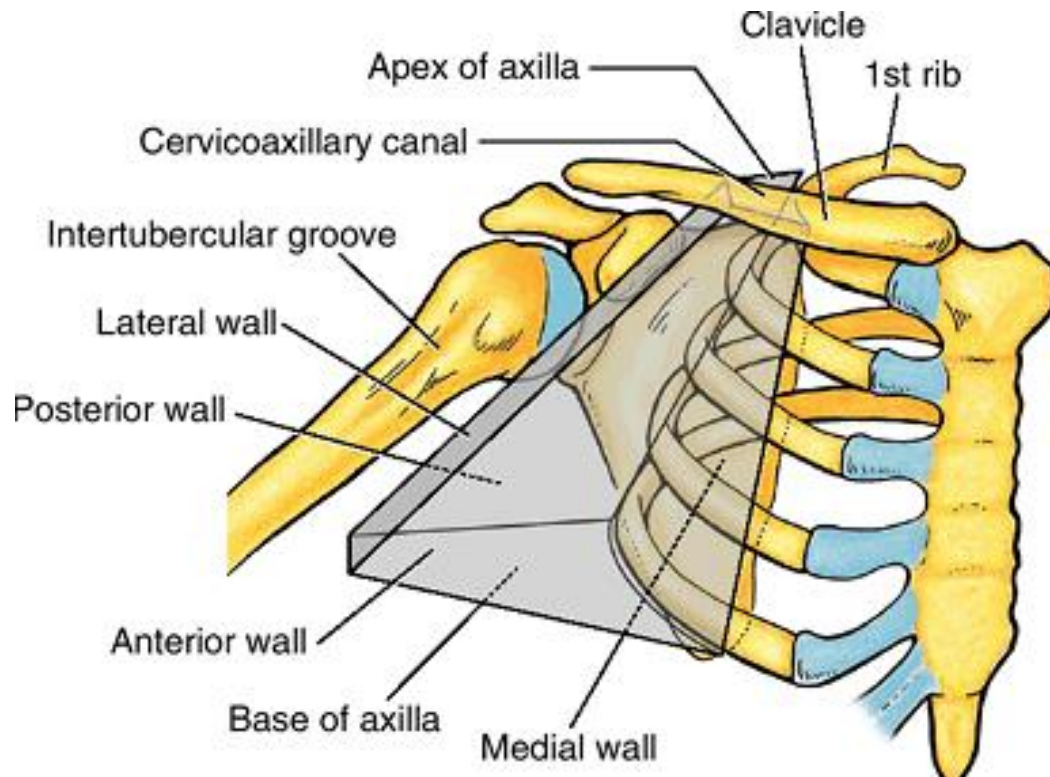
❖ Has an apex, a base, and four walls

✓ **Apex:** Is cervicoaxillary canal: passageway between neck and axilla.

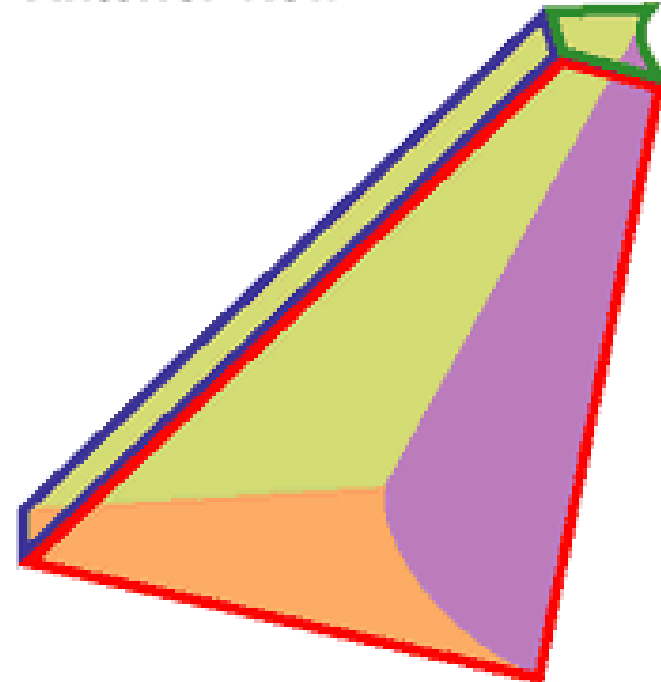
▪ Bounded by 1st rib, clavicle and superior edge of scapula

✓ **Base:** Formed by concave skin, subcutaneous tissue, and axillary fascia.

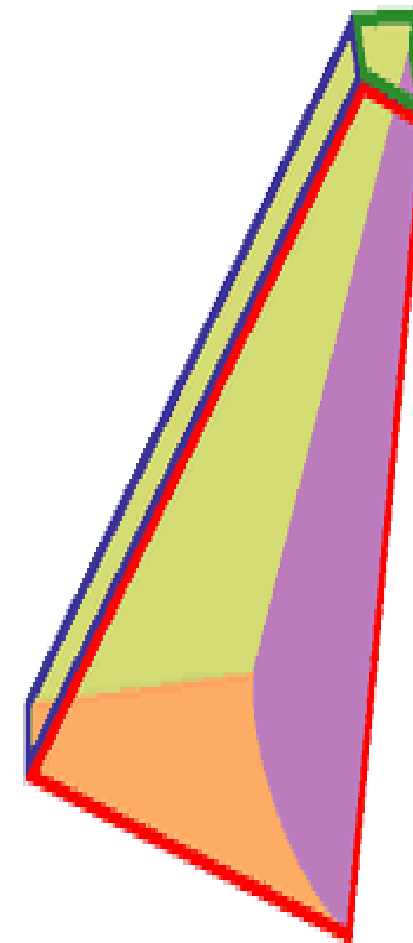
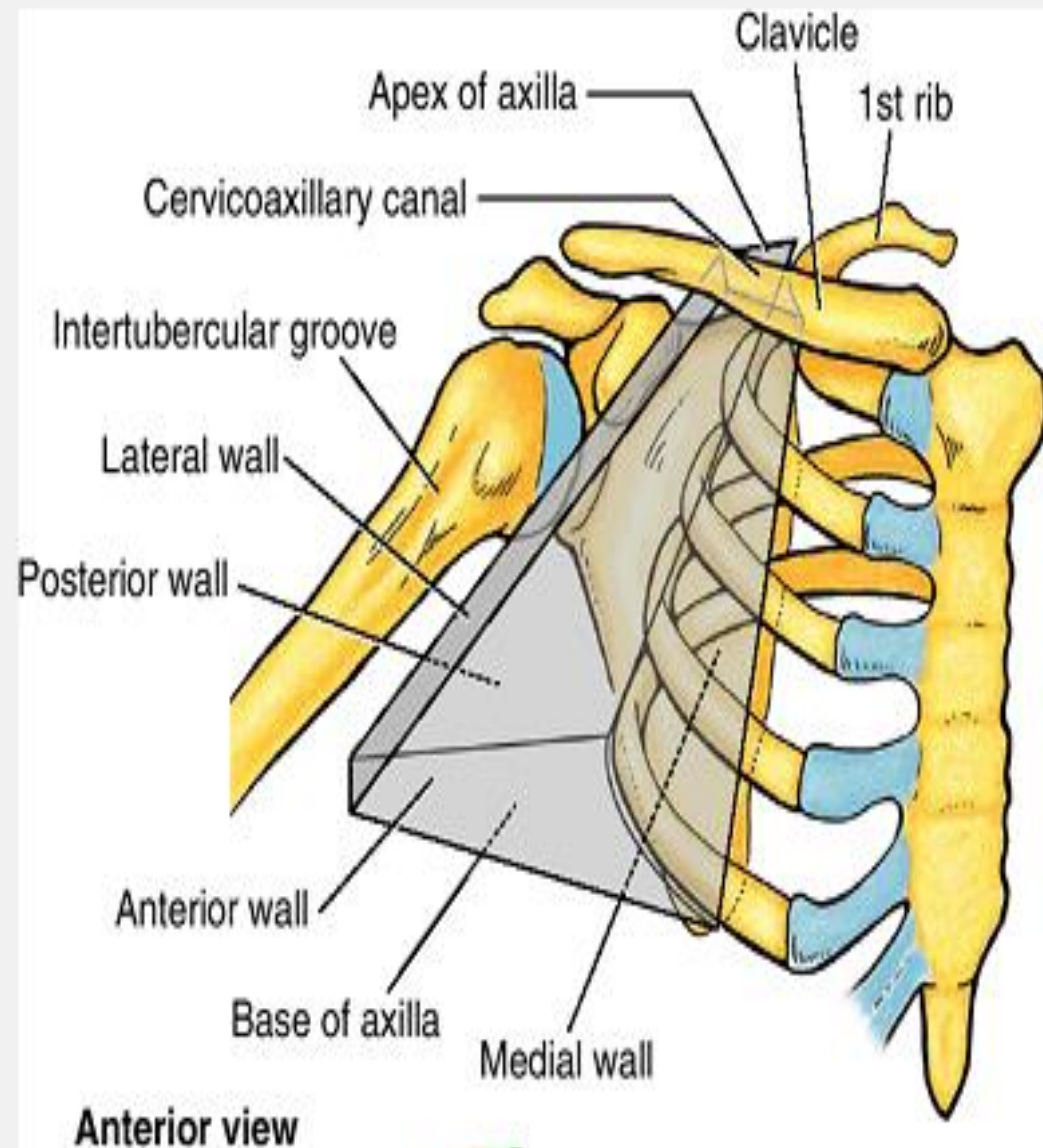
➔ Is bounded by anterior & posterior axillary folds, thoracic wall, & medial aspect of the arm .



Anterior view



Axilla,,,

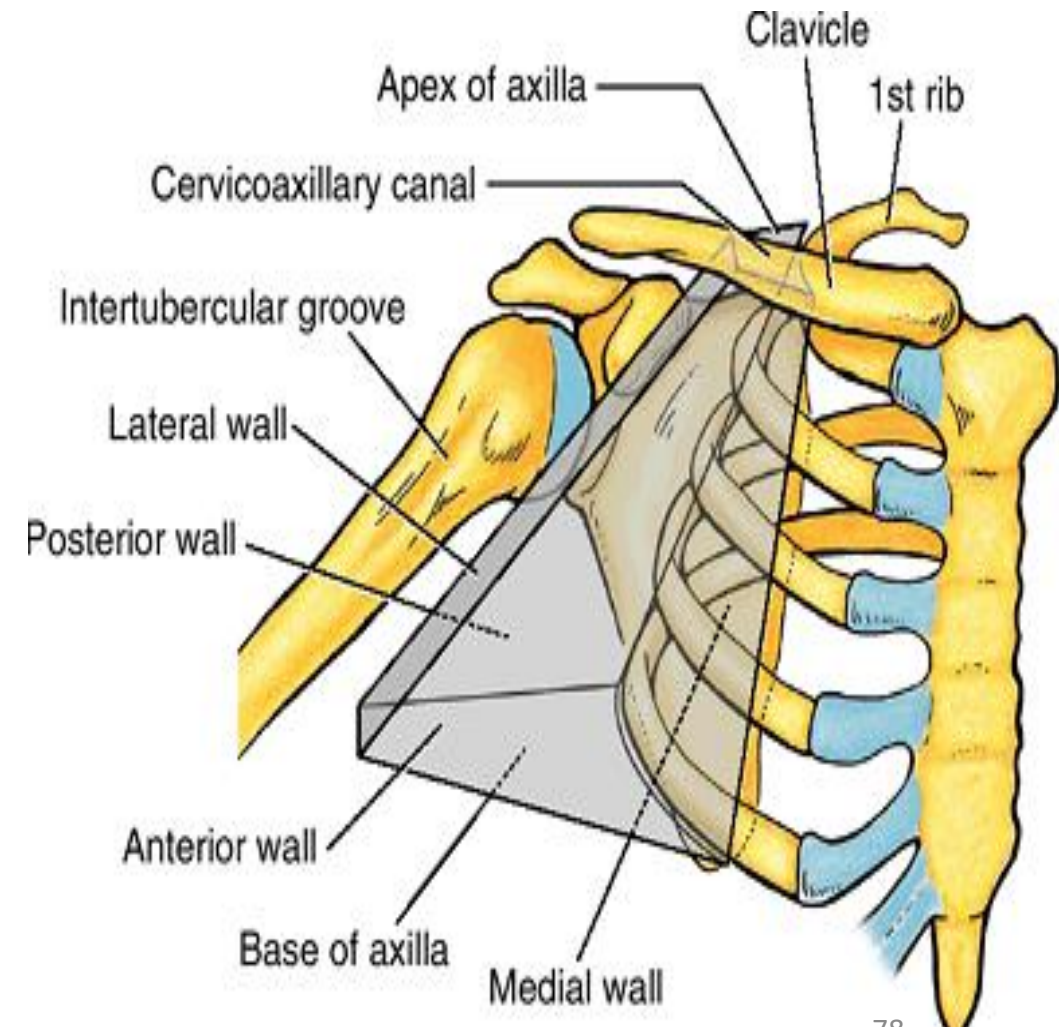


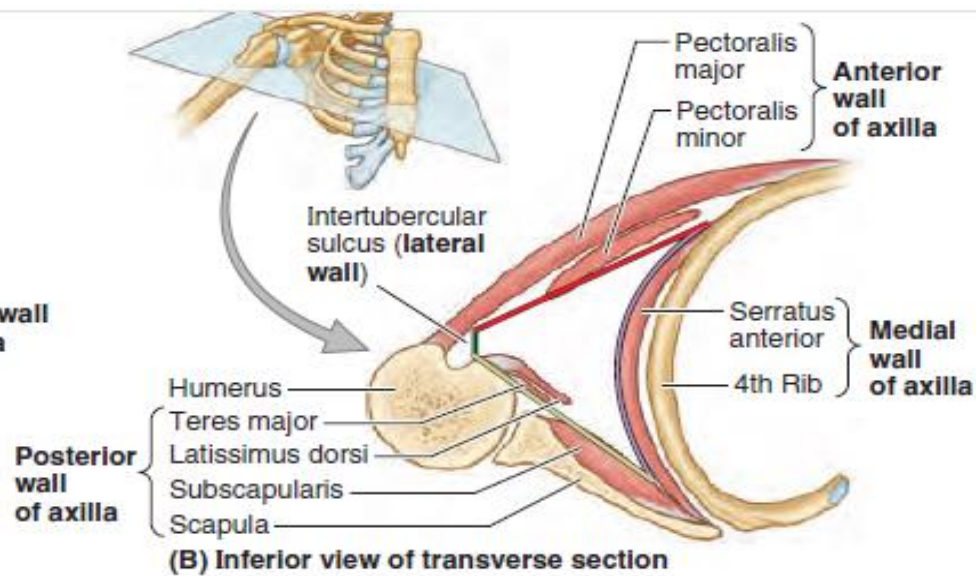
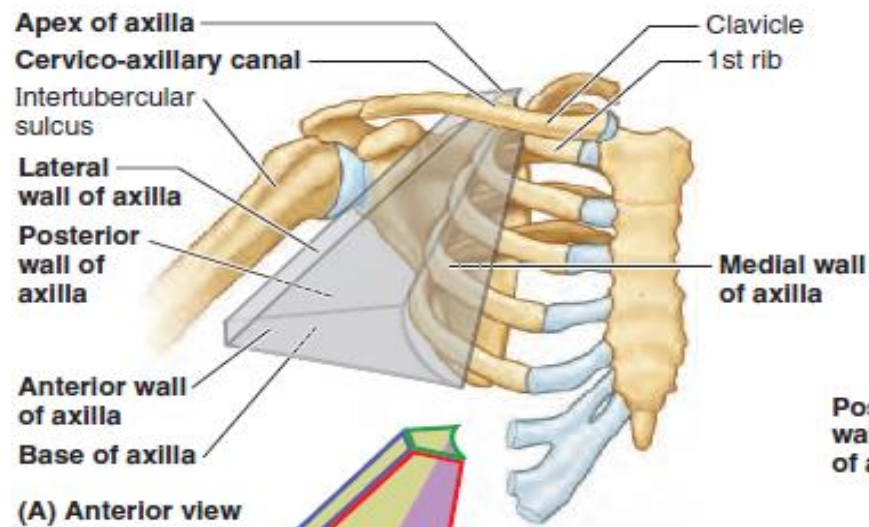
Axillary boundaries

- Apex
- Base
- Anterior wall
- Lateral wall
- Medial wall
- Posterior wall

Axilla,,,

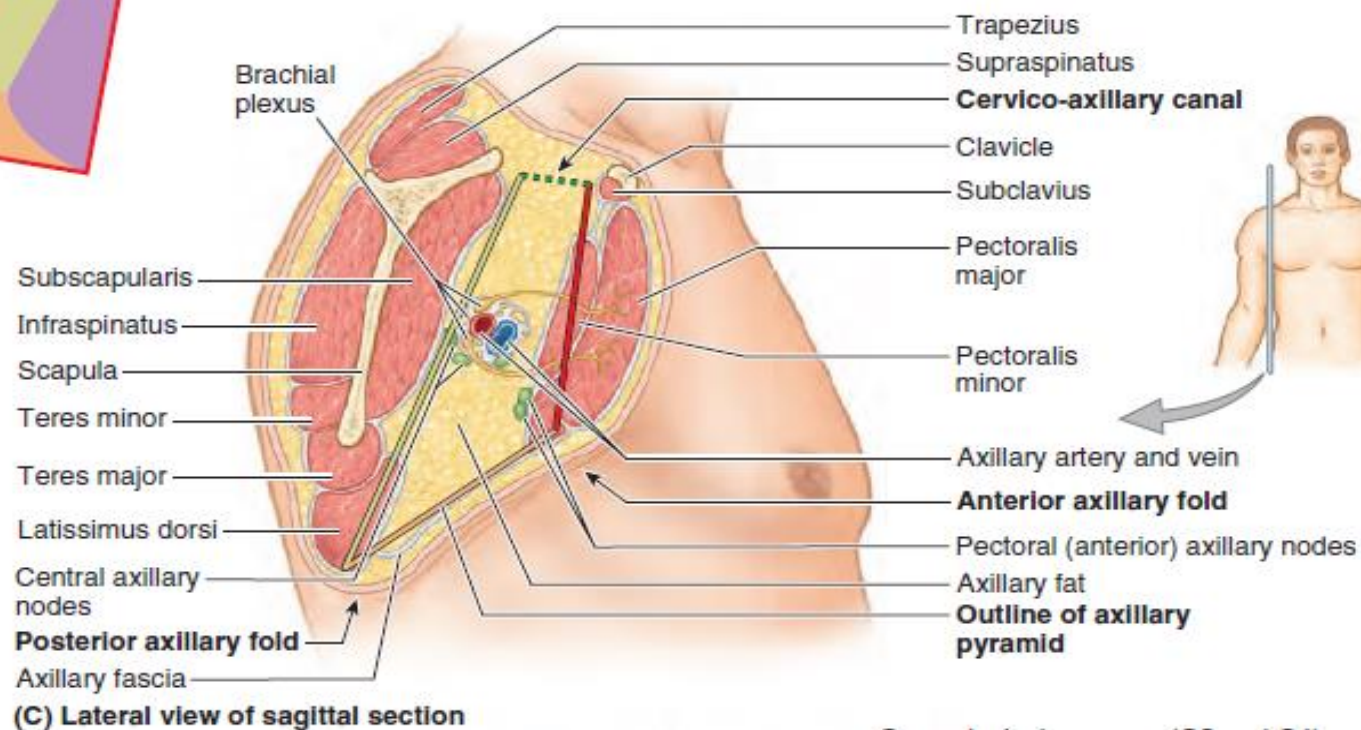
- **Medial wall:** Formed by the thoracic wall (1st-4th ribs and intercostal muscles) and serratus anterior muscle
- **Lateral wall:** humerus.
- **Posterior wall:** subscapularis, teres major, and latissimus dorsi muscles and inferiorly contains (Posterior axillary fold)
- **Anterior wall:** pectoralis major and pectoralis minor muscles with their fascia and inferiorly contain (Anterior axillary fold)





Axillary boundaries

- Apex
- Base
- Anterior wall
- Lateral wall
- Medial wall
- Posterior wall



Contents of the Axilla

❖ The axilla contains:

- axillary artery and its branches

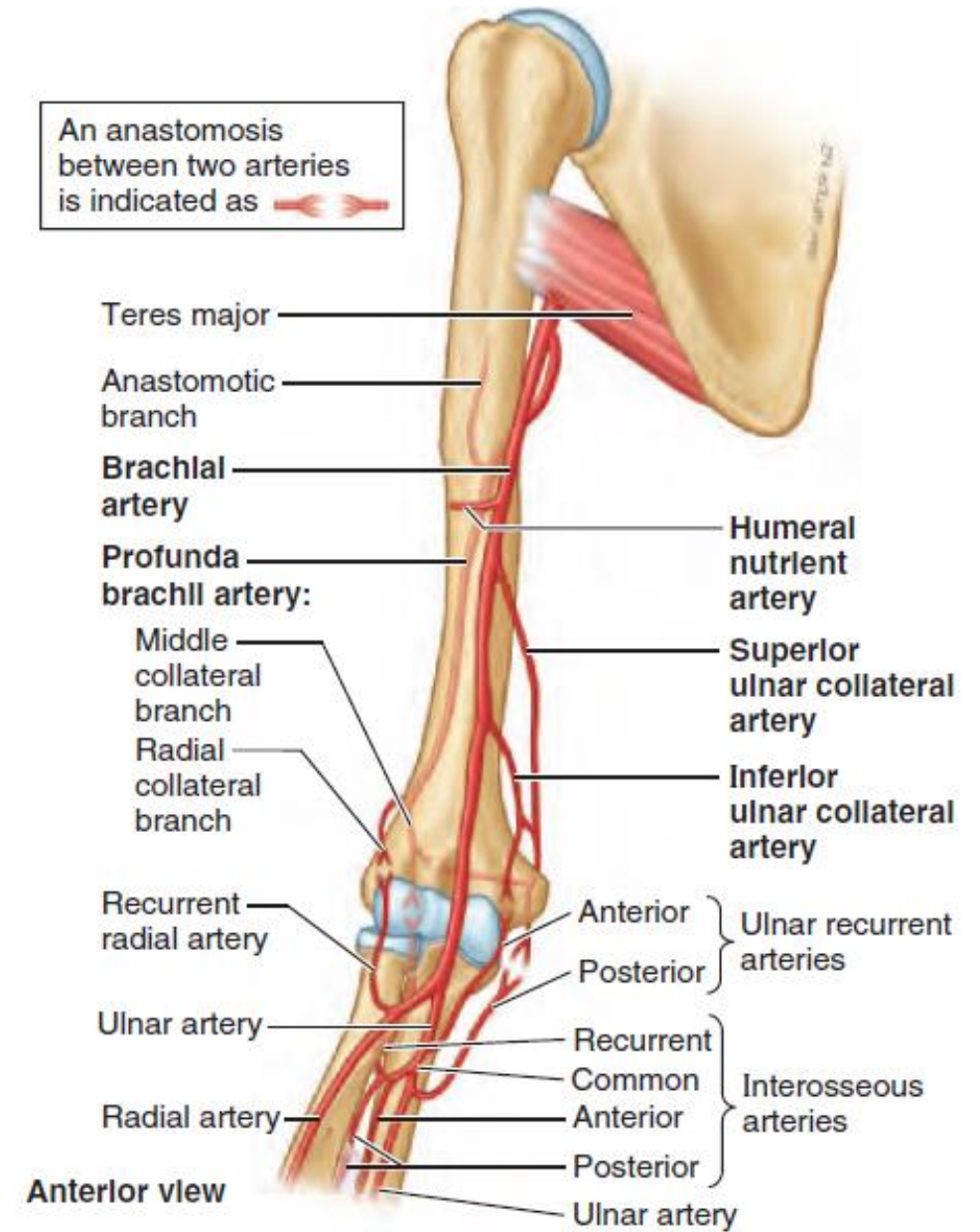
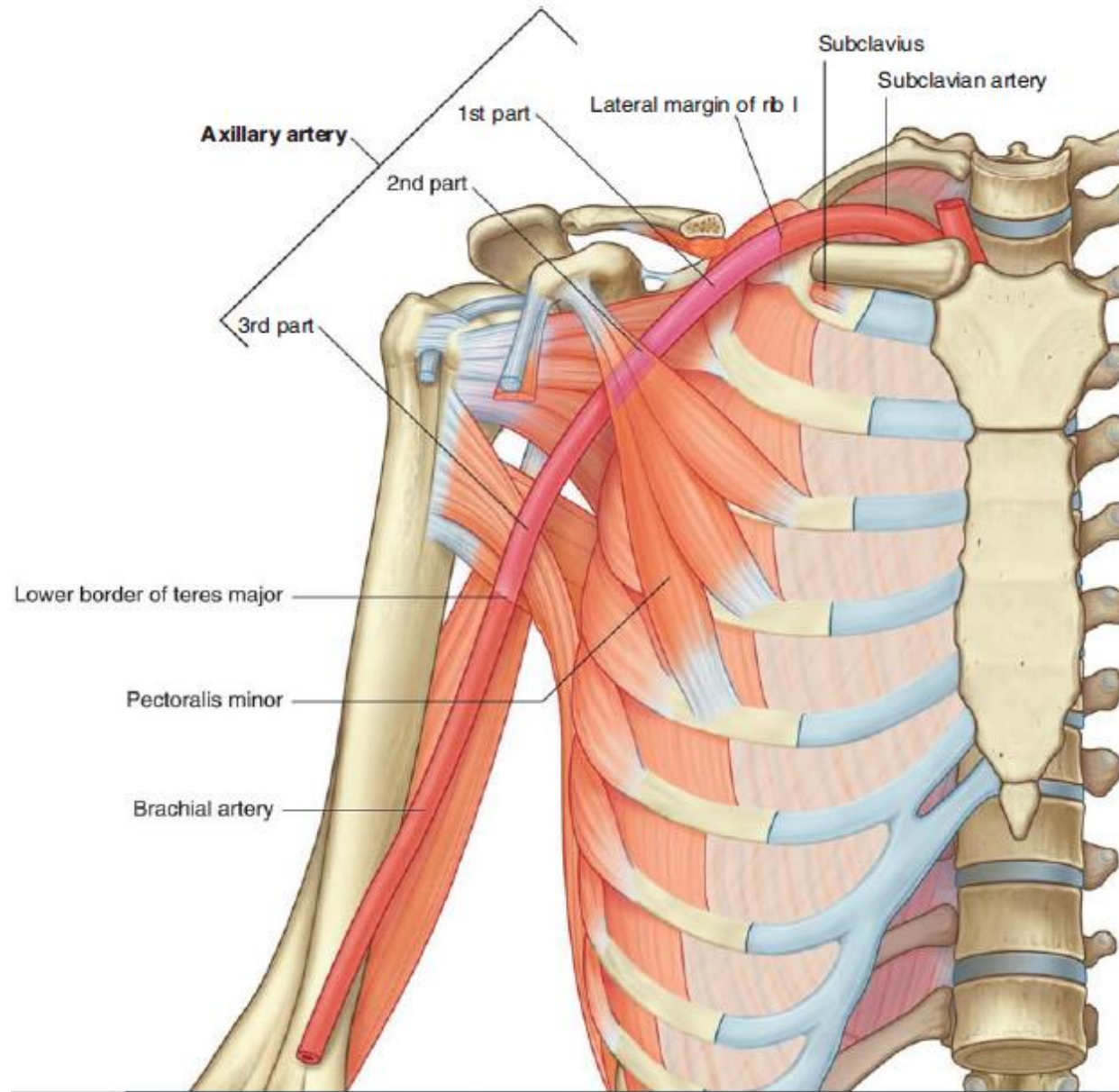
- axillary vein and its tributaries

- nerves of the cords and branches of the brachial plexus

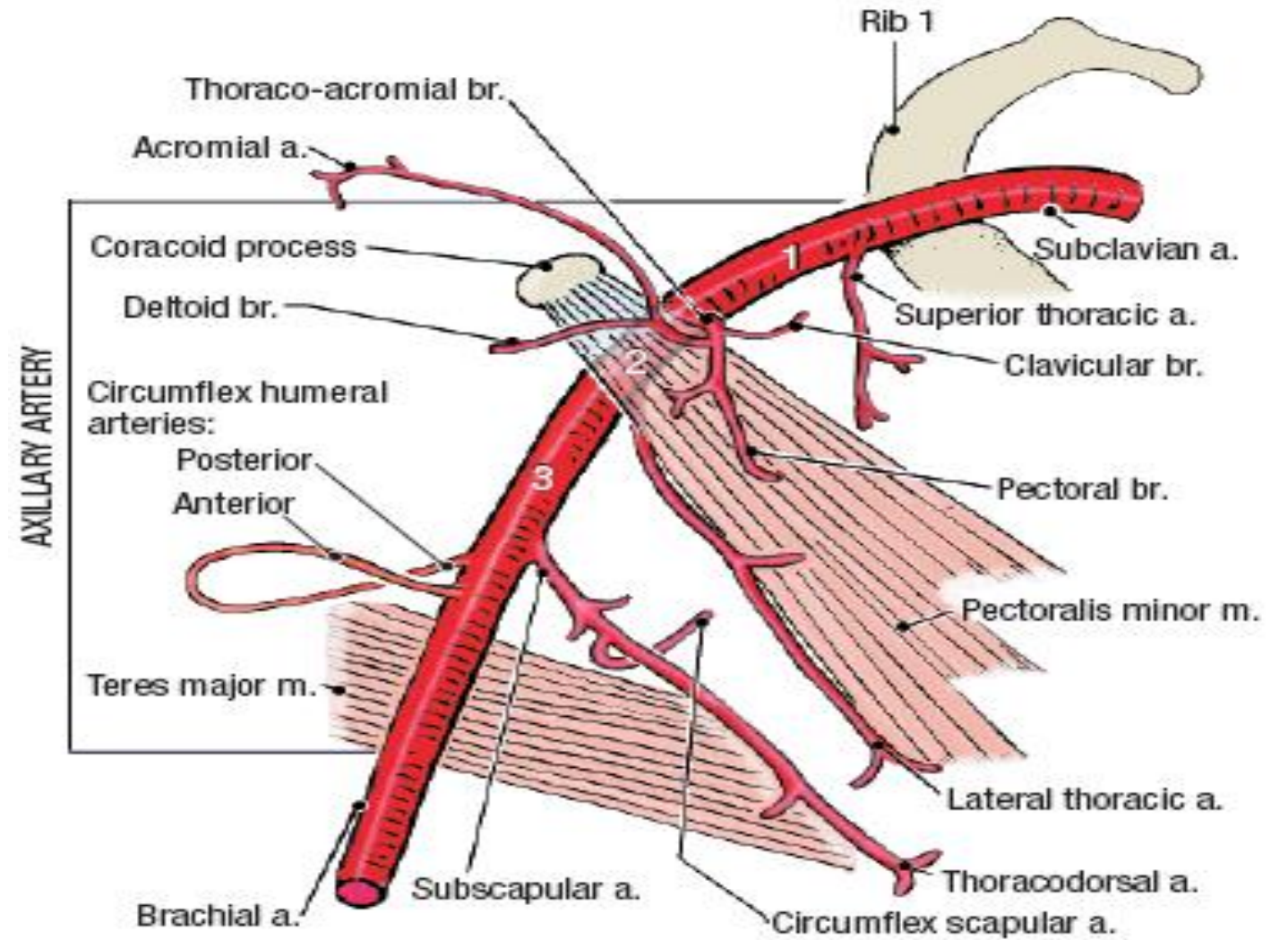
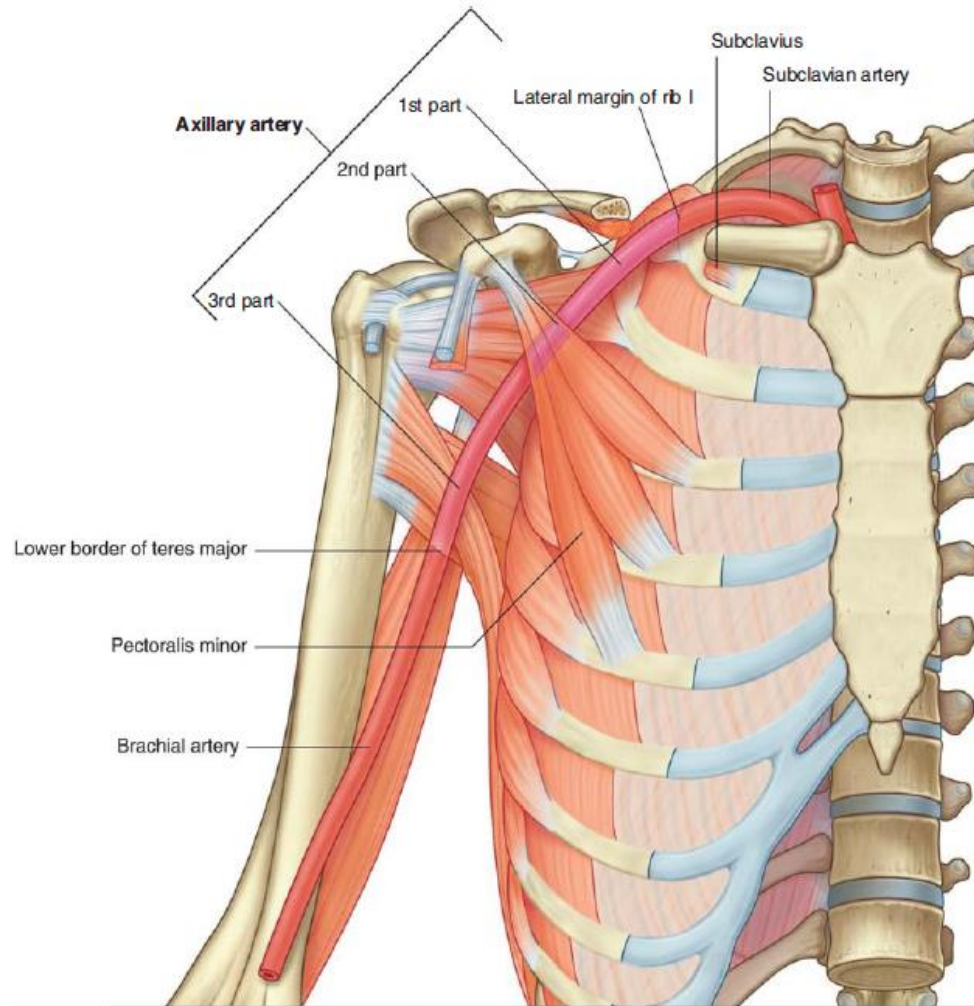
- lymphatic vessels

- groups of axillary lymph nodes

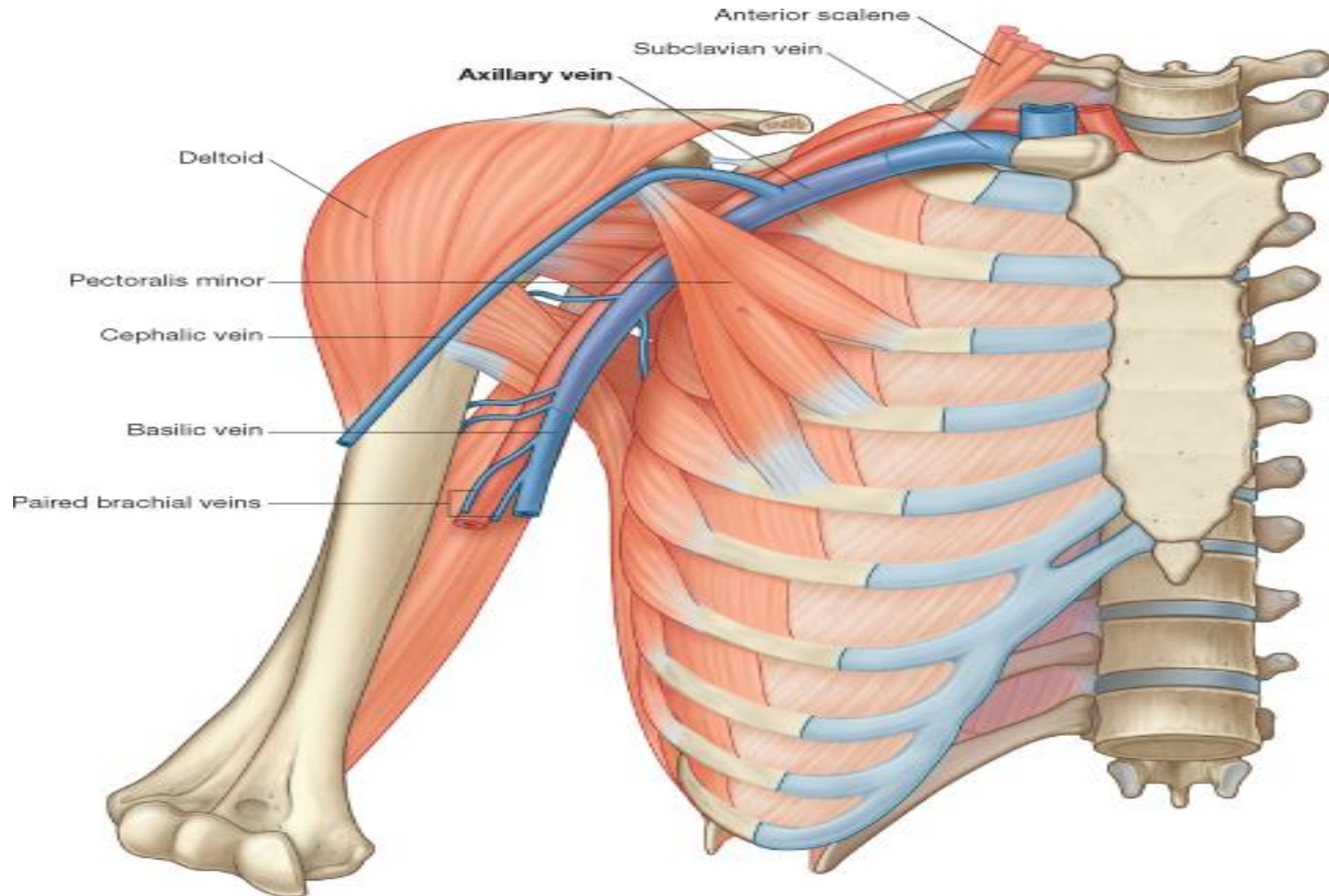
Contents of the Axilla,,,



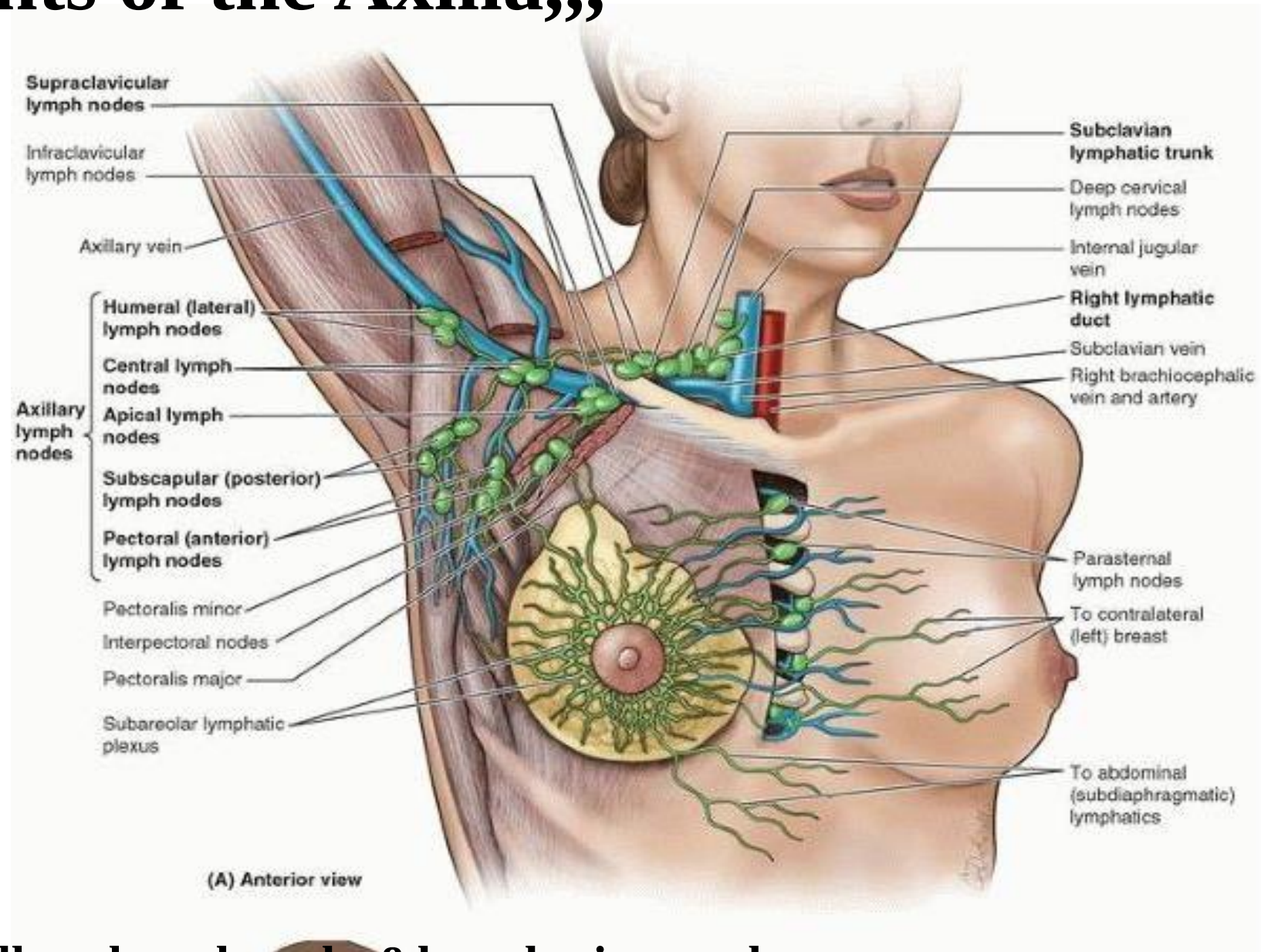
Contents of the Axilla,,



Contents of the Axilla,,,



Contents of the Axilla,,,



Axillary lymph nodes&lymphatic vessels

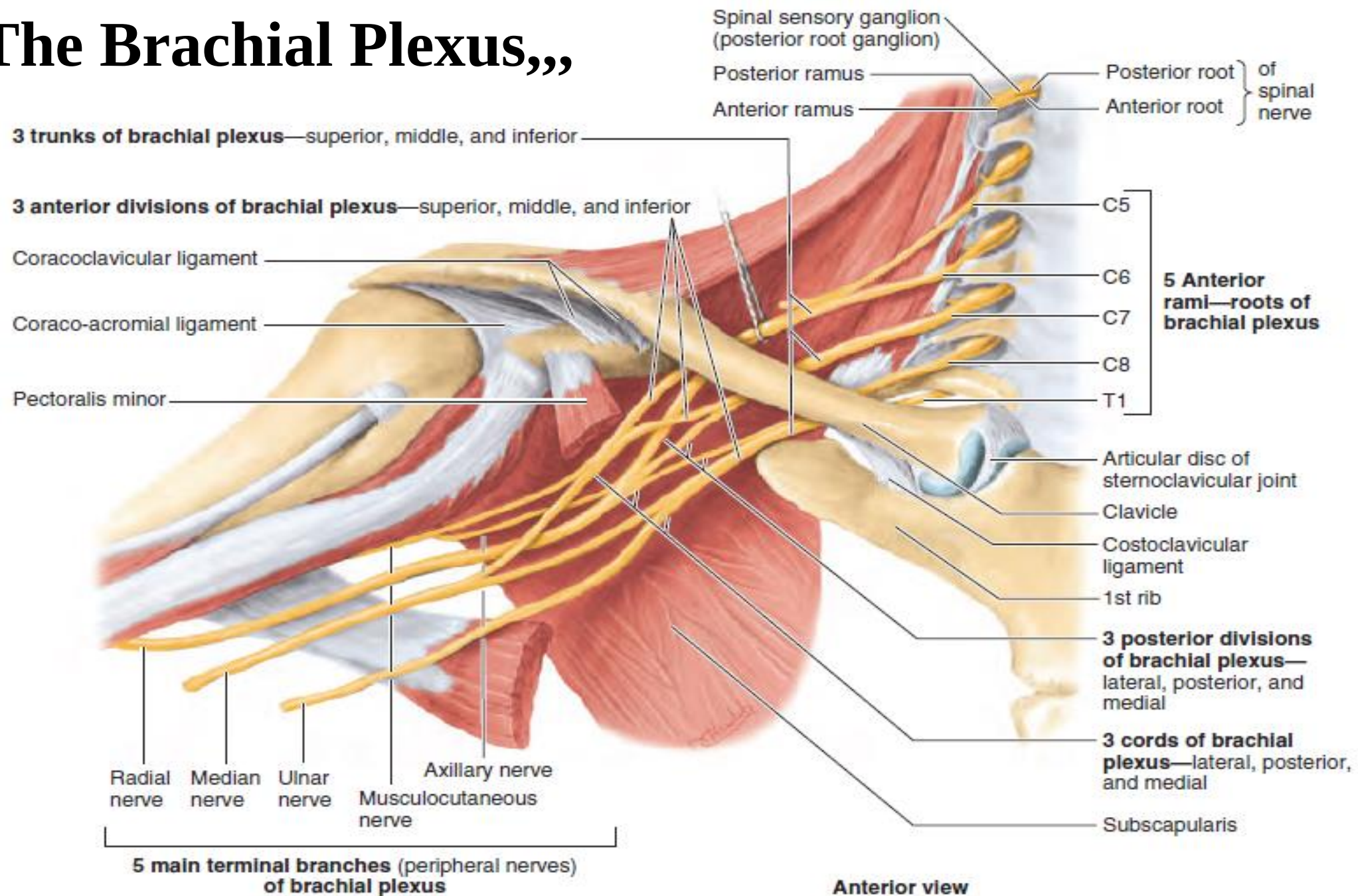
The Brachial Plexus

- ✓ **The brachial plexus** a major nerve network supplying the **upper limb**
- ✓ It begins in the neck and extends into the axilla
- ✓ Is formed by the **union of the anterior rami** of the last four cervical (**C5 through C8**) and the first thoracic (**T1**) nerves.

□ **The roots of the brachial plexus unite to form three trunks:**

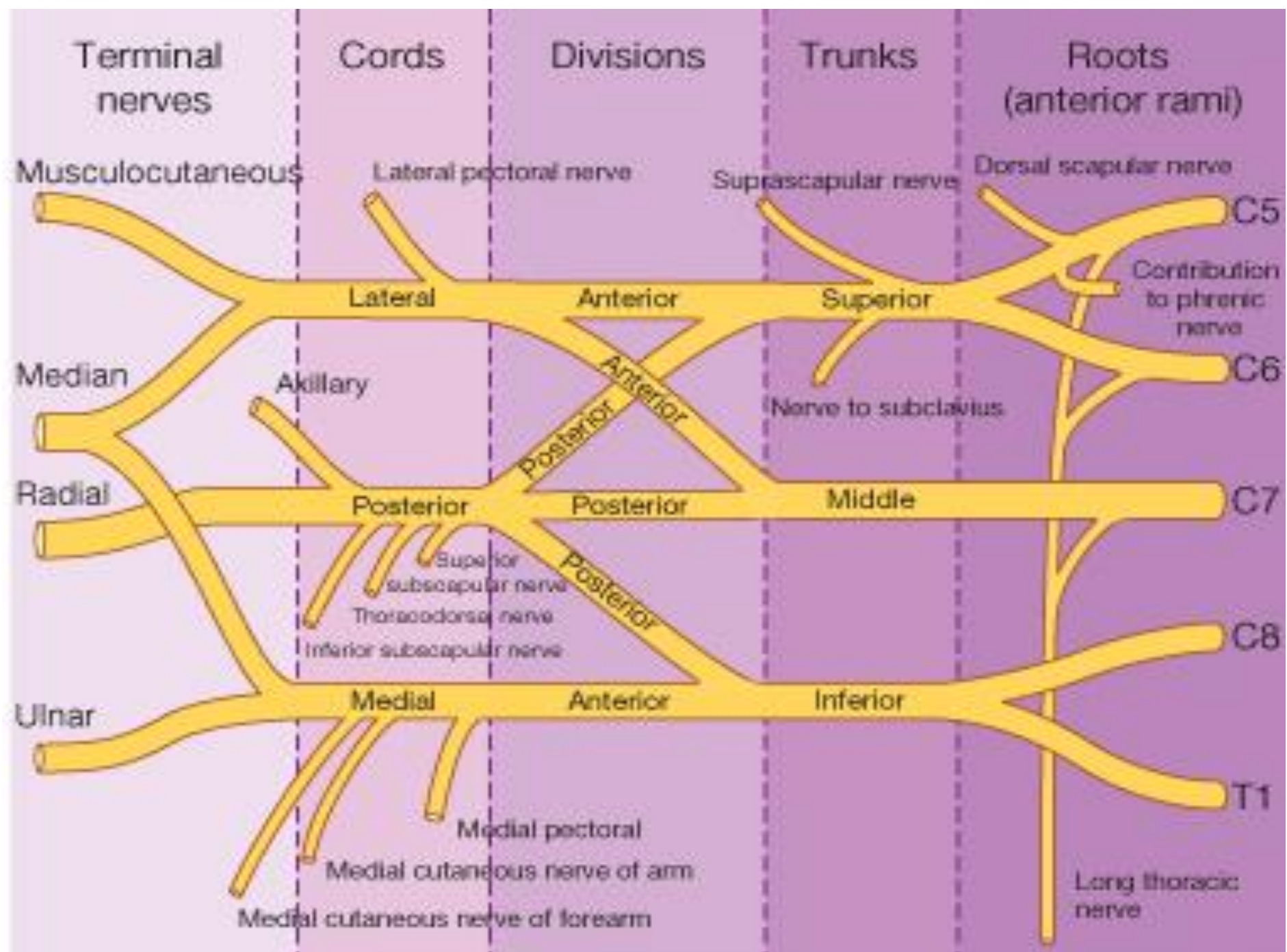
- **A superior trunk:** from the union of the **C5** and **C6** roots.
- **A middle trunk:** is a continuation of the **C7** root.
- **An inferior trunk:** from the union of the **C8** and **T1** roots.

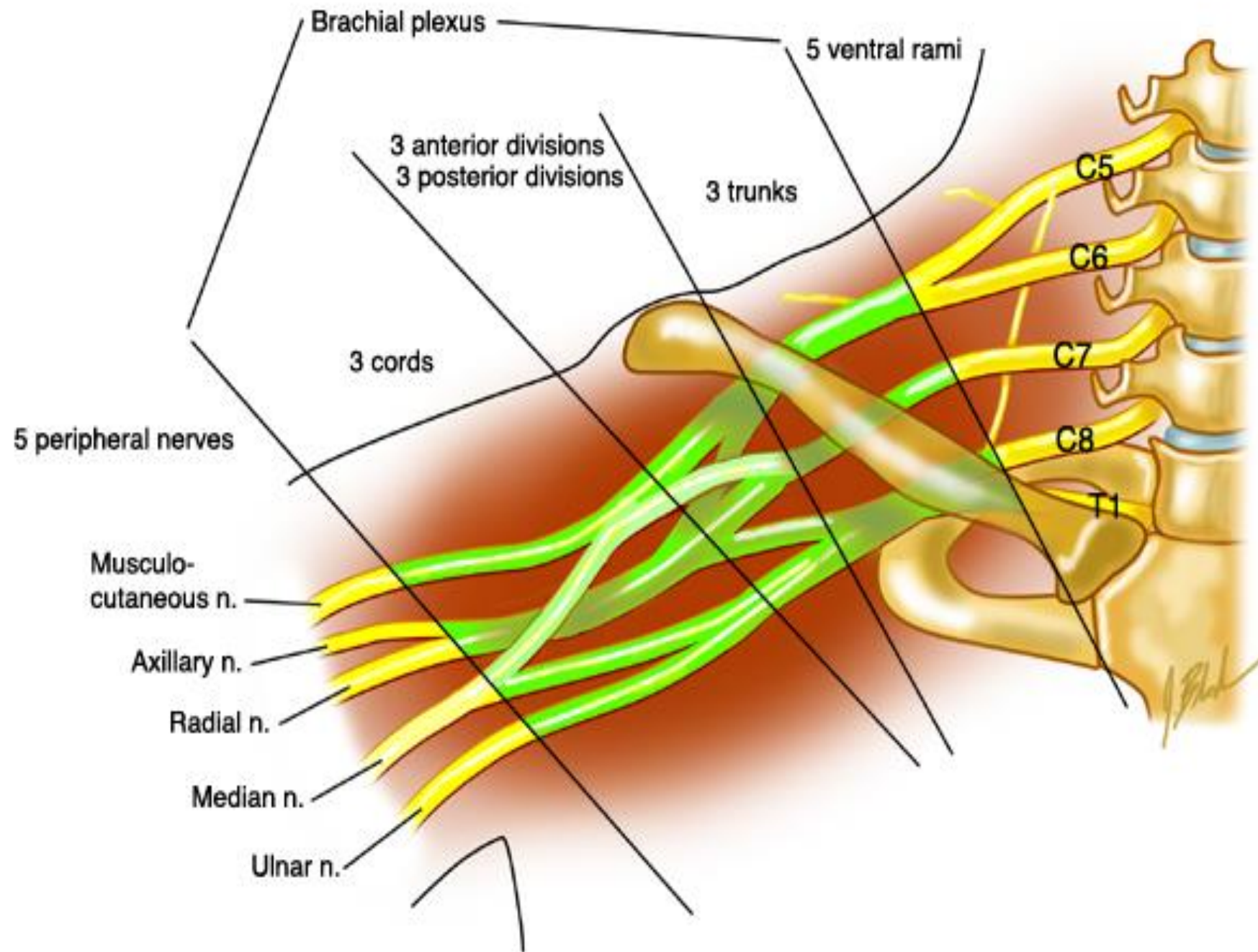
The Brachial Plexus,,,



The Brachial Plexus,,,

- ➡ Each trunk of the brachial plexus divides into **anterior** and **posterior divisions** as the plexus passes through the **cervicoaxillary canal** posterior to the clavicle.
- ➡ **Anterior divisions** of the trunks supply **anterior (flexor)** compartments of the upper limb
- ➡ **posterior divisions** of the trunks supply **posterior (extensor)** compartments





Branches from Roots

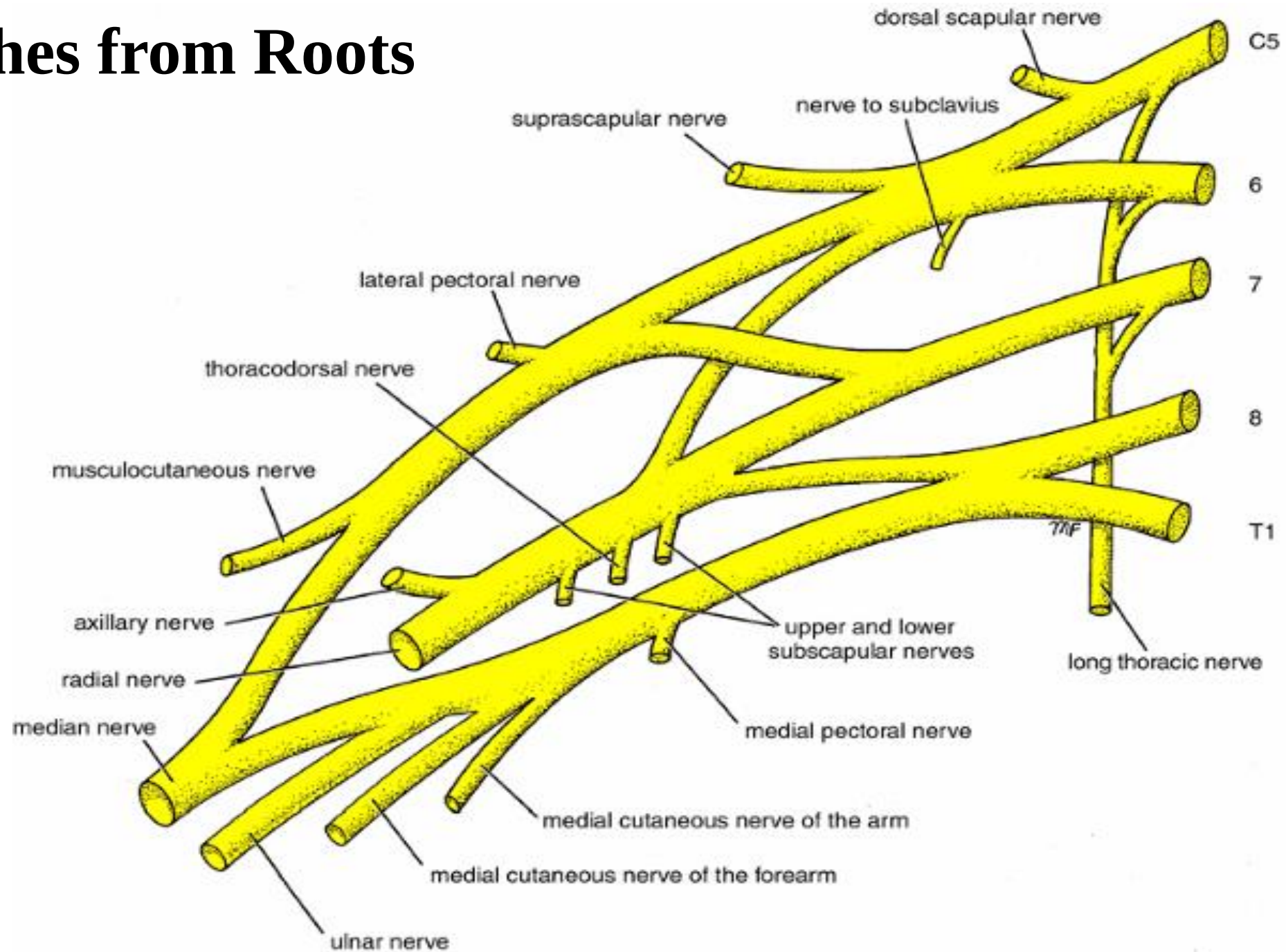
❑ Dorsal scapular nerve (C5)

- Pierces the scalenus medius muscle to reach the posterior cervical triangle
- **supply Rhomboids and levator scapulae.**

❑ Long thoracic nerve (C5, 6, and 7):

- Passes through cervico-axillary canal & behind the brachial plexus
- supply **serratus anterior muscle.**

Branches from Roots



Branches from the upper trunk

1) **Suprascapular nerve:-** Innervate supraspinatus and infraspinatus muscles; glenohumeral (shoulder) joint

- Passes through the scapular notch under the superior transverse scapular ligament, whereas the suprascapular artery passes over the ligament.

2) **Subclavian nerve;** Descends in front of the brachial plexus and the subclavian artery and behind the clavicle to reach the subclavius muscle.

- Innervates Subclavius , sternoclavicular joint

Branches from lateral cord

1. Lateral pectoral nerve: **Innervate pectoralis major**

- Pierces the costocoracoid membrane of the clavipectoral fascia.
- **Is accompanied by the pectoral branch of the thoracoacromial artery**

2. Musculocutaneous(C5-C7).

- Exits axilla by piercing coracobrachialis and descends between **biceps brachii and brachialis**
- Supplying **coracobrachialis, biceps brachii and brachialis both;**
- Continues as lateral cutaneous nerve of forearm.

Branches from medial cord

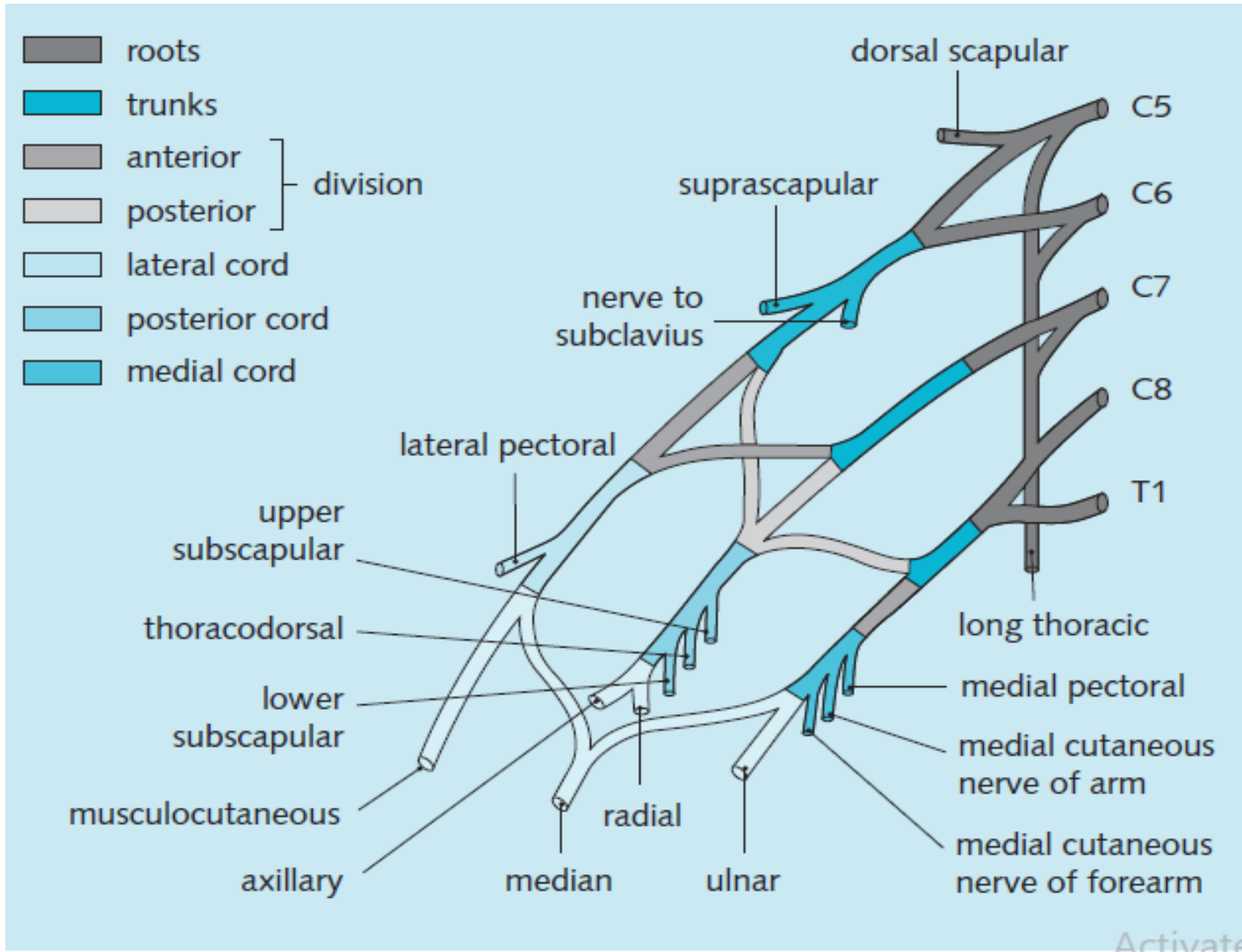
1. **Medial pectoral nerve:**

- Innervate **pectoralis minor** and **sternocostal part of pectoralis major**.
- Passes forward between the axillary artery and vein.

2. **Medial cutaneous nerve of arm:** Skin of medial side of arm, as far distal as medial epicondyle of humerus and olecranon of ulna.

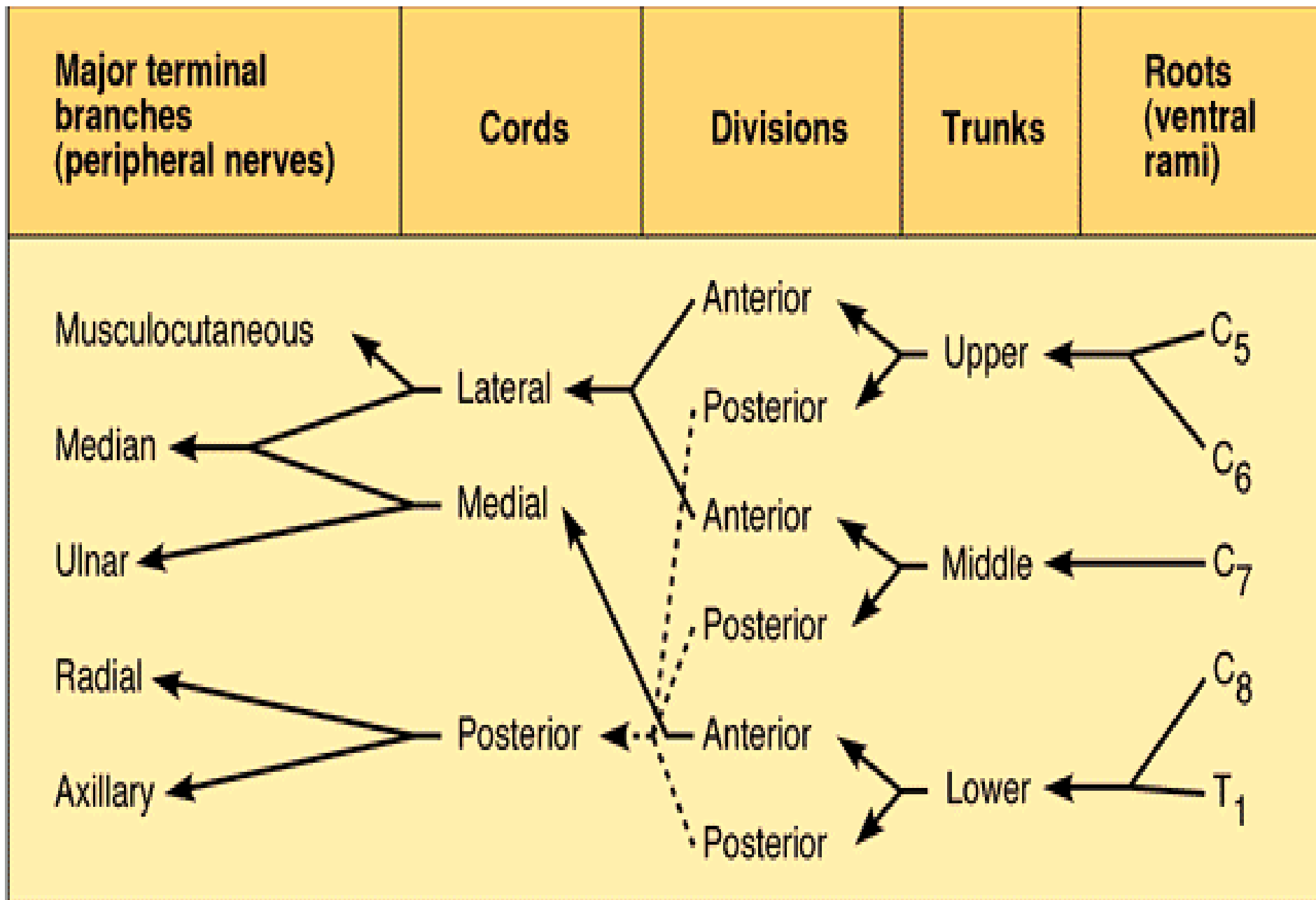
3. **Median cutaneous nerve of forearm:** Skin of medial side of forearm, as far distal as wrist.

Branches from medial cord,,



Branches from medial cord,,

- 4. Ulnar nerve (C8, T1 and C7):** Descends medial arm; passes posterior to medial epicondyle of humerus; then descends ulnar aspect of forearm to hand.
- Has palmar cutaneous branch, dorsal branch, superficial and deep branches.
 - Innervate flexor carpi ulnaris m., flexor digitorum profundus m. (ulnar half), abductor digiti minimi m., flexor digiti minimi brevis m., opponens digiti minimi m., ulnar 2 lumbrical mm., palmar and dorsal interosseous mm.
 - skin of the medial side of the wrist and hand; skin of the medial 1 1/2 digits



Branches from both lateral & medial cord

❖ **Median nerve:-** Has anterior interosseous n., palmar branch., recurrent (motor) branch, common palmar digital nn. (for digits 1-3)

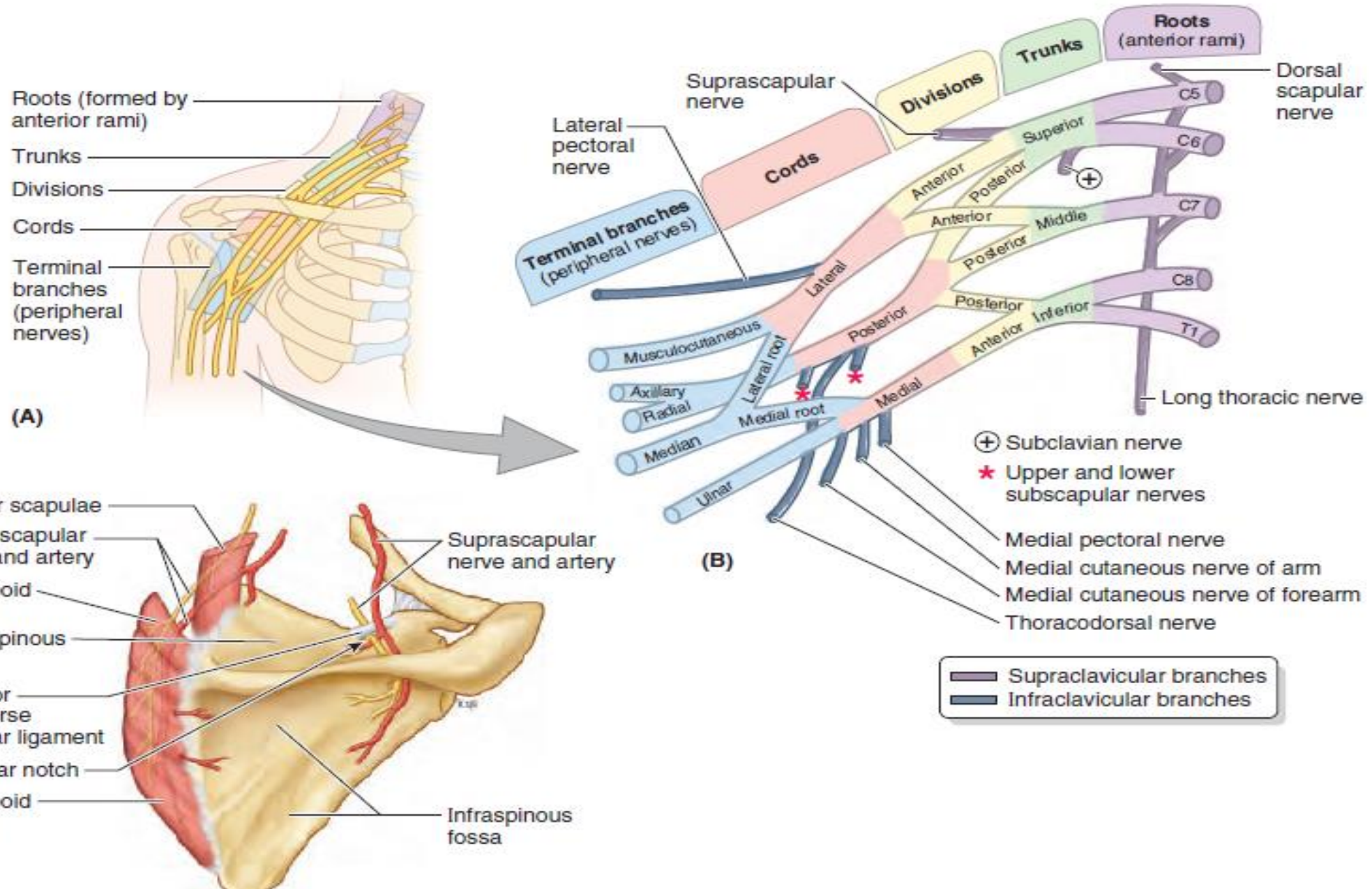
• **Innervates:-**

- ✓ pronator teres m., flexor carpi radialis m., palmaris longus m.,
- ✓ flexor digitorum superficialis m., flexor digitorum profundus m. (lateral half),
- ✓ flexor pollicis longus m., pronator quadratus m., abductor pollicis brevis m.,
- ✓ flexor pollicis brevis m., opponens pollicis m., lateral **2 lumbrical mm.**
- ✓ skin of the radial half of the palm and palmar side of the lateral 3 1/2 digits (and nail bed for these digits)

Branches from posterior cord

- 1. Upper subscapular nerve:** Superior portion of subscapularis
- 2. Lower subscapular nerve:** Inferior portion of subscapularis and teres major
- 3. Thoracodorsal nerve:** Latissimus dorsi
- 4. Axillary nerve(C5,C6) Innervation:** Glenohumeral joint; teres minor and deltoid muscles; skin of superolateral arm

Branches from posterior cord,,



Branches from posterior cord,,

5. Radial nerve(C5-T1):Exits axillary fossa posterior to axillary artery;

- Passes posterior to humerus **in radial groove with deep artery of arm, between lateral** and medial heads of triceps;
- *Perforates lateral intermuscular septum; enters cubital fossa, dividing into superficial and deep branches.*
- **Innervation:** All muscles of **posterior compartments of arm and forearm;** skin of posterior and inferolateral arm, posterior forearm, and dorsum of hand lateral to axial line of digit 4

Branches of the brachial plexus and their distribution

Branches of the brachial plexus and their distribution.

Branches	Distribution
<i>Roots</i>	
Dorsal scapular nerve (C5)	Rhomboid major, rhomboid minor, and levator scapulae muscles
Long thoracic nerve (C5–C7)	Serratus anterior muscle
<i>Upper trunk</i>	
Suprascapular nerve (C5, C6)	Supraspinatus and infraspinatus muscles
Nerve to subclavius (C5, C6)	Subclavius muscle
<i>Lateral cord</i>	
Lateral pectoral nerve (C5–C7)	Pectoralis major muscle
Musculocutaneous nerve (C5–C7)	Coracobrachialis, biceps brachii, brachialis muscles, and the skin along the lateral border of the forearm (lateral cutaneous nerve of the forearm)
Lateral root of median nerve (C5–C7)	Joins the medial root (C8, T1) to form the median nerve (see below)

Branches of the brachial plexus and their distribution,,

<i>Posterior cord</i>	
Upper subscapular nerve (C5–C6)	Subscapularis muscle
Thoracodorsal nerve (C6–C8)	Latissimus dorsi muscle
Lower subscapular nerve (C5–C6)	Subscapularis and teres major muscles
Axillary nerve (C5–C6)	Deltoid and teres minor muscles. Skin over the lower half of the deltoid muscle (upper lateral cutaneous nerve of arm)
Radial nerve (C5–C8, T1)	Triceps, brachialis, anconeus, and posterior muscles of forearm. Skin of the posterior aspects of arm, forearm, the lateral half of the dorsum of the hand, and dorsal surface of the lateral three-and-a-half digits.

Branches of the brachial plexus and their distribution,,

Medial cord

Medial pectoral nerve (C8, T1)	Pectoralis major and minor muscles
Medial cutaneous nerve of the arm (C8, T1)	Skin of the medial side of the arm
Medial cutaneous nerve of the forearm (C8, T1)	Skin of the medial side of the forearm
Ulnar nerve (C8, T1)	In the forearm, flexor carpi ulnaris and medial half of flexor digitorum profundus. In the hand, hypothenar muscles, adductor pollicis, third and fourth lumbricals, palmar and dorsal interossei, palmaris brevis muscles. Skin of the medial half of the dorsum and palm of hand, skin of the palmar, and dorsal surfaces of the medial one-and-a-half digits (palmar and dorsal digital cutaneous branches).
Median nerve (C5–C8, T1)	In the forearm, pronator teres, flexor carpi radialis, flexor digitorum superficialis (median nerve). Flexor pollicis longus, lateral half of flexor digitorum profundus and pronator quadratus (anterior interosseous branch). In the hand, thenar muscles, first two lumbricals (median nerve). Skin of lateral half of palm and palmar surface of lateral three-and-a-half digits (palmar and digital cutaneous branches).

Brachial plexus injury

- Injuries to superior parts of the brachial plexus (C5 and C6) usually result from an excessive increase in the angle between the neck and the shoulder.
- Lesion the upper trunk brachial plexus (C5,6) occurs commonly during difficulty labor.
- It leads to paralysis of the **deltoid, supraspinatus, infraspinatus, biceps, brachialis, brachioradialis and supinator.**

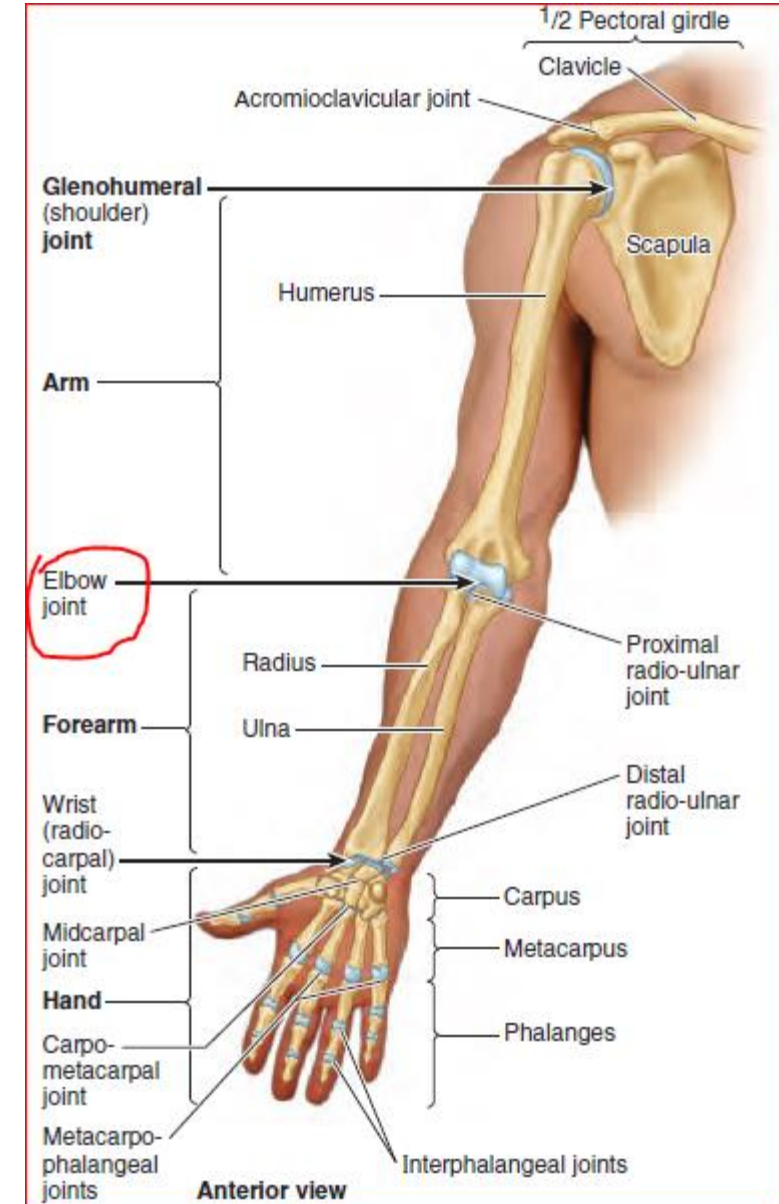
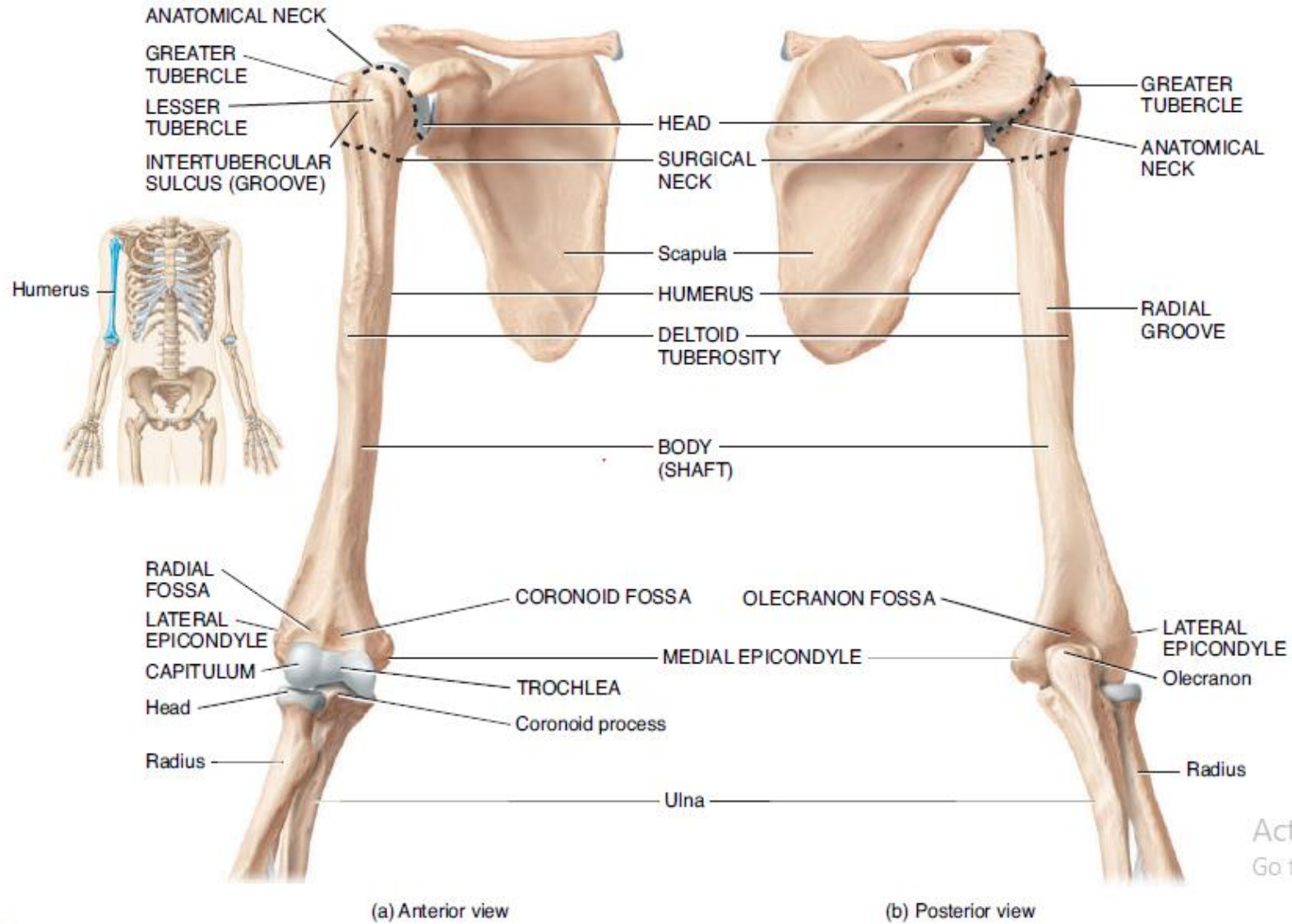
❑ **As a result:**

- the arm is extended
- The forearm is extended and pronated.
- The shoulder is medially rotated. Such deformity is called “ Police man’s (waiter's) tip position”

The brachium(arm)

- The arm lies between the shoulder and elbow joint.
- The upper part of the arm receives tendons from the shoulder and scapular region, whereas the lower part sends tendons to the forearm.
- This unique arrangement enables the arm to take part in flexion, extension, abduction, adduction, and circumduction at the glenohumeral joint.
- **Flexion and extension at the elbow joint.**
- The arm is supported by the **humerus** (arm bone) the largest & longest bone in the upper limb.

The brachium(arm),,,

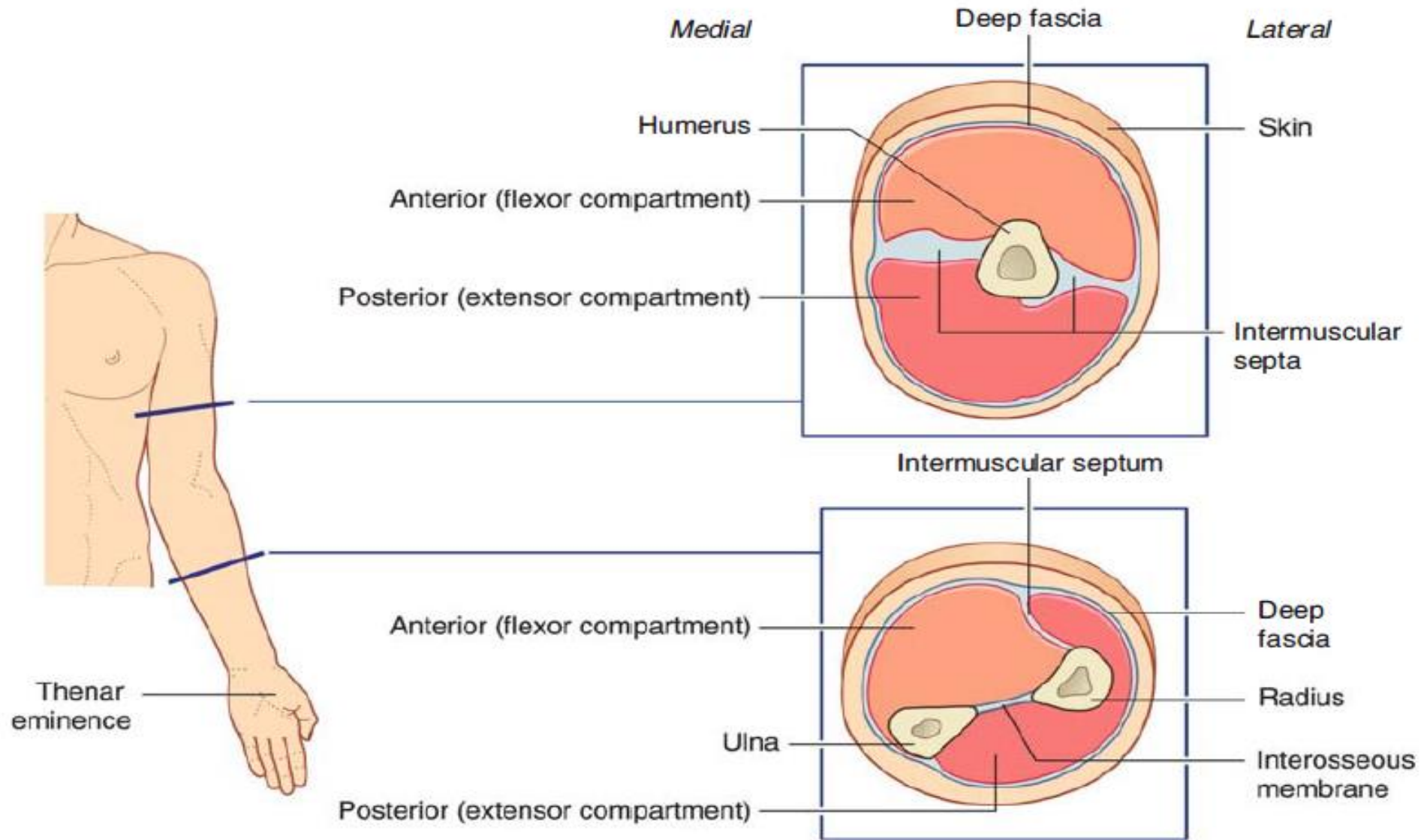


Skeletons of the arm!

The brachium(arm),,,

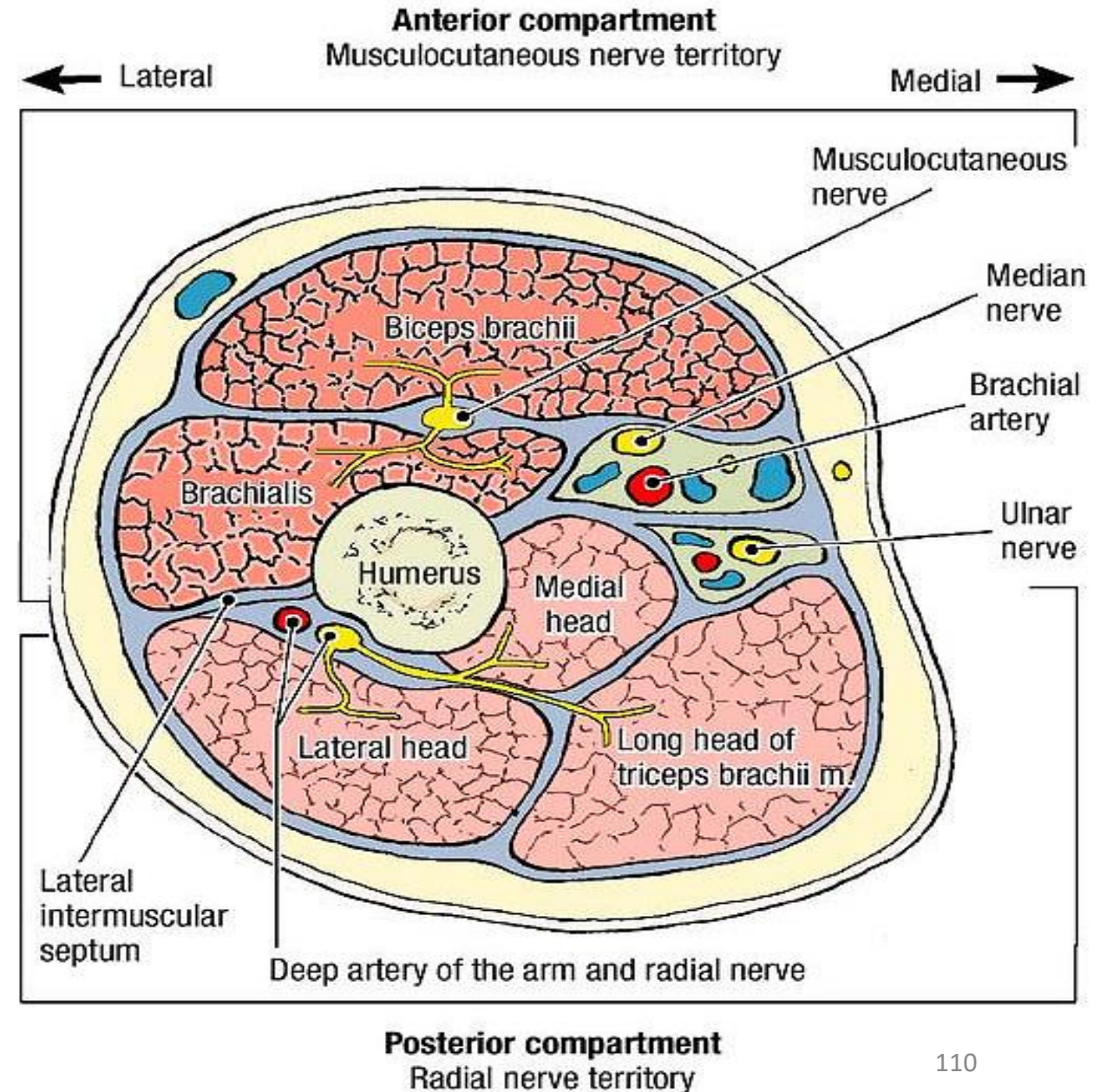
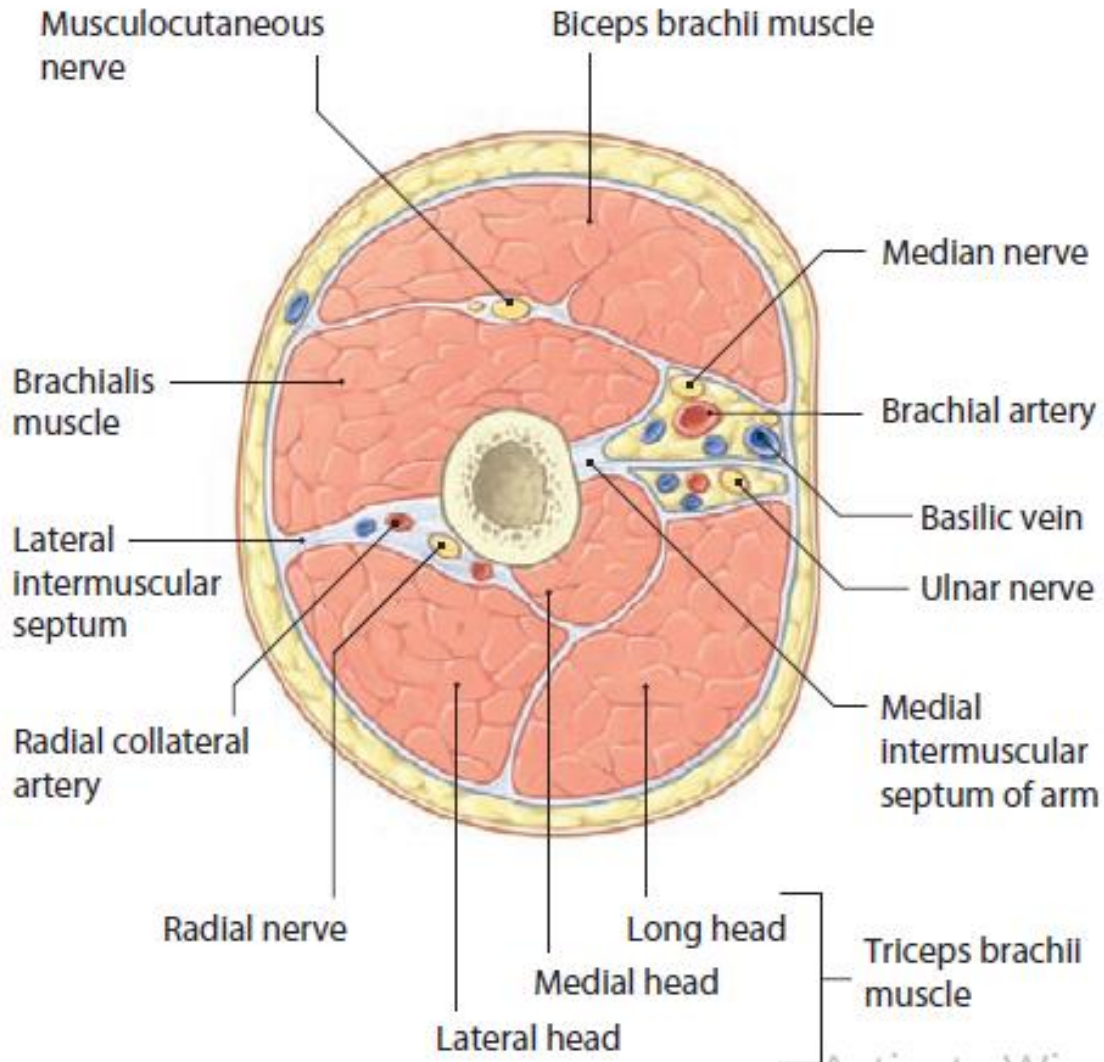
- Two types of movement occur between the arm and the forearm **at the elbow joint**:
 - Flexion and extension.
 - Pronation and supination.
- ❖ The muscles performing these movements are clearly divided **into anterior** and **posterior** groups by **intermuscular septa** and **humerus**:
 - The anterior compartment
 - Posterior compartment
- **Compartmentalization** of the arm is completed by the medial and lateral intermuscular septa of the deep fascia.

The brachium(arm),,,



Muscle components in the arm and forearm.

The brachium(arm),,,



Muscles of anterior compartment of the arm

- ❖ The **flexor muscles** of the anterior compartment are almost **twice** as **strong as the extensors** in all positions as a result we are better **pullers than pushers**

❑ Three muscles of anterior compartment:

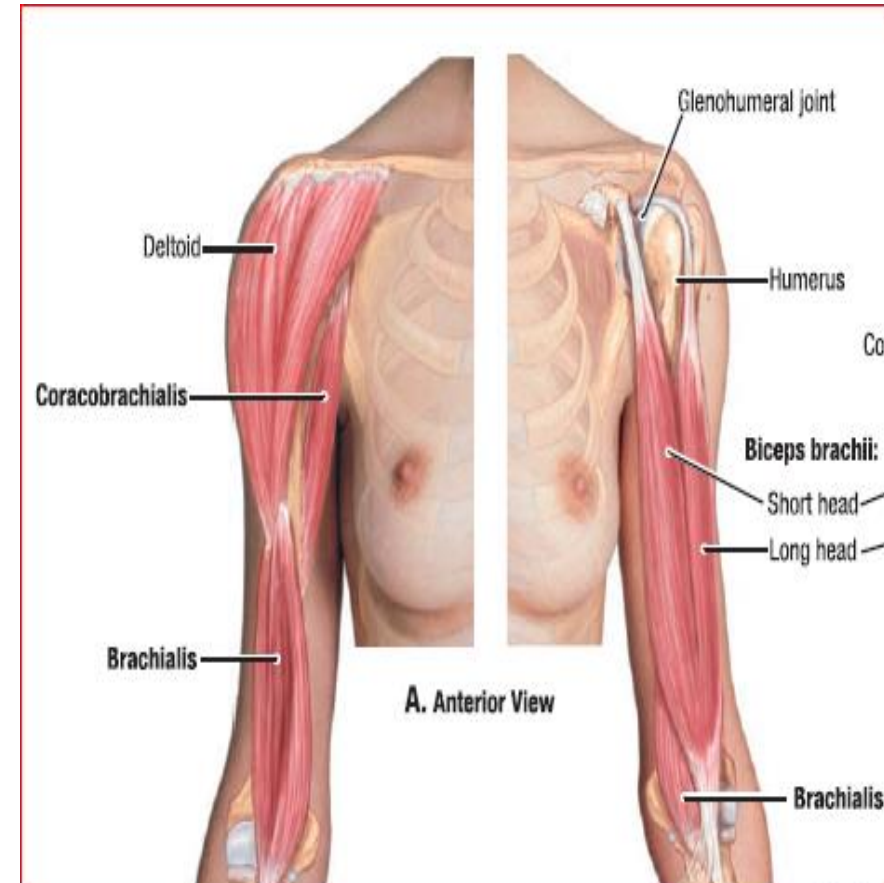
- Biceps brachii
- Brachialis
- Coracobrachialis

Innervation: Musculocutaneous nerve (C5, C6, C7)

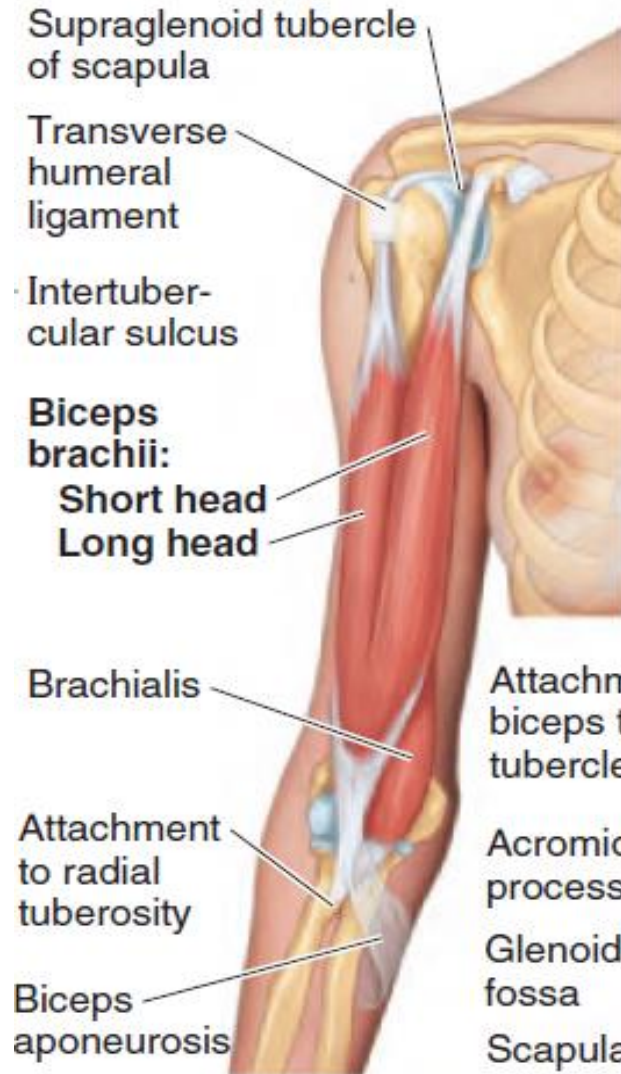
Biceps brachii

Origin: has two heads (bi, two + L. caput, head).

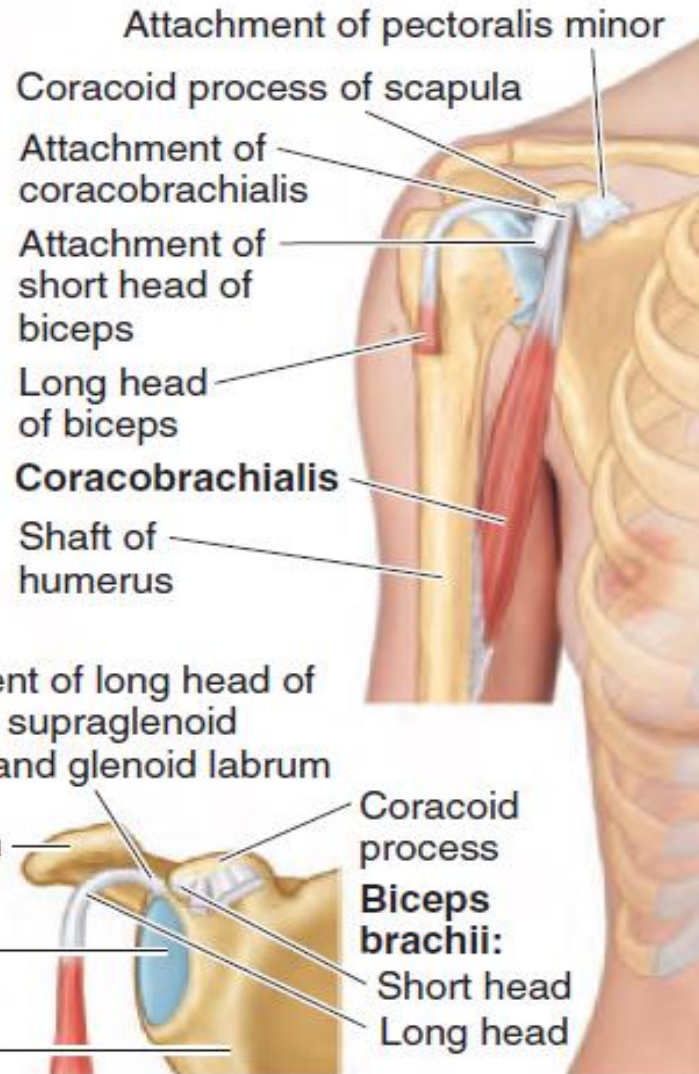
1. Long head (lateral head)
2. Short head (medial head)



Muscles of anterior compartment of the arm,,,

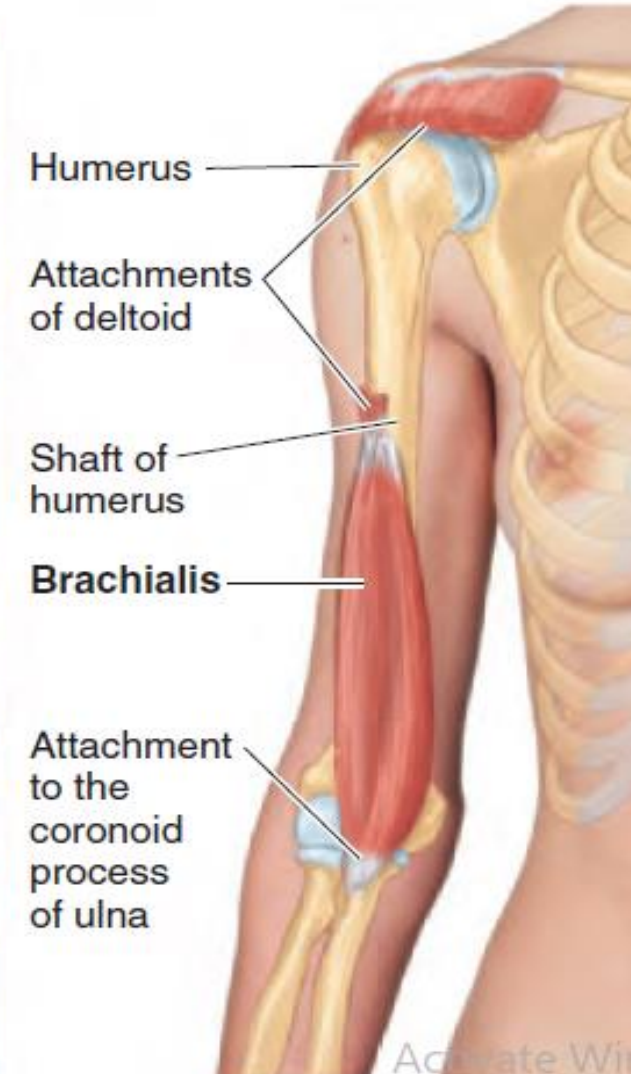


(B) Biceps brachii



Anterior views

(C) Coracobrachialis



(D) Brachialis

Muscles of anterior compartment of the arm,,

MUSCLES OF ARM

Muscle	Proximal Attachment	Distal Attachment	Innervation ^a	Muscle Action
Biceps brachii	<i>Short head:</i> tip of coracoid process of scapula <i>Long head:</i> supraglenoid tubercle of scapula	Tuberosity of radius and fascia of forearm via bicipital aponeurosis	Musculocutaneous nerve (C5, C6, C7)	Supinates forearm and, when it is supine, flexes forearm; short head resists dislocation of shoulder
Coracobrachialis	Tip of coracoid process of scapula	Middle third of medial surface of humerus		Helps flex and adduct arm; resists dislocation of shoulder
Brachialis	Distal half of anterior surface of humerus	Coronoid process and tuberosity ulna	Musculocutaneous nerve ^b (C5, C6) and radial nerve (C5, C7)	Flexes forearm in all positions

Muscles of Posterior compartment of the arm

- One muscle of this compartment is the **Triceps brachii**:
- **It** is a large fusiform muscle in the **posterior compartment of the arm**

Origin: by three head

- **Long head**: from **infraglenoid tubercle** of scapula
- **Lateral head**: from posterior surface of humerus superior to radial groove
- Covers the nerve and profundus vessels in radial groove
- **Medial head**: a very wide origin from posterior surface of humerus inferior to radial groove corresponding to the origin of brachialis anteriorly

❑ **Anconeus** is a small triangular muscle on the posterior aspect of the elbow, covers the posterior aspect of the ulna proximally

Muscles of Posterior compartment of the arm,,

❑ **Insertion:** Posterior surface of the proximal end of olecranon of ulna and fascia of forearm

❑ **Innervation:** Radial nerve (C6, C7, C8)

❑ **Action:** Chief extensor of forearm, long head resists dislocation of humerus

long head resists dislocation of humerus

- Helps in adduction but it is actually the least active head

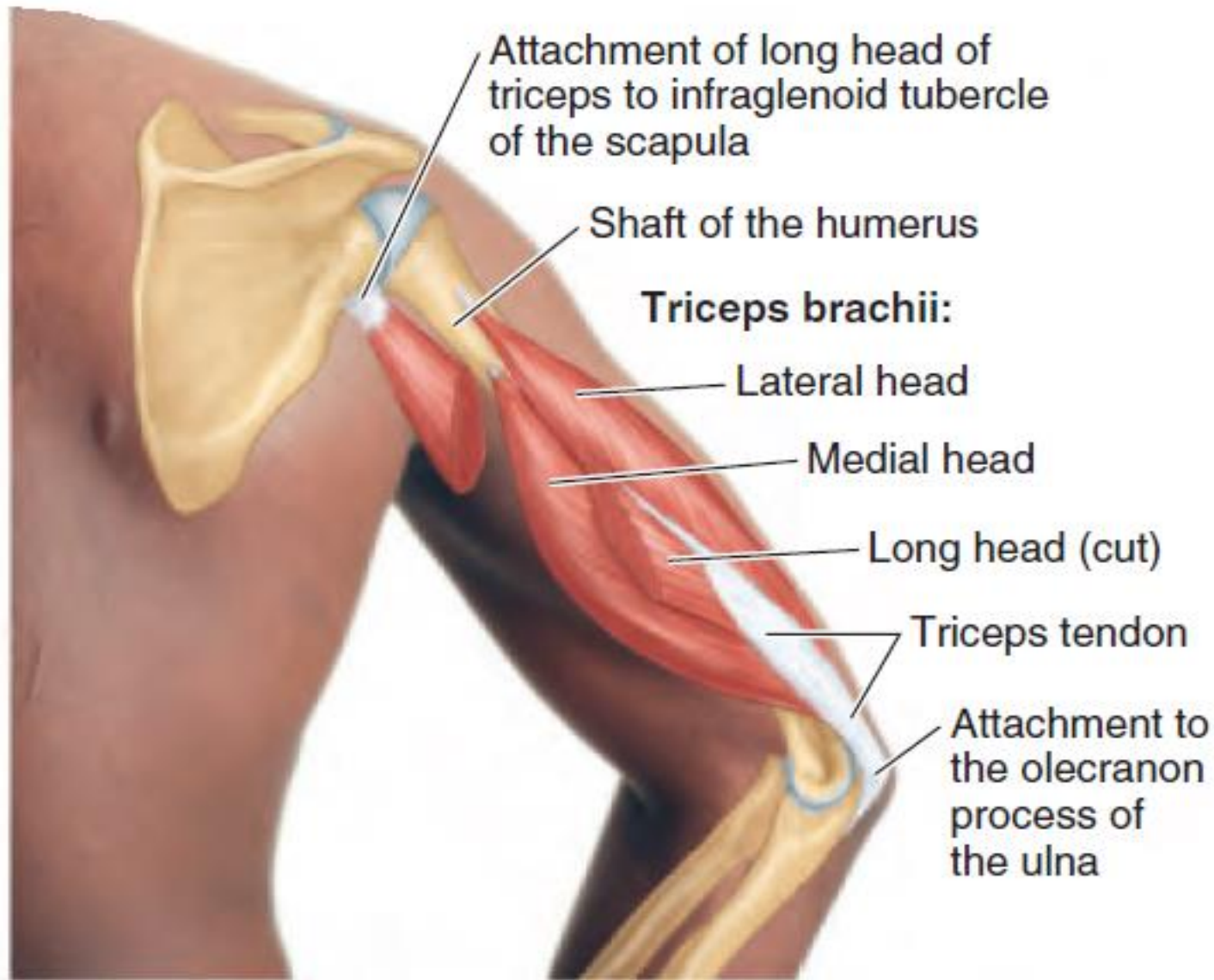
❑ **The medial head** is the workhorse of forearm extension, active at all speeds and in the presence or absence of resistance

❑ **The lateral head** is strongest but is recruited into activity primarily against resistance

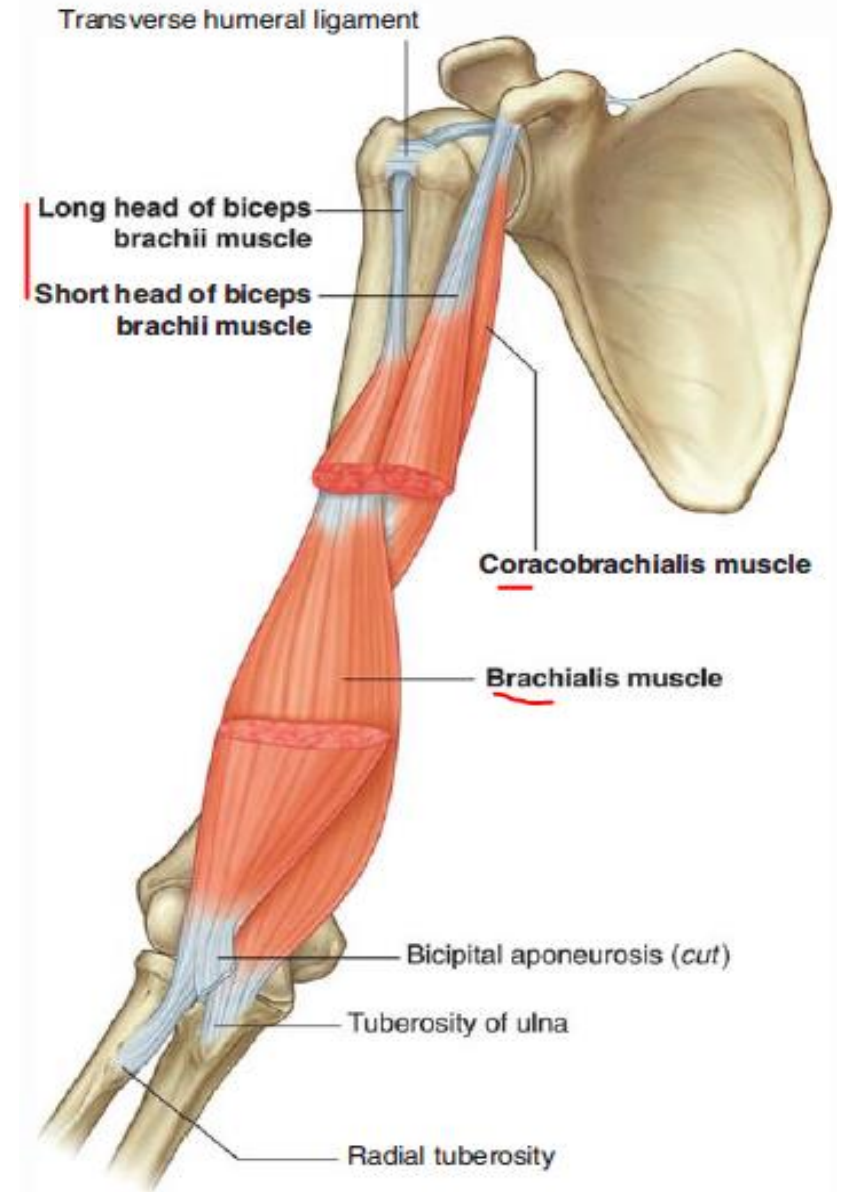
❖ **The anterior and posterior compartments** of the arm are innervated by the **musculocutaneous and radial nerves.**

❖ In addition, two major nerves from the brachial plexus (the median and ulnar nerves) pass through the arm to innervate structures in the forearm and hand.

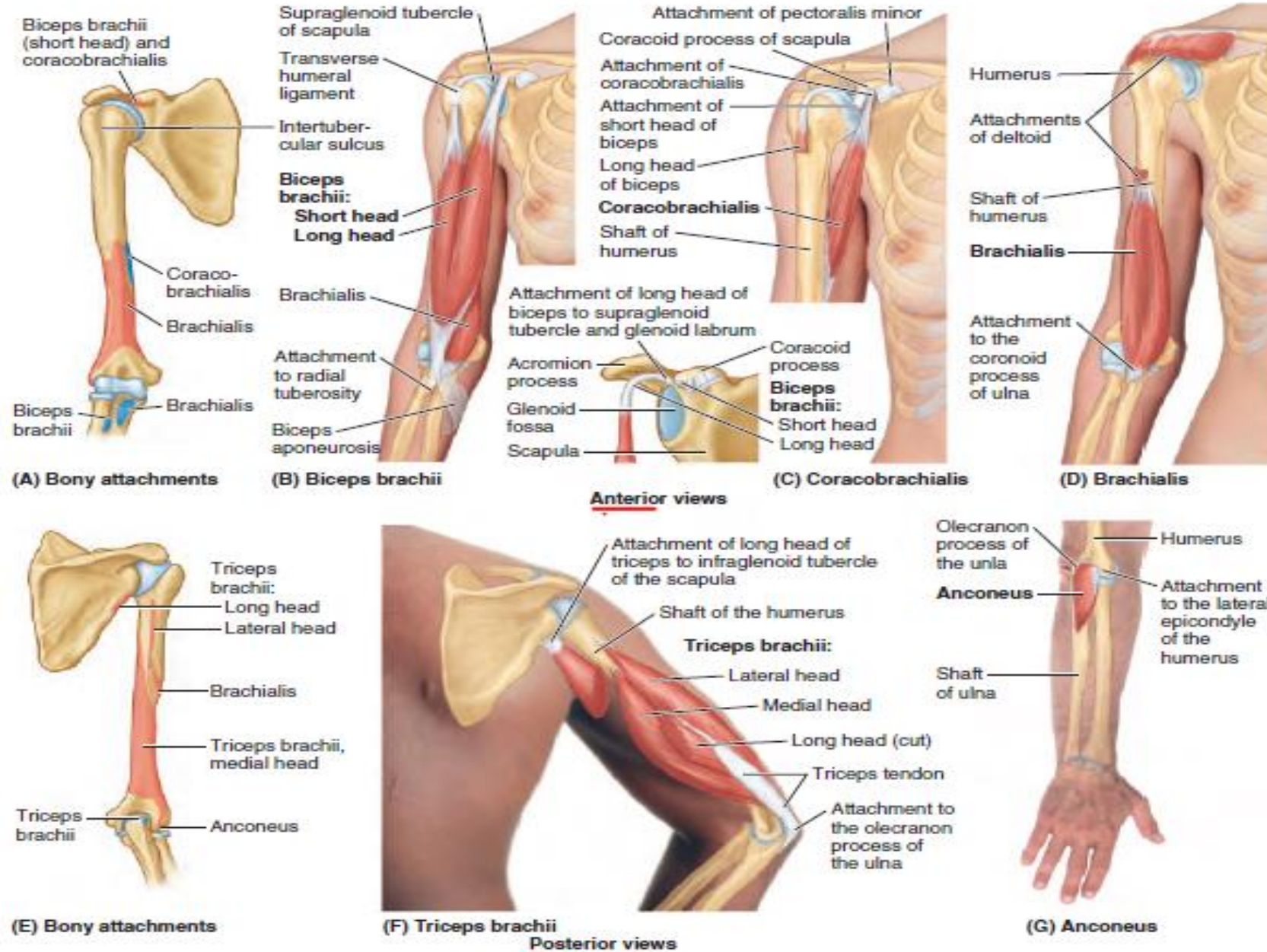
Muscles of Posterior compartment of the arm,,,



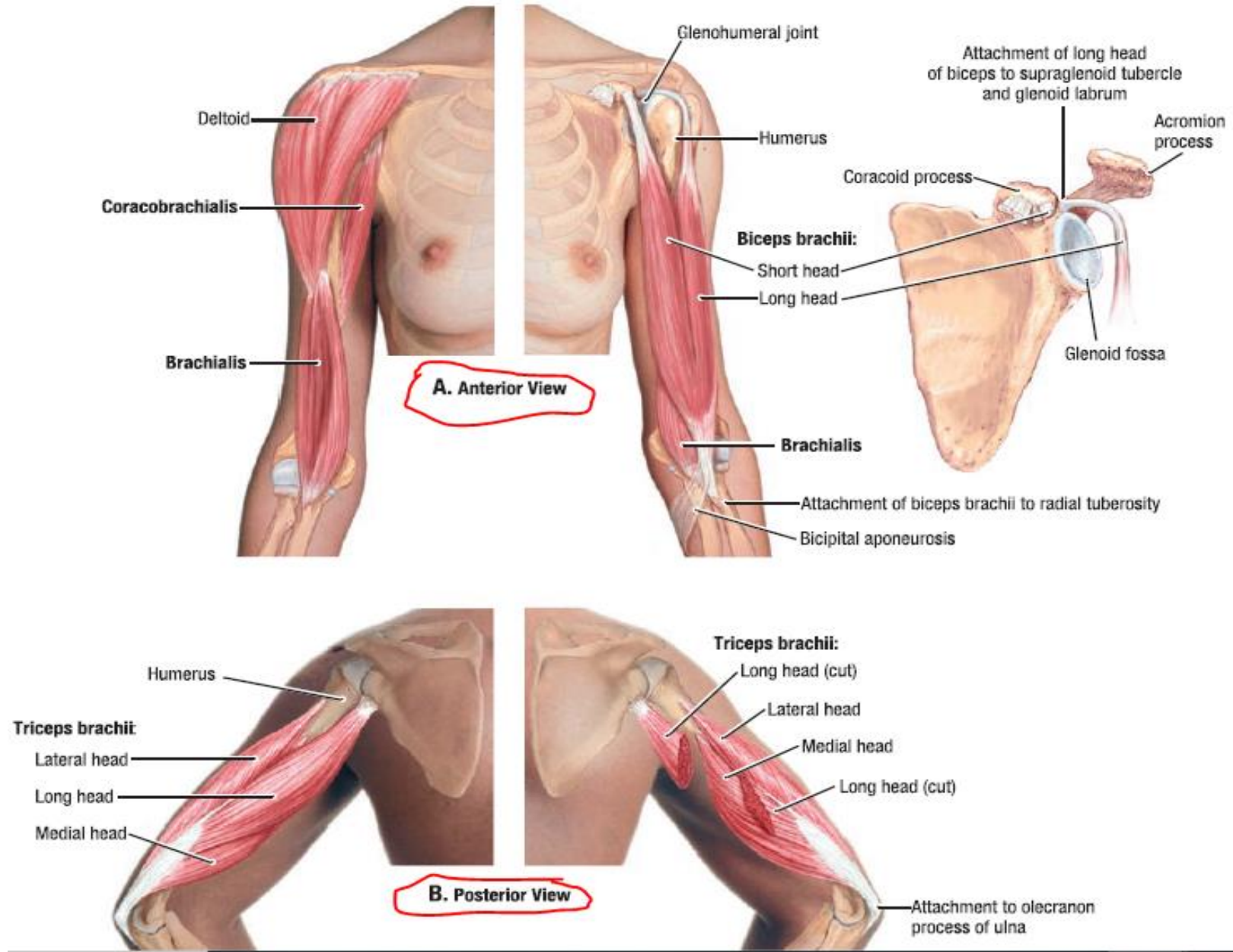
Triceps brachii
Posterior views



Muscles of Posterior compartment of the arm,,



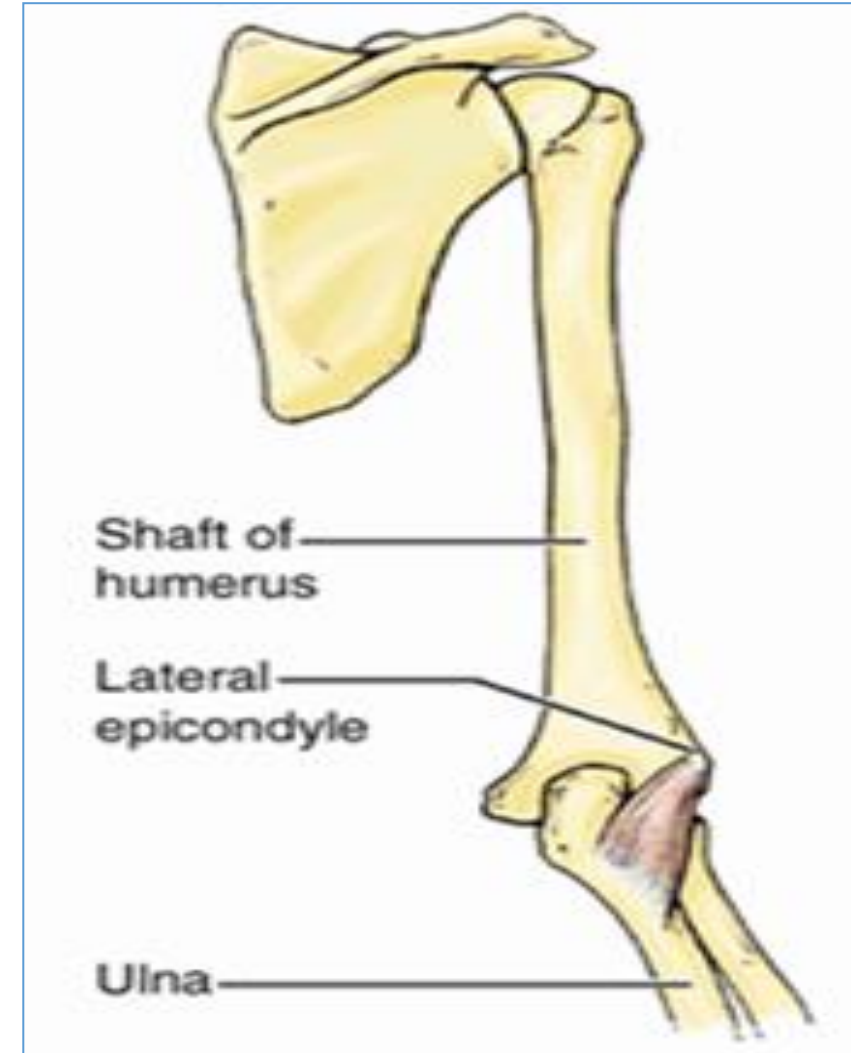
Muscles of Posterior compartment of the arm,,,



Muscles of Posterior compartment of the arm,,

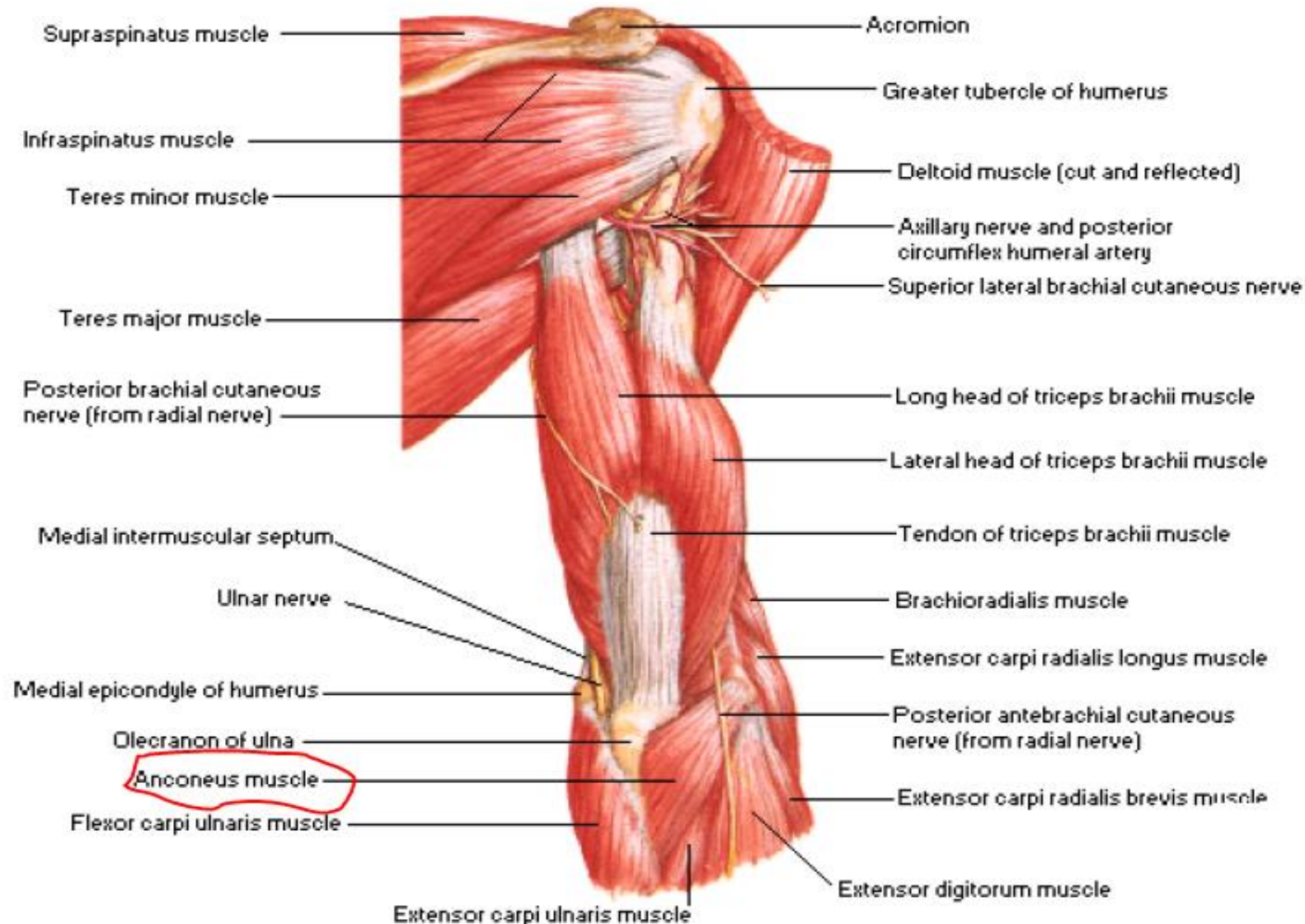
❖ **Anconeus:**

- **Origin:** Lateral epicondyle of humerus
- **Insertion:** Lateral surface of olecranon and superior part of posterior surface of ulna
- **Innervation:** Radial nerve (C7, C8, T1)
- **Action:** Assists triceps in extending forearm; stabilizes elbow joint; may abduct ulna during pronation.



Muscles of Posterior compartment of the arm,,

Posterior View - Superficial Layer

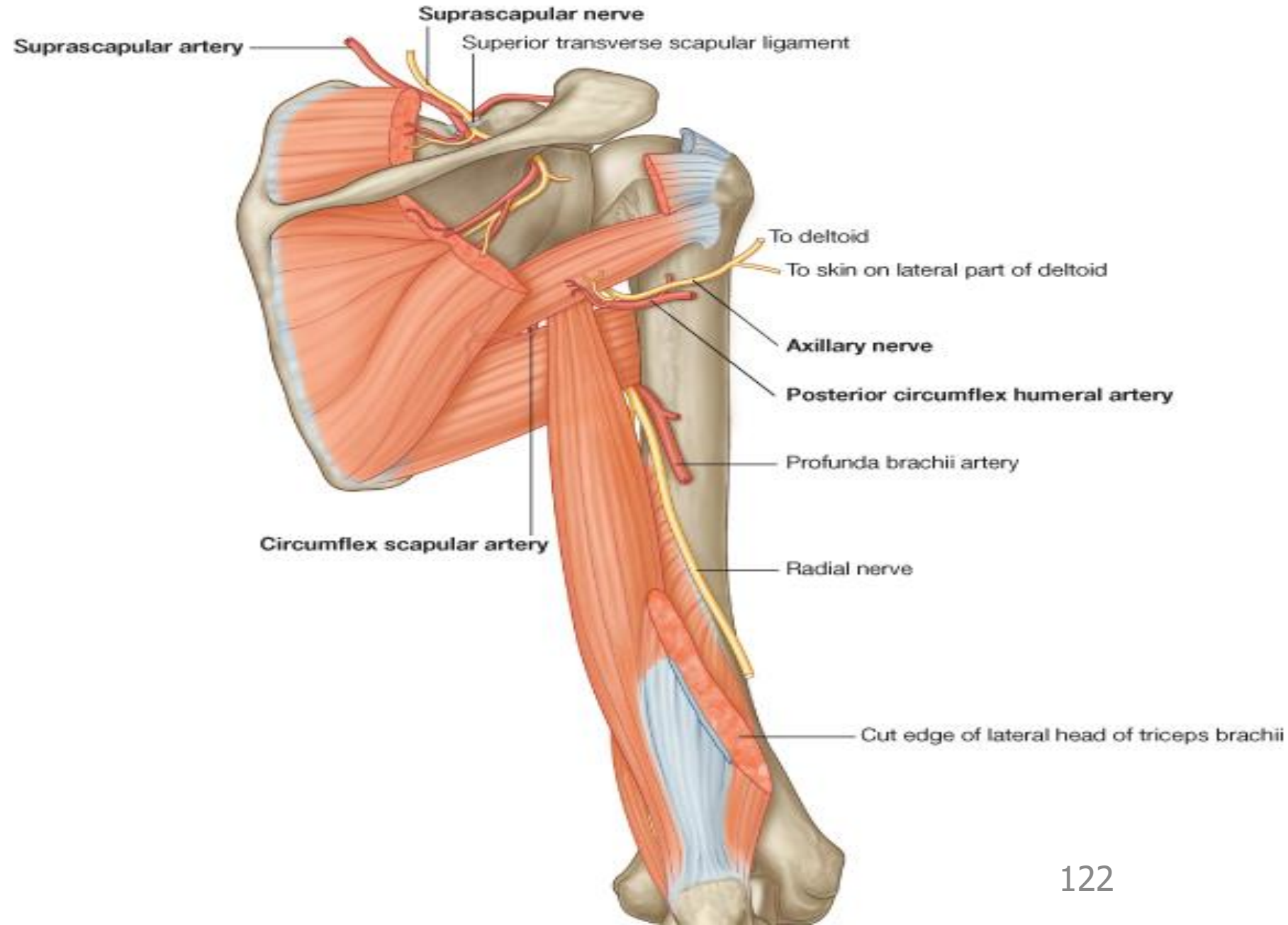


Muscles of Posterior compartment of the arm,,

❑ Quadrangular space:

- It is the passageway for nerves and vessels passing the axilla to the posterior scapular region.
- ❖ It is boundaries **are formed by:**
 - **Superiorly** : *teres minor*;
 - **Laterally** : *surgical neck of the humerus*;
 - **Inferiorly** : *teres major*;
 - **Medially** : *the long head of triceps brachii*.
- Contents : **axillary nerve and the posterior circumflex** humeral artery and vein pass through it.

Muscles of Posterior compartment of the arm,,



Muscles of Posterior compartment of the arm,,,

- **Triangular space**

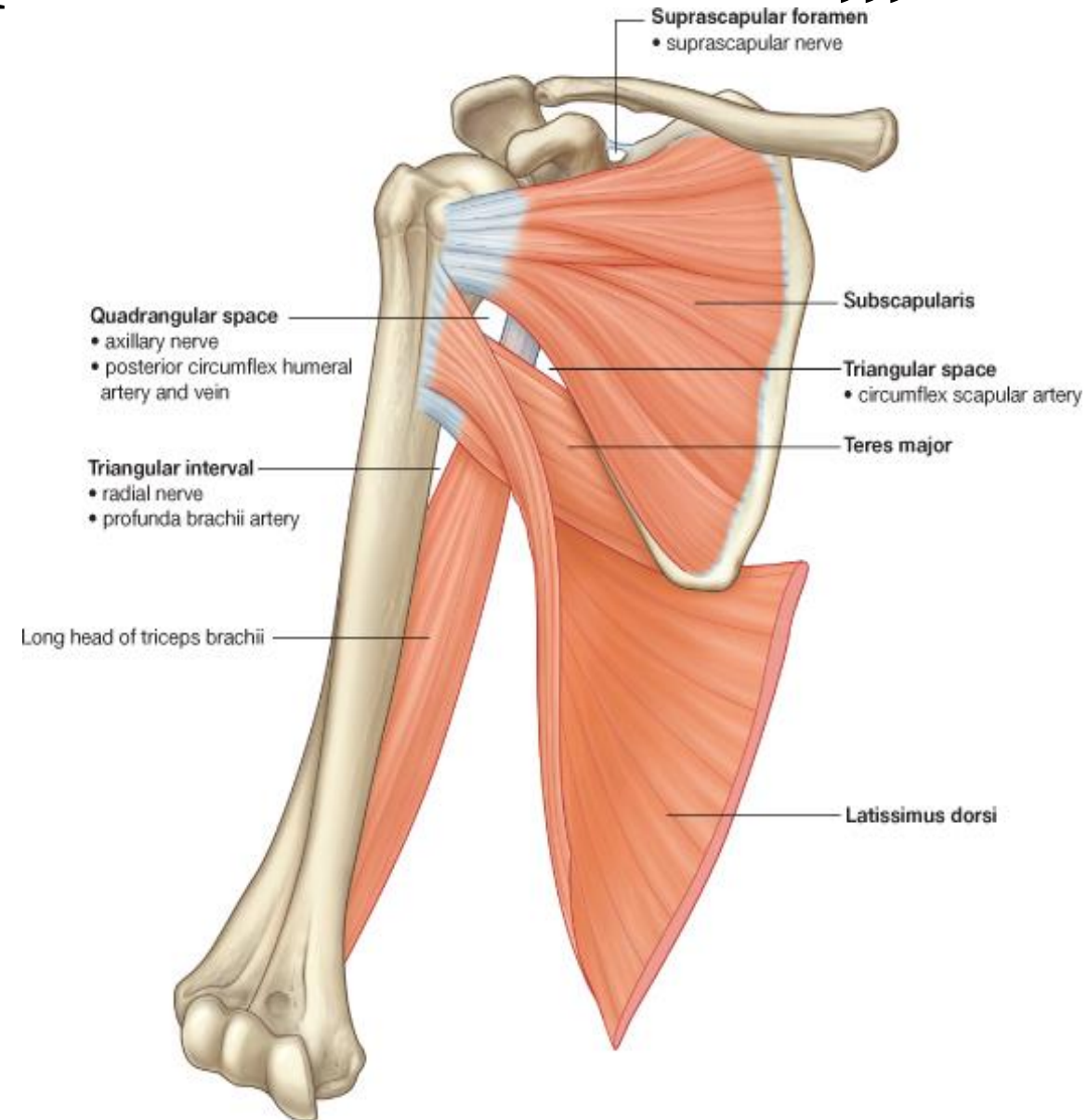
- the triangular space **is formed by:**

- **Laterally** : *long head of triceps brachii;*

- **Inferiorly** : *teres major;*

- **superiorly** : *teres minor.*

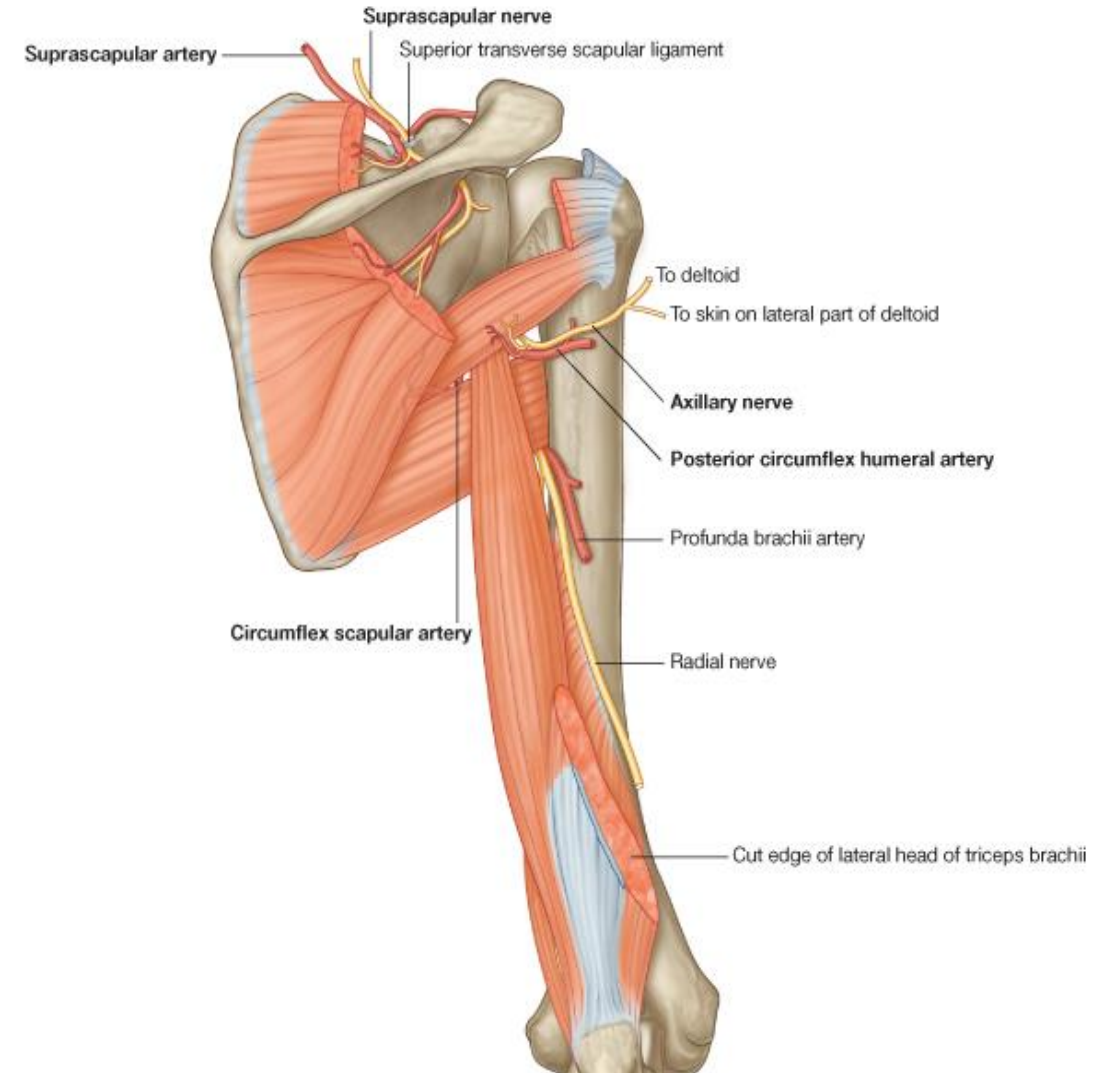
- Content : *circumflex scapular artery*
and *vein* pass through this gap



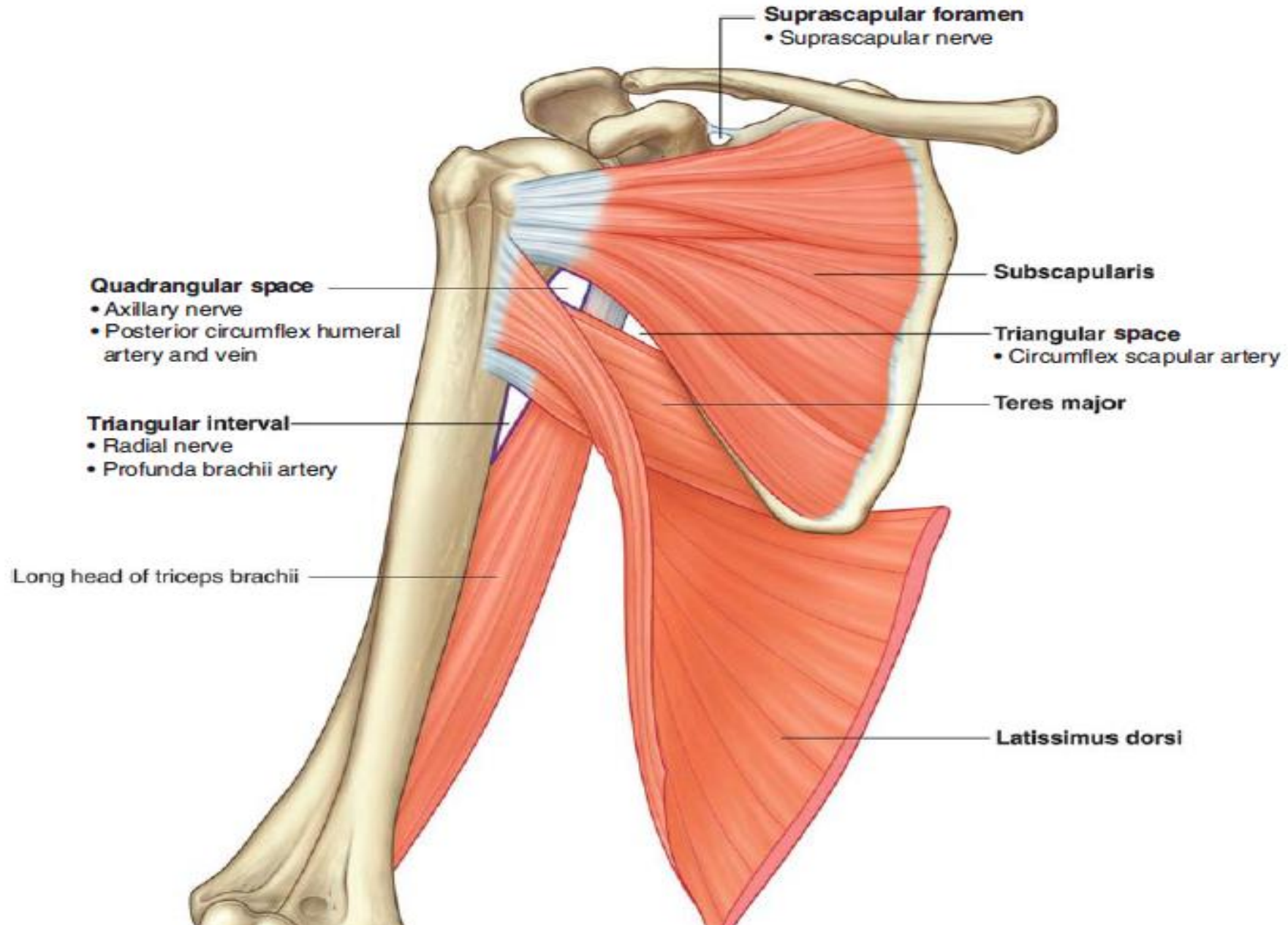
Muscles of Posterior compartment of the arm,,,

- The triangular interval is **formed by:**
- **Medially:** *lateral margin of the long head of triceps brachii;*
- **Laterally:** *the shaft of the humerus;*
- **Superiorly:** *inferior margin of teres major*

Contents : *radial nerve, the profunda brachii artery (deep artery of arm), and associated veins pass through it*



Muscles of Posterior compartment of the arm,,,



Muscles of Posterior compartment of the arm,,

Muscle	Proximal Attachment	Distal Attachment	Innervation	Main Actions
Biceps brachii	<i>Short head:</i> tip of coracoid process of scapula <i>Long head:</i> supraglenoid tubercle of scapula and glenoid labrum	Tuberosity of radius and fascia of forearm through bicipital aponeurosis	Musculocutaneous nerve (C5, C6 , C7)	Supinates forearm and, when forearm is supine, flexes elbow joint; short head flexes shoulder joint; long head helps to stabilize should joint during abduction.
Brachialis	Distal half of anterior surface of humerus	Coronoid process and tuberosity of ulna	Musculocutaneous nerve (C5–C7) and radial (C5–C7)	Flexes elbow joint in all positions
Coracobrachialis	Tip of coracoid process of scapula	Middle third of medial surface of humerus	Musculocutaneous nerve (C5, C6 , C7)	Assists with flexion and adduction of shoulder joint
Triceps brachii	<i>Long head:</i> infraglenoid tubercle of scapula <i>Lateral head:</i> posterior surface of humerus, superior to radial groove <i>Medial head:</i> posterior surface of humerus, inferior to radial groove	Proximal end of olecranon of ulna and fascia of forearm	Radial nerve (C6, C7 , C8)	Extends the elbow joint; long head steadies head of humerus when shoulder joint is abducted
Anconeus	Lateral epicondyle of humerus	Lateral surface of olecranon and superior part of posterior surface of ulna	Radial nerve (C7–T1)	Assists triceps in extending elbow joint; stabilizes elbow joint; abducts ulna during pronation

The cubital fossa

- Shallow triangular depression on anterior surface of the elbow.
- Deeply, it is a space filled with a variable amount of fat anterior to the most distal part of the humerus and the elbow joint.

❖ Boundaries:

- **Base:** an imaginary line connecting the medial and lateral epicondyles.
- **Medially:** the pronator teres.
- **Laterally:** the brachioradialis.
- **The floor:** the brachialis and supinator muscles of the arm and forearm, respectively.
- **The roof:** skin, superficial fascia containing median cubital vein, deep fascia and the bicipital aponeurosis
- **Apex:** over lapping of brachioradialis and the pronator teres

The cubital fossa,,,

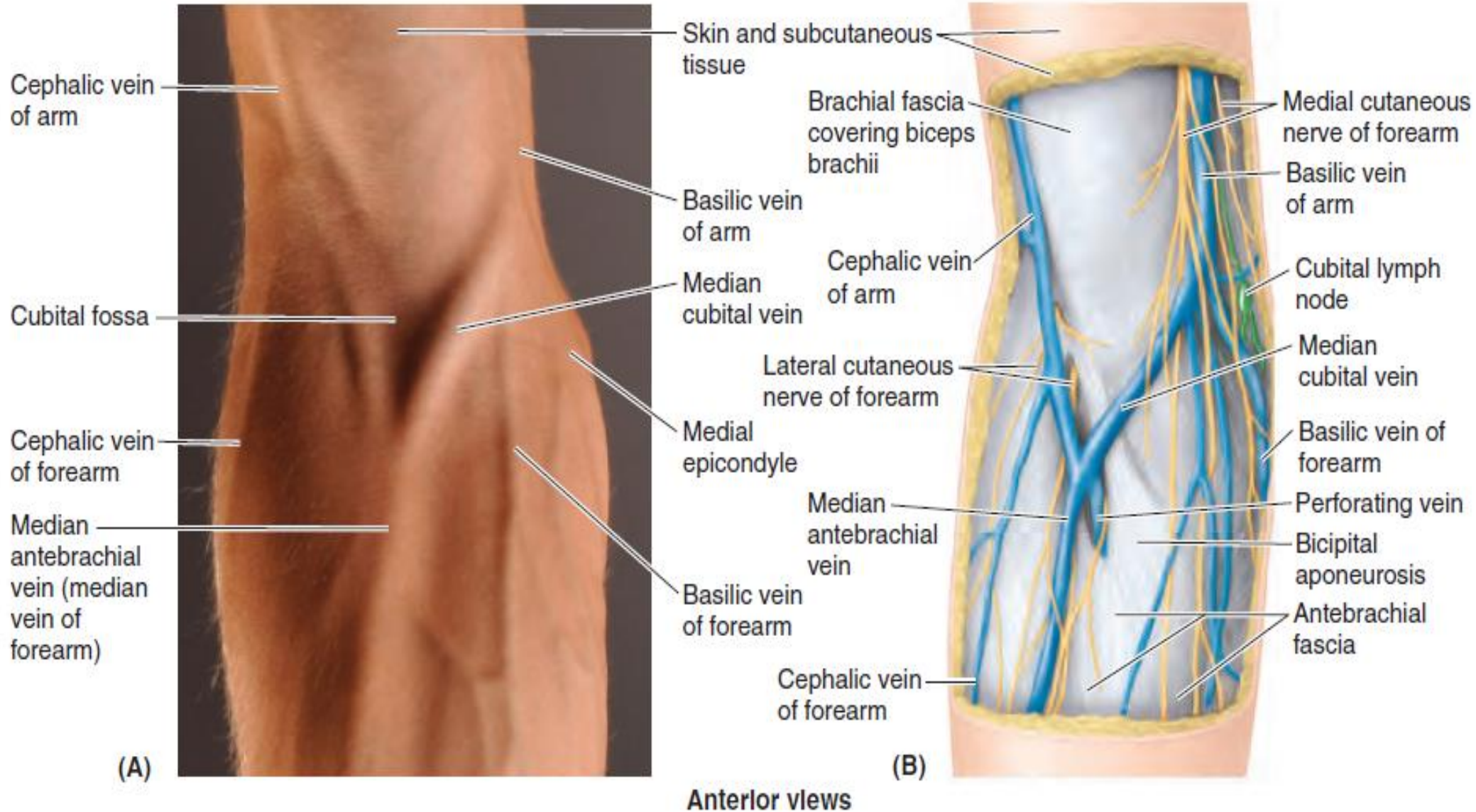


The cubital fossa,,,

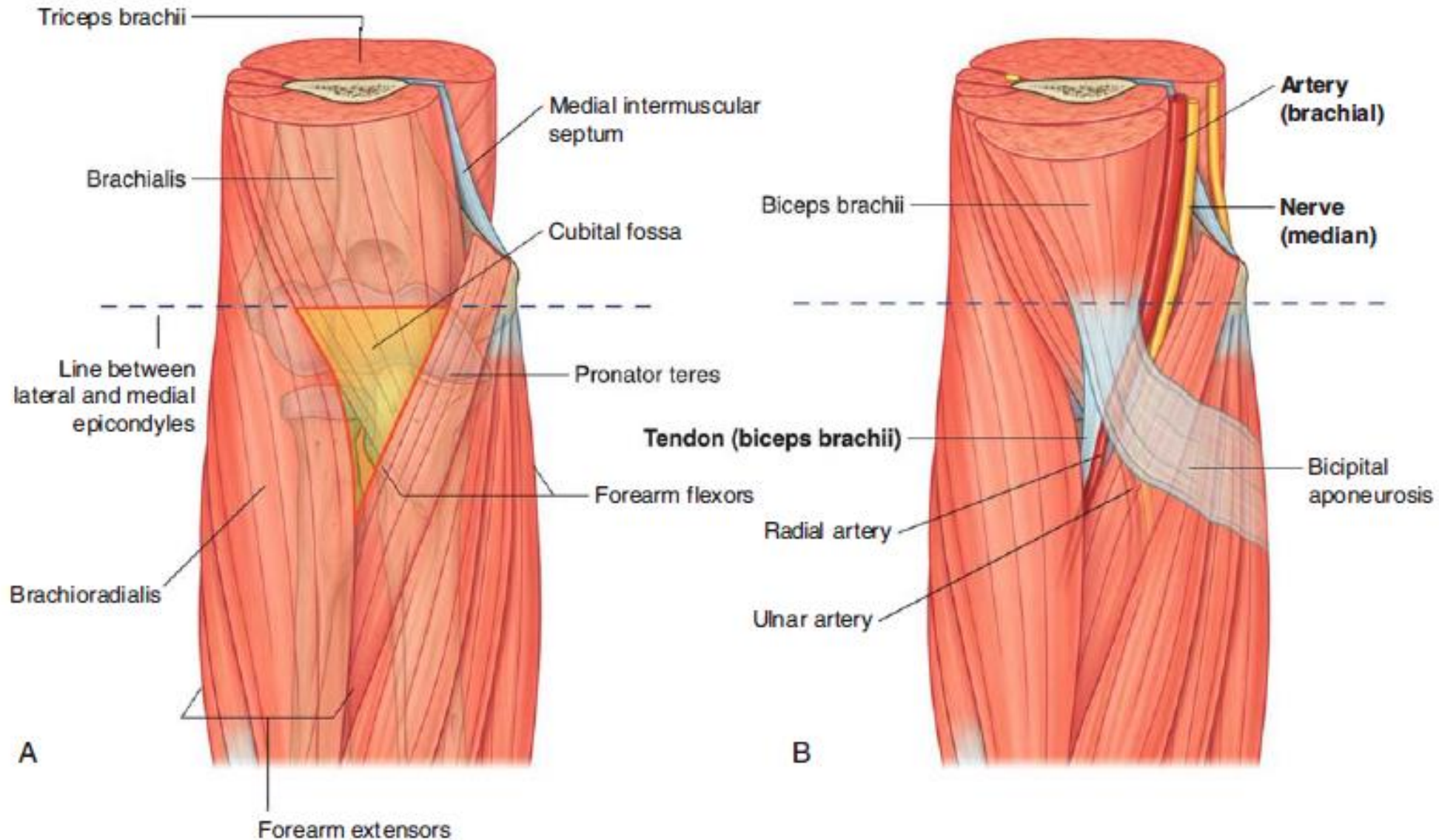
□ Contents

- ✚ Terminal part of brachial artery and branches, radial and ulnar arteries
 - ✚ Accompanying veins of the arteries
 - ✚ Biceps brachii tendon
 - ✚ Median nerve
 - ✚ Radial nerve, dividing into its superficial and deep branches
 - ✚ Median cubital vein
 - ✚ Medial and lateral cutaneous nerves of the forearm
- ❖ The **cephalic vein** is lateral, the **basilic vein** is medial, and the **median cubital vein** joins the two.
- ❖ The median vein of the forearm ends in the median cubital vein

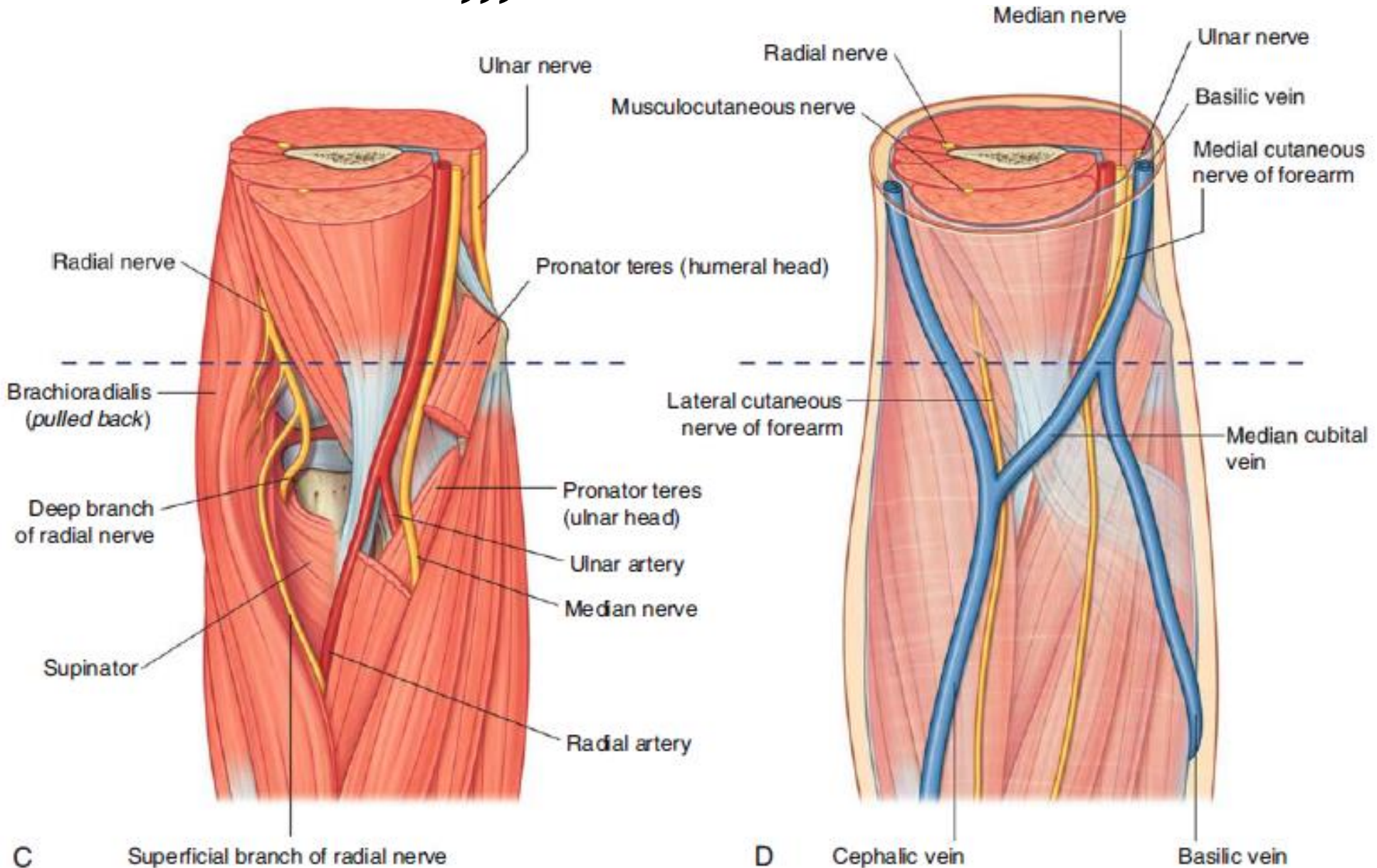
The cubital fossa,,,

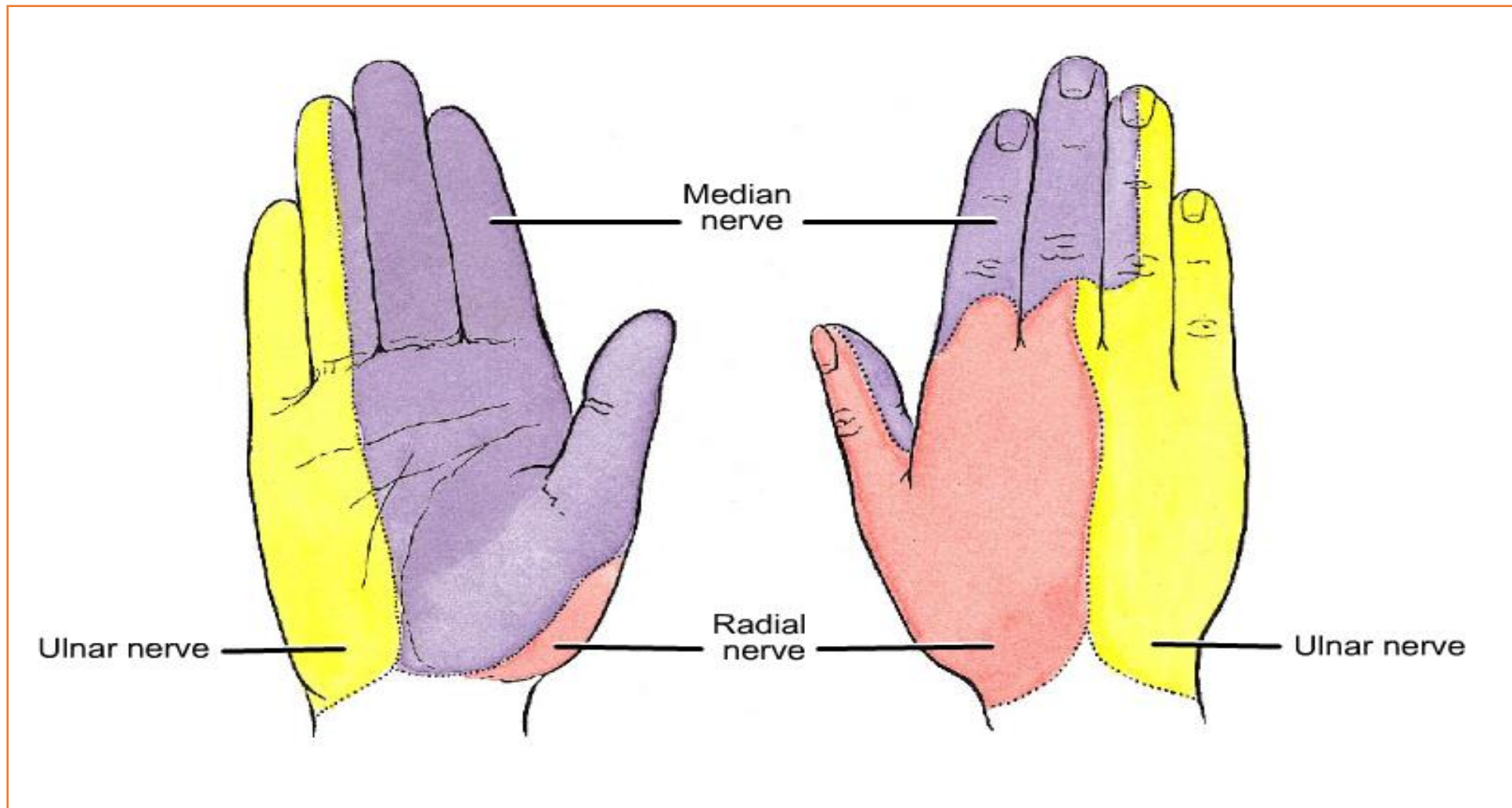


The cubital fossa,,,



The cubital fossa,,





Thank you for your attention!!!!